

Efficiency optimization of the push-belt CVT by variator slip control

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Thesis

Efficiency optimization of the push-belt CVT by variator slip control

B. Bonsen

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Efficiency optimization of the push-belt CVT by variator slip control

PROEFSCHRIFT

ter verkrijging van de graad van doctor aan de
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Preface

This thesis is part of the result of a project jointly executed by three PhD students. In the following text the project and the project context will be briefly introduced, as well as the overall goals of the project and the strategy to attain these goals.

The project

The project is started in cooperation with three parties: the Technische Universiteit Eindhoven (TU/e), Van Doorne's Transmissie (VDT) and the University of Twente (UT). Each of these parties undertakes a part of the project. The project is organized within the BTS (Bedrijfs Technologische Samenwerking) framework. In this framework companies and universities work together on projects that can stimulate economic development in the future.

The project is divided into three sub-projects. Each sub-project focusses on one research topic and is carried out by one of the participants:

1. develop a model that describes the micro-slip behavior in a variator and translate this into a dynamic measurement method. The aim of this topic is to improve the efficiency and durability of the variator (TU/e),
2. develop new materials that combine high breaking strength with good fatigue resistance and are ground breaking for both properties (VDT),
3. develop a failure mode model and a wear prediction model for a boundary lubrication contact (UT).

The work in this thesis is part of the first topic, the micro-slip research performed by the TU/e. This research, performed in cooperation with VDT, focusses on the control and actuation of the variator. In this thesis only this part of the project will be discussed.

Goals

Most developments in the field of Continuously Variable Transmissions can be divided into five areas of attention:

1. efficiency,
2. durability,
3. maximum torque,
4. costs and
5. driveability.

In this project the main focus is on efficiency, durability and maximum torque.

The overall aim of the first research topic is to improve the efficiency and operability of the pushbelt type CVT. To achieve this, three subtargets can be formulated:

- Investigate methods to detect slip in a pushbelt type variator.
- Develop a control method to control slip in a pushbelt type variator.
- Investigate alternative actuation methods that will reduce the losses associated with the actuation system in a CVT and increase the controllability of the variator.

This thesis addresses modeling and measurements characterizing slip in a pushbelt type variator, methods to detect slip in a pushbelt type variator as well as a control method to control slip in a variator. This thesis does not describe the Electro-Mechanical Pulley Actuation CVT (EMPAct CVT) [43] [86], which was also developed within the scope of this project.

Contents

1	Introduction	1
1.1	CVT Drivelines	2
1.1.1	CVT components	2
1.1.2	Belt types	4
1.1.3	Transmission load	6
1.2	Driveline efficiency	7
1.2.1	Driveability versus fuel economy	8
1.3	CVT efficiency	8
1.3.1	Mechanical system efficiency	9
1.3.2	Actuation system	9
1.4	Improvement strategy	11
1.4.1	Alternative actuation system	11
1.4.2	Clamping force reduction	12
1.4.3	Improvement potential	12
1.5	Contribution of this thesis	12
1.6	Outline of this thesis	13
2	Variator Modeling	15
2.1	Introduction	16
2.1.1	Variator geometry	17
2.1.2	Models	19
2.2	Stationary Variator Modeling	22
2.2.1	Continuous belt model	22
2.2.2	Pushbelt model	33
2.2.3	Forces in the pushbelt	34
2.2.4	Conclusion stationary model	40
2.3	Variator Transient Model	41
2.3.1	Creep-mode shifting	41
2.3.2	Slip-mode shifting	41
2.3.3	Models	42
2.3.4	Measurements	43
2.3.5	Comparison	47
3	Slip in the variator	49
3.1	Slip and Traction	49
3.1.1	Traction curve	50
3.1.2	Play in the belt	51
3.1.3	Results	54
3.2	Dynamic Variator Slip Model	55
3.3	Ratio and Slip Estimation	58
3.3.1	Estimation vs Measurement	58
3.3.2	Position measurement	59

3.3.3	Input/Output torque	59
3.3.4	Beltspeed	61
3.3.5	Modulation methods	62
3.3.6	Results	66
4	Variator system losses	69
4.1	Losses	69
4.1.1	Efficiency of the variator	70
4.1.2	Losses in the hydraulic actuation system	70
4.1.3	Losses in the variator	70
4.2	Measurements	71
4.3	Torque loss	71
4.4	Results	74
4.5	Efficiency improvement potential	74
4.6	Conclusions variator efficiency	76
5	CVT Control	79
5.1	Control problem	80
5.2	Classic Clamping-force Control	80
5.3	Ratio control	82
5.4	Slip Control	82
5.5	Simulation	86
5.6	Conclusions and recommendations variator control	88
6	First generation slip controlled variator	91
6.1	Experimental setup	91
6.2	Control implementation	93
6.2.1	Introduction	93
6.2.2	System identification	93
6.2.3	Controller tuning	94
6.3	Experimental results	95
6.4	Conclusions and recommendations first implementation	96
7	Gain scheduled PI control of slip in a CVT	99
7.1	Transmission testrig	99
7.2	Control implementation	100
7.2.1	Slip model	100
7.3	Experimental results	107
7.4	Conclusions and recommendations gain scheduled PI control	109
8	Implementation of slip control in a production vehicle	111
8.1	Nissan Primera	111
8.1.1	Engine	111
8.1.2	Transmission control	112
8.1.3	Clamping force control	112
8.1.4	Variator Properties	113
8.2	Variator Robustness	113
8.3	Comfort	116
8.4	Conclusions and recommendations Nissan Primera tests	117
9	Conclusions and Recommendations	119
9.1	Conclusions	119
9.2	Recommendations	121

A	Slip control using Linearizing and decoupling feedback	123
A.1	Linearizing and decoupling feedback	123
A.1.1	Linearizing feedback	123
A.1.2	Decoupling feedback	124
A.2	Controller design	124
A.3	Conclusions and recommendations linearizing and decoupling feedback	126
B	Equations	127
B.1	System matrices for the linearized model	127
B.2	Iterative calculation of the wrapped angle and running radii	128
B.3	Quadratic approximation of the wrapped angle and running radii	129
C	Robust PI Control	131
D	Drive cycles	133
D.1	NEDC	133
D.2	FTP75	135
E	Ratio Setpoint Strategy	137
E.1	Optimal Operation Line	138
E.2	Drivability	138
E.2.1	Shiftspeed	140
	Bibliography	143
	Nomenclature	151
	Summary	157
	Samenvatting	159
	Curriculum Vitae	161

Introduction

The Continuously Variable Transmission (CVT) is increasingly used in automotive applications. It has an advantage over conventional automatic transmissions, with respect to the large transmission ratio coverage and absence of comfort issues related to shifting events. This enables the engine to operate at more economic operating points. For this reason, CVT equipped cars are more economical than cars equipped with planetary gear automatic transmissions. Despite these advantages, V-belt type CVT's still have rather large potential in transmission efficiency, as can be seen in Figure 1.1. Also torque capacity (currently at about 350 Nm) needs expansion.

Recent developments in the field of automatic transmissions include Automated Manual Transmissions (AMT), Dual Clutch Transmissions (DCT) and the Electronic CVT (eCVT) developed for the Toyota hybrid cars. Although the AMT is very efficient with respect to transmission efficiency, the AMT has a disadvantage with respect to CVTs with regard to driveability. Shifting in AMTs causes a discontinuity in the driving torque. The DCT solves this problem by using an additional clutch and driveshaft. This however, increases the cost of the system. The eCVT uses an electric motor and a generator combined with a planetary gear to drive the wheels. This is an interesting

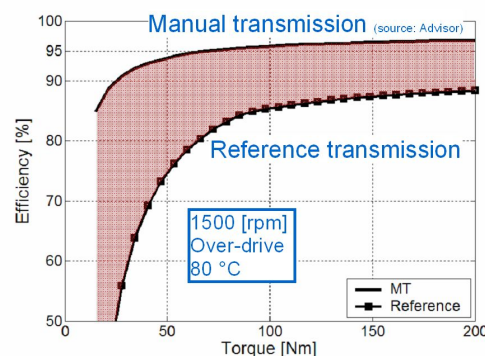


Figure 1.1: The efficiency of a reference CVT set against the efficiency of a manual transmission.

option, because there is an additional benefit of an electric power buffer. This greatly improves fuel economy, but at the same time also greatly increases the cost of the system.

New systems with increasing numbers of gears of up to 8 gears in the latest automatic gearbox decrease the advantage of the CVT with respect to the transmission ratio coverage and optimal engine operation (i.e. fuel economy). However, the increasing number of gears present additional costs and increased size of the transmissions. The pushbelt type CVT already has a large ratio coverage and unlimited number of gears.

As stated before, the main disadvantages of current CVTs are transmission efficiency and torque capacity. Therefore development of these systems is mainly focussed in these areas.

This thesis proposes to use control of slip in a pushbelt type Continuously Variable Transmission. It will be shown that control of slip in such a CVT can be used to optimize the efficiency of a CVT.

Models will be given for the forces, shifting behavior, slip and efficiency of the variator. The variator is the ratio changing device of the CVT. Several methods to estimate slip in a CVT will be discussed. Furthermore, a controller will be proposed that can be used to control slip in a CVT. A functional prototype has been produced and the results from tests with this prototype will also be discussed. A general introduction on CVT drivetrains will be given in order to describe the main focus points of this research.

1.1. CVT Drivelines

The CVT will be discussed in the form in which it is used the most: transversely mounted in a front wheel driven car. The driveline contains an internal combustion engine, a transmission, and drive shafts to the wheels. (Figure 1.2)

1.1.1. CVT components

The CVT consists of several components, i.e. an actuation system, a launching device, a drive-neutral-reverse set, a variator and a final drive reduction. These parts are discussed briefly here.

Launching device

A CVT needs a separate device for launching. Often a torque converter (TC) is used for this purpose. After vehicle launch, the TC can be locked by engaging the lock-up clutch. It then forms a fixed connection between the engine and the rest of the transmission. These components and their models are described in more detail in Serrarens [70], Lechner et al. [48] and B. Bertsche et al. [6].

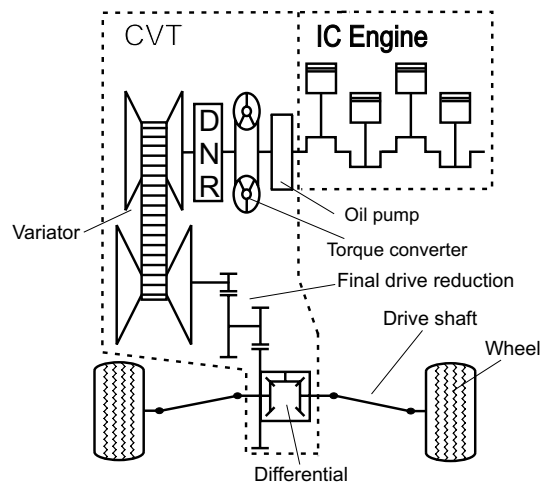


Figure 1.2: Driveline Components

Drive-Neutral-Reverse set

To enable the driver to put the driveline in neutral or to select forward or reverse driving, the CVT contains a Drive-Neutral-Reverse (DNR) set. The DNR consists of a planetary gear set and two wet plate clutches. The clutches can either couple the planet carrier to the transmission housing (Reverse) or the ring gear to the planet carrier (Drive). If none of the clutches is engaged, the transmission is in neutral.

Variator

The variator consists of a segmented steel V-belt and two shafts with conical pulleys, that forms the heart of the CVT. The belt is clamped between two pairs of conical sheaves. In the variator the transmission ratio is determined by simultaneous adjustment of the running radii of the belt on the pulleys. On each shaft, there is one fixed and one axially moveable sheave. Axial movement of the moveable sheave adjusts the gap between the sheaves and thereby the belt running radius. The input shaft of the variator is called the primary shaft, the output shaft is the secondary shaft. In Figure 1.3 the working principle of the variator is illustrated. In this figure the shifting process from the low ratio to overdrive ratio is shown. Here the front shaft is the input shaft.

Final drive reduction

The secondary shaft of the variator is connected to the differential gear via a number of gears. The differential gear distributes the torque between the two drive shafts. Together, the gears on the secondary variator shaft, the intermediate shaft and the differential gear form the Final Drive Reduction, or FDR. The wheels rotate in the same direction as the engine when the DNR is in Drive.

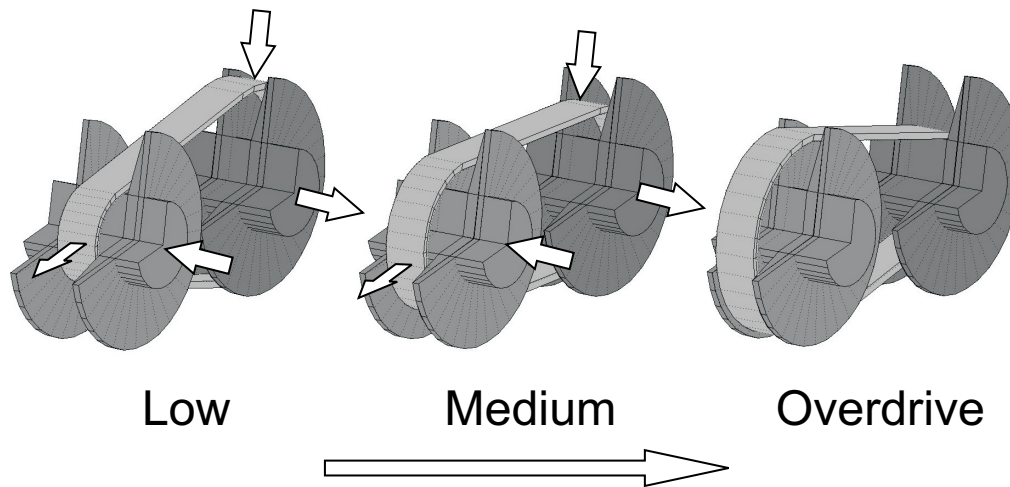


Figure 1.3: The working principle of the V-belt type variator illustrated by the shifting process from low ratio to overdrive ratio.

Actuation system

Early CVTs used a mechanical system to control clamping force and ratio [24]. The efficiency and driveability however were very poor. To achieve better controllability, hydraulic systems were developed. The advantages are a high power density and control of the speed ratio independent of the shaft speeds.

In current CVT systems the clamping force is delivered by a hydraulic actuation system. Also mechanical systems (torque cams) and electromechanical systems, for example the EMPAct CVT [86], are possible.

For a hybrid vehicle application, a CVT with electro-hydraulic actuation was developed [33]. In this setup, an electric motor drives the hydraulic system's oil pump. This is necessary in hybrid vehicles, since the IC engine is occasionally shut down and can not be used to continuously drive the oil pump.

Recently attention has been given to electro-mechanical systems. Aichikikai developed such a system for dry hybrid belts [83]. The EMPAct, developed at the TU Eindhoven, also is an electro-mechanically actuated CVT for metal pushbelts.

1.1.2. Belt types

The belt appears in several forms. The most important belt types are:

- the dry belt,
- the chain and
- the pushbelt.



Figure 1.4: Dry belt CVT as produced by Aichikikai

Dry Belt

The development of the V-belt type CVT began with rubber V-belts [24]. This type of CVT is still being developed and used. However, rubber V-belt CVTs are not well suited for automotive applications, because of their limited torque capacity. Nevertheless, there are some interesting concepts on the market, for example the Bando Avance system [83]. Dry belts are of interest because a much higher friction coefficient is established between belt and pulleys than in lubricated variants. A dry belt CVT therefore needs less clamping force and can be much smaller and lighter. This is interesting for low power applications, such as light motorcycles and small cars.

A problem that arises from the non-lubricated belt pulley contact is that there is no cooling of this contact by lubrication oil. This greatly limits the torque capacity of this type of variator. A picture of a dry V-belt CVT [83] is given in Figure 1.4.

Chain

CVT chains as developed by Luk [51] or GCI [91] consist of pins and segments. The pins are typically crowned to enhance the chain-pulley contact. Most CVT chains only have rolling contacts between the pins and have static contacts between the pins and the segments.

The chains have very little internal friction due to their internal rolling contacts (i.e. no sliding contacts). The efficiency of the chains compared to pushbelts is therefore generally higher, especially for low input torques.

Figure 1.5 gives a picture of the Luk CVT chain. This chain uses two pins per section. The pins of the chain are grouped two by two. This enables the chain to transmit the tension force by rolling and static contacts only, i.e. no sliding occurs in the chain. However, the pins need to rotate in order to change the configuration of the chain. When compressed between the pulley sheaves, this will cause a counteracting friction between the pulley and the pins. The GCI chain solves this problem by shortening one pin slightly, eliminating this friction. A disadvantage is that the number of pins is effectively halved, lowering the stiffness and the strength.

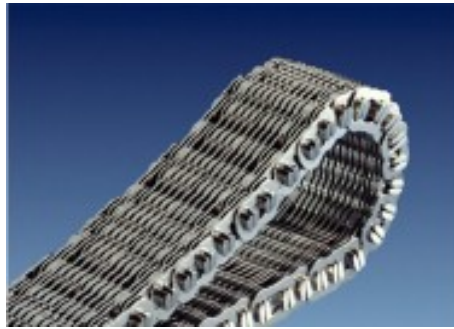


Figure 1.5: A Luk CVT chain

Typically chains produce more noise than pushbelts. This is caused by the relatively small number of pins that continuously run into the pulley. Luk overcomes this problem by varying the length of the segments. This causes the system to make noise in a wider frequency band, but with a lower amplitude [38]. GCI claims that their chain causes less noise than the Luk chain, because of the eliminated rotation under load.

Pushbelt

The Van Doorne's Transmissie pushbelt consists of blocks and bands as shown in Figure 1.6. The bands, normally 2 sets of between 9 and 12 bands each, fit tightly together, holding the blocks together. The bending stiffness of the bands is very small and may be neglected, so that only a tension force can be present in the bands. The blocks can transmit torque when they are under compression, hence the name pushbelt. The compression force can never exceed the tension in the bands, otherwise the contact between the pushbelt and the pulleys could be lost and buckling could occur.

Apart from the losses in the bearings of the shafts and losses due to slip, there are friction losses in the pushbelt. The bands and the blocks do not run at the same radius, causing a speed difference between the blocks and the bands. This results in friction losses in the pushbelt, which lowers the efficiency of the pushbelt.

Because of the continuous bending and stretching of the bands, fatigue issues are important. Fatigue resistance specifications limit the torque capacity of the variator, because the maximum clamping forces are limited.

There are much more blocks in a pushbelt than there are pins in a chain. This results in more quiet operation and higher axial stiffness. Also better resistance to wear is achieved due to the lower surface pressure between pushbelt and pulley.

1.1.3. Transmission load

At the input or engine side, the transmission is loaded with the engine torque. In a typical medium-size car, the maximum engine torque can vary between different types, but is roughly somewhere in



Figure 1.6: A metal pushing V-belt

the 150-350 Nm range. Engine torque changes can be highly dynamic, from no load to maximum torque in half a crankshaft rotation. The engine torque is filtered by a torsion damper. The engine torque can also be amplified by a torque converter.

At the output side, the transmission is loaded by the torque in the drive shafts. Apart from the reaction torque of up to 2500 Nm that results from the engines driving torque, torque peaks can occur resulting from hitting a curbstone, abruptly stopping of spinning wheels or other events occurring in the road-wheel contact.

1.2. Driveline efficiency

Figure 1.7(a) shows the energy flow in a CVT driveline. The output in the CVT driveline is about the same as the output of a manual transmission driveline [62]. For this driveline the energy flow is shown in Figure 1.7(b). The CVT losses are higher, but the engine losses are less than the losses with a manual gearbox. The transmission is assumed to have an average efficiency of around 80% in the CVT driveline. CVT main loss components are the variator loss and the hydraulic loss [35].

The main reason for the low efficiency of modern production CVTs is the high clamping force necessary to transfer the engine torque. To prevent belt slip at all times, the clamping forces in modern production CVTs are usually much higher (at least 30% or more) than needed for normal operation, i.e. without disturbances. Higher clamping forces result in higher losses in both the hydraulic and the mechanical system, i.e. increased pump losses and increased friction losses because of the extra mechanical load that is applied on all variator parts.

Excess clamping forces also reduce the endurance of the belt, since the net pulling force in this element is larger than strictly needed for the transfer of the engine power. Also the contact pressure between V-belt and sheaves is higher than strictly needed, leading to increased wear. This excess loading leads to heavier components, thereby compromising power density.

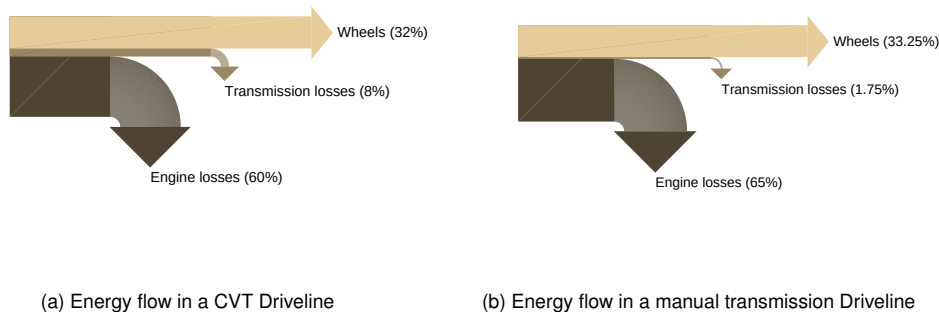


Figure 1.7: Energy flows in different drivelines

1.2.1. Driveability versus fuel economy

For every power level in an internal combustion engine, there is one speed-torque combination which achieves optimal fuel efficiency. If for all the engines power levels these points of optimal efficiency are connected, a line appears that is called the optimal operation line, or OOL. With a CVT, using its continuously variable range of transmission ratios, the OOL can be followed for high driveline efficiency. If the driveline is operating in a low power situation and the driver demands more power by pressing the accelerator pedal, the engine can be in a higher torque level within half a crankshaft rotation. However, the torque reserve on the OOL is low, and much higher power levels cannot be reached without increasing engine speed. This takes far more time than the earlier mentioned increase in engine torque, because the CVT has to be shifted and the engines inertia has to be accelerated. This time lag compromises the cars sporty feel and is regarded as a driveability disadvantage. To counteract this, the CVT is usually operated in a lower transmission ratio, letting the engine run below the OOL in a region with higher engine speed, larger torque reserve and lower engine efficiency.

1.3. CVT efficiency

Improving CVT efficiency is a key factor in improving the fuel efficiency of CVT equipped vehicles. Factors that influence the efficiency of the CVT are:

- Mechanical system efficiency,
- Actuation system losses,
- Control strategy.

In Figure 1.8 the use of engine power in a typical CVT is shown when driving at constant moderate speed. It can be seen that the actuation system takes a large part of the power and also that the

variator is not very efficient. If the clamping forces are lowered to the lowest possible value, the efficiency of the CVT improves [10] [59]. This is shown in Figure 1.9. Not only the efficiency of the variator improves, but also the power needed for the actuation system is decreased. However, the actuation system still requires a significant amount of power. If the actuation system is made more efficient this power can be reduced. The resulting power chart is shown in Figure 1.10. In total the efficiency in the chosen operating point can be increased from circa 69% to circa 87%.

1.3.1. Mechanical system efficiency

The mechanical system of a CVT consists of shafts with bearings, gears, a variator and a launching device, mostly a torque converter. The mechanical efficiency depends on the design and construction of these components, the clamping forces and the transferred torque.

Unlocked torque converters have a very limited power transmission efficiency. To overcome this problem mostly a lock-up clutch is added. When this clutch is closed, the efficiency is equal to the efficiency of the lock-up clutch. The Driveline Management System (DMS) therefore has to engage the lock-up clutch as soon as possible after vehicle launch.

The lock-up clutch as well as the clutches in the DNR set are commonly of the wet-plate type. This type usually has some slip when torque is transmitted. Although this slip is very small, it causes the clutches to have a smaller than 100% efficiency.

The bearings on the primary and secondary shafts are heavily loaded in radial direction. This load is caused by the tension in the belt, which in turn depends on the clamping forces. This load results in friction losses in these bearings.

The torque transmission in the variator is based on friction and, similar to the clutches, this results in some slip. The total amount of this speed-loss is the result of slip between the blocks and the pulleys and between the blocks and the bands in the pushbelt. The power losses in the variator depend linearly on the clamping forces.

The gears between the secondary shaft with the differential gearbox and the differential gears itself cause some power losses in the transmission. All transmissions contain a differential gearbox, so these losses are no issue if the performance of CVTs is compared to other types of transmissions.

1.3.2. Actuation system

The power to operate the actuation system of the CVT is taken from the drivetrain, so the power requirement of this system decreases the efficiency of the CVT. Several actuation principles exist:

- Hydraulic actuation,
- Electro-hydraulic actuation,
- Electro-mechanical actuation.

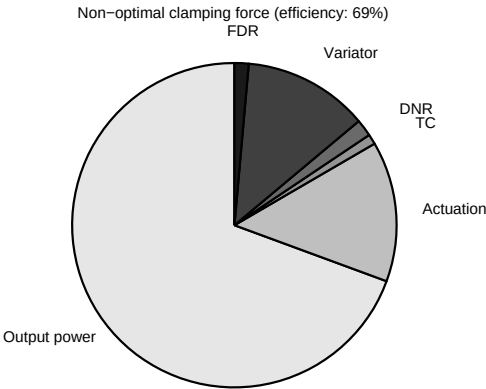


Figure 1.8: Flow of engine power in a current CVT at part-load

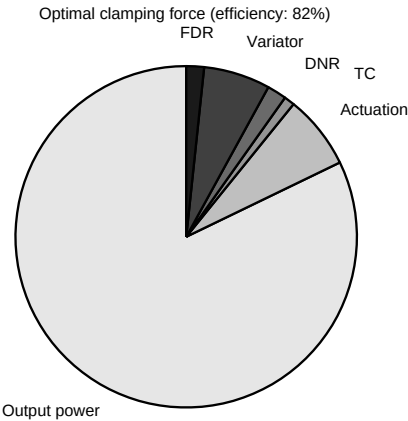


Figure 1.9: Flow of engine power in a CVT with optimal clamping force at part-load

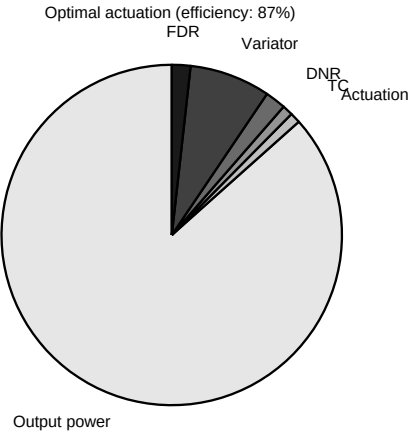


Figure 1.10: Flow of engine power in a current CVT with optimal clamping force and highly efficient actuation system at part-load

Most present day production CVTs operate on hydraulic power, generated by an oil pump connected directly to the input shaft. Normally this will be a pump with a constant flow per revolution. The flow has to meet certain specifications in all driving situations and especially should be large enough to operate the CVT at idle engine speed. Higher engine speeds cause the pump to generate too much flow for the CVT, increasing the actuation losses of the CVT. A better solution is found in dual flow pumps, which can switch between a high- and a low-flow mode.

Another possibility is using an electro-hydraulic actuation system [72]. An electro-hydraulic actuation system generates just enough flow to operate the CVT, but it needs an extra elektromotor.

Eliminating the hydraulic system altogether is possible using an electro-mechanical actuation system [86]. This solution not only eliminates the excess oil-flow, but also eliminates the leakage in the hydraulic cylinders.

1.4. Improvement strategy

To increase the efficiency of the CVT the most prominent power losses will be addressed in this project. These are:

- Actuation system losses,
- Mechanical system losses.

As mentioned before, the actuation system losses can be lowered by lowering the clamping forces and/or changing the design of the actuation system. Whereas the mechanical system losses can be lowered by lowering the clamping forces and/or changing the design of the variator.

The design of the belt and the variator lies not within the scope of this project and therefore the design of the actuation system and the level of the clamping forces are the issues to be investigated in this project.

1.4.1. Alternative actuation system

To eliminate the excess flow from the hydraulic unit and to eliminate oil leakage from the hydraulic cylinder in the variator, the approach is taken to use electro-mechanical instead of hydraulic actuation. The fuel saving potential of this approach is greater than that of electro-hydraulic and alternative hydraulic actuation systems [45] [46].

An electro-mechanical actuation system can be designed to hold a prescribed ratio without power consumption, using a self-braking (worm-wheel) transmission or to hold the ratio with a certain motor torque. This last method has less friction and a faster response, but uses some power even when not shifting.

Electro-mechanical systems can be much stiffer than low-pressure hydraulic systems. This is a benefit for ratio and clamping force control. The electro-mechanical CVT developed by van de Meerakker et al. [86], the EMPAct CVT, is an example of such a CVT.

1.4.2. Clamping force reduction

Current CVTs use over-clamping to avoid slip. Over-clamping means that more clamping force is used than is necessary to transfer the input torque to the output shaft without excessive slip. This over-clamping gives a margin of safety to overcome a priori unknown disturbances. Using more clamping force than necessary also means more friction than necessary due to the higher forces. Furthermore, the actuation system has to provide a higher force. In the case of hydraulic actuation system this means flow at a higher pressure and thus higher losses.

To avoid over-clamping, an alternative method must be used to deal with the disturbances. If no over-clamping is used, then every disturbance will cause severe slip events which could potentially damage the variator [89]. Alternatively, measurements of the actual torque or of the slip in the variator can be used to control the clamping force with lower or no over-clamping while reducing the risk of excessive slip.

Lowering the safety margin and using feedback control will demand higher bandwidth for the clamping force actuation system.

1.4.3. Improvement potential

The potential improvement in terms of fuel economy depends on the driving conditions. In this project the NEDC cycle (see also Appendix D) is used as a benchmark. The transmission efficiency, however, is optimized over the whole working range of the CVT.

The improvement goals are:

- 10% overall improvement on the NEDC cycle,
- 25% less mechanical losses by clamping force reduction,
- 75% less actuation power needed by combining lower clamping forces with a highly efficient actuation system.

The reference transmission used for the improvement comparison is the CK2 transmission from Jatco [40]. This transmission is sold in high volumes throughout the world.

1.5. Contribution of this thesis

In this thesis the use of slip as the major control variable in a pushbelt type variator will be proposed as a method to reduce the required clamping forces. If slip control is to be used, then a number of questions need to be answered: what happens when a variator starts slipping, how can slip in a CVT be measured or detected, how does slip influence the efficiency of the variator and can slip in a variator be controlled in currently available automotive CVTs? These are questions that are addressed in this thesis.

Modeling and simulation combined with experimental data will be used to gain insight in the working principles of the variator, especially with respect to slip in this variator. Methods for estimating the slip in a pushbelt type variator will be evaluated and a control law will be designed using these models and the measurement data.

Models are shown for the torque transmission, for the variator transient behavior and the for slip. Experimental data is obtained by numerous experiments on transmission testtrigs and in test vehicles. These tests give insight into the working principles of the V-belt type variator, provide input data for the models and make it possible to evaluate these models.

The measurement data is used to determine the optimal operating point of a pushbelt type variator. Measurements of the traction curve, i.e. plots of the transmitted torque versus slip, are combined with measurements of the variator efficiency. Control of slip in a variator is enabled by accurate estimation of slip and accurate models of the variator. Using the models, a method is developed to accurately control the slip.

Simulations are used to evaluate the control method. These simulations give insight into the possibilities of slip control without the limitation of one particular implementation. Apart from the simulations, the proposed control method is implemented and tested in laboratory conditions and in two test vehicles. Some results from these implementations will also be shown.

1.6. Outline of this thesis

The flowchart of the development process of the slip controlled CVT is graphically shown in Figure 1.11. First, fairly realistic, complex models are derived and evaluated, using data from experiments on the reference transmission. Next, simplified models are used to design a controller for slip in a variator. The designed controller is tested in simulations with the models and implemented first on a beltbox testrig, then in a prototype transmission, and finally in a test vehicle, which are all tested in the laboratory.

In Chapter 2 models for several aspects of the V-belt type variator will be discussed. Stationary models for the forces in the belt and on the pulleys in stationary conditions (no shifting) and transient models describing the ratio changing behavior are presented.

In Chapter 3 a model for the slipping behavior of the variator will be discussed and several possible methods are described to estimate the slip. These methods include pulley position measurement, measurement of the running radius of the belt on the pulleys, torque measurement, belt speed measurement and speed modulation methods.

In Chapter 4 an efficiency model describing the efficiency of the variator in all operating points will be shown. The parameters for this model are derived from experimental data. From this data and the model the optimal value for the slip in the variator can be obtained.

Chapter 5 describes the variator control problem. The classic clamping force control system is reviewed. Simulations are shown illustrating the problems associated with clamping force control.

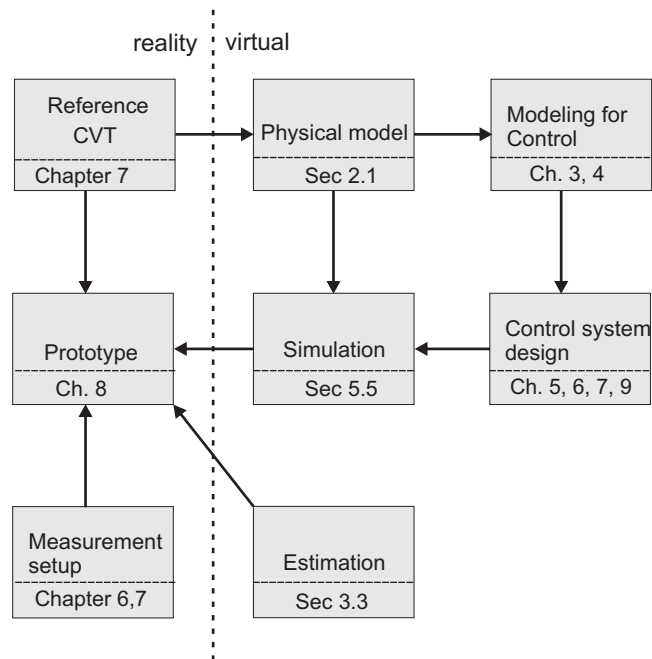


Figure 1.11: Information use in the design process

In Chapter 6 the first proposed slip control algorithm is evaluated. The test setup is discussed and the design and implementation are explained. Some results are shown from the measurements.

In Chapter 7 the implementation on the transmission testrig in a production CVT is explained. The design and implementation in this transmission is explained and the results are discussed.

Chapter 8 gives the experimental results of the implementation of slip control in the test vehicle, the Nissan Primera, and gives an evaluation of the results.

Finally, some conclusions and recommendations are given in Chapter 9.

Variator Modeling

Understanding of the basic mechanical properties of the V-belt variator is essential to improve its performance. Modeling the variator can give this insight and the resulting models can be used to study the influences of the design parameters of the variator. For controller design and verification, especially for the control of the transmission ratio and for variator simulation, models of the stationary and transient behavior of the variator are needed. The clamping forces and especially the ratio of clamping forces are of interest here, therefore the forces in the variator will be studied in this chapter.

In Section 2.1 a general introduction is given to the principles of the V-belt type variator. An overview of the literature on this topic is given. Next, the geometry of the variator is discussed. Because of the importance of friction in variators, a few friction models are discussed. These friction models will be compared in the belt model in Section 2.2.

In Section 2.2 the clamping force ratio for stationary conditions, i.e. when the variator is not shifting, is studied. Two models will be presented, a continuous belt model and a pushbelt model. A parameter sensitivity analysis and an experimental model verification will be given for both models. In Section 2.3 the shifting behavior of the variator is discussed. Several models from literature will be described. A qualitative comparison of these models will be given and experimental data will be shown.

Models for other drivetrain components found in common CVT drivetrains are described for example by Lechner and Naunheimer [6] and by Serrarens [70].

Although models that describe continuous belts and chains will be discussed, all measurements have been performed on the metal pushing V-belt of Van Doorne's Transmissie. Although focus of this research is on this type of belt, the continuous belt model that will be described, is also applicable to other types of belts or chains.

2.1. Introduction

The variator of a CVT is a device that uses friction to transmit power from a driving pulley set to a belt and then from the belt to the driven pulley set. The interaction between pulleys and belt determines the forces acting on the pulleys and the belt. These forces cause bending of the pulleys and shafts and elongation of the belt.

This deformation of pulleys and belt influence the efficiency and power transmission of the variator. For controller design and verification the clamping forces needed for torque transfer and ratio changing and the relation between clamping force and efficiency are important.

Because of the friction and of the various finite stiffnesses involved, the power transmitting mechanism of a variator poses a mathematically complex modeling problem. Most models take only a part of the mechanism into account. For example Gerbert [31] and Van Rooij [92] assume the pulleys and belts to be rigid. Also they assume the friction force to behave like Coulomb friction. Kobayashi [47] and Asayama [5] have proposed models for pushbelt type variators. In their models they assume rigid pulley sheaves and rigid blocks and bands in the pushbelt. The main difference with the continuous belt models is the compression force that is present between the blocks of the pushbelt.

Others include a model for the flexibility of the pulleys like Srnik and Pfeiffer [79], Tenberge [84] and Sattler [67]. Including the flexibility of the pulley increases the complexity of the model, but also increases the accuracy of the results compared to measured data [84]. Modeling pulleys with finite stiffness can be very complex and thus cause time consuming calculations. Tenberge [84] uses finite element analysis for calculating the bending of the pulley. Doing the finite element calculations during the iterative calculation process would make this calculation very slow. To overcome this problem he calculated the resulting bending for one block element separately for different tension forces. During the calculation of the tension force distribution he sums all the precalculated pulley deformations of the individual blocks. Using iterations he obtains a solution assuming the deformations are independent. Srnik [79] uses a rigid pulley, but simplifies the pulley bending using linear springs in the shaft and the moveable pulley.

Models of the transient variator behavior, i.e. shifting, include the work of Ide [36], [37], Shafai [71], Sorge [78] and Carbone [18]. Ide showed for the shifting behavior that the rate of change of speed ratio is dependent on the rotational speed of the input shaft. Shafai on the other hand assumes no relation with the speed of the input shaft. He assumes a viscous damping related to the pulley movement. Sorge [78] and Carbone [18] studied the influence of pulley bending on the transient behavior of the CVT. The latter argues that the rate of change of speed ratio depends on the bending of the pulley, the input shaft speed and a logarithmic function of the clamping forces.

In this chapter an analysis of existing models of the forces acting in a (push)belt type variator will be made. A continuous belt model will be compared to a (continuous) pushbelt model. A parameter sensitivity study is made on both models. The parameters of the models are estimated using mea-

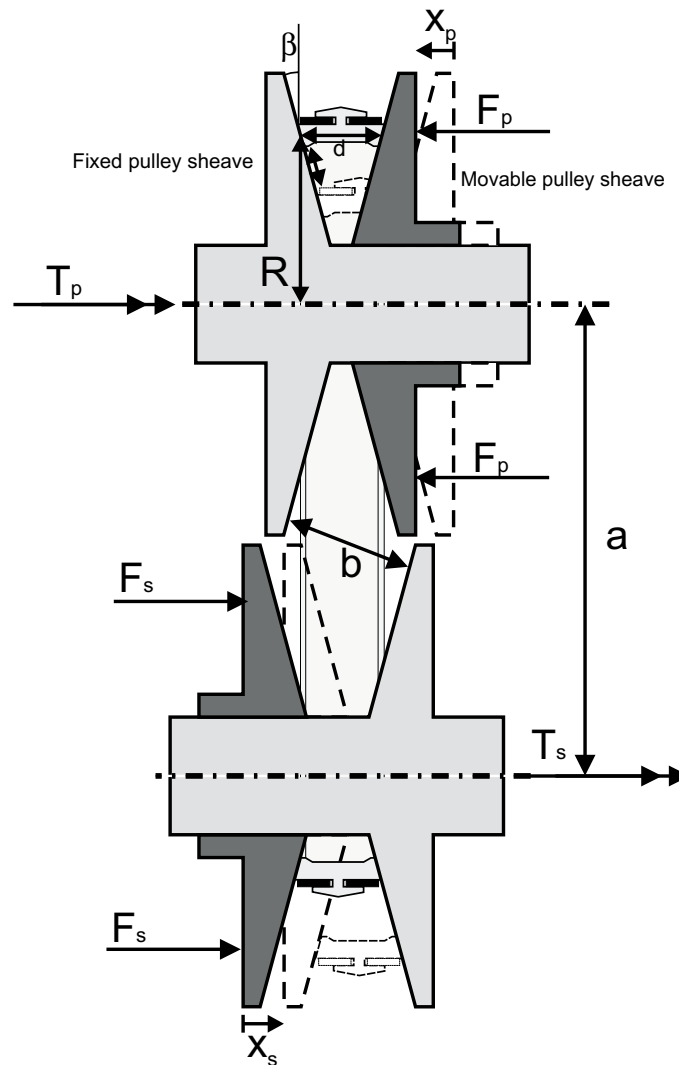


Figure 2.1: Pulley arrangement

sured data. The outcome of the models is also compared this data to evaluate their effectiveness. Furthermore, several variator transient models are compared in this chapter. Also the outcome of these models is compared to measured data.

2.1.1. Variator geometry

The V-belt type variator appears in a few different forms. The difference is mostly the shape and materials used in the belt or chain and the shape of the pulleys.

Pulleys

The pulley set on the input shaft, i.e. the engine side of the transmission, is referred to as the primary pulley, the pulley set on the output shaft is called the secondary pulley. Each pulley consists of a fixed and a moveable pulley sheave, as shown in Figure 2.1. The primary and secondary moveable sheaves are on opposite sides of the belt, as also shown in Figure 2.1. Because mostly

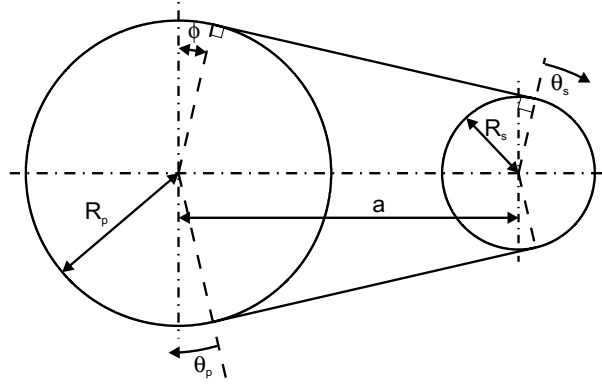


Figure 2.2: Geometric belt configuration

only one part of the pulley moves, the axial position of the belt is not constant. Moreover, the movement x_p of the primary moveable sheave is not exactly the opposite of the movement x_s of the secondary moveable sheave. Geometric relations of the belt are given in Equations (2.1) through (2.4). In these equations it is assumed that the path of the belt on the pulleys is part of a circle. This assumption implies that the pulleys and belt are assumed to be rigid.

The length of the belt can now be calculated with:

$$L = R_p(\pi + 2\phi) + R_s(\pi - 2\phi) + 2a \cos \phi \quad (2.1)$$

In this equation L represents the length of the belt, R_p represents the primary running radius, R_s represents the secondary running radius, ϕ is the angle between the centerline connecting the pulley shafts and the straight part of the belt as shown in Figure 2.2 and a represents the shaft center distance between the primary and secondary shafts.

For ϕ in Equation (2.1) the relation can be found to be:

$$R_p - R_s = a \sin \phi \quad (2.2)$$

Furthermore, the position of the primary and secondary pulley are given by:

$$x_p = 2(R_p - R_{mp}) \tan \beta \quad (2.3)$$

$$x_s = 2(R_s - R_{ms}) \tan \beta \quad (2.4)$$

In this equation R_{mp} and R_{ms} represent the minimum running radii of the primary and secondary axis respectively. The difference in the movement of the primary and secondary moveable pulley sheaves, $x_p - x_s$, causes the axial position of the belt to be slightly misaligned at both pulleys.

Two dimensionless variables can be defined that describe the state of the variator, the geometric and speed ratio respectively.

Definition 1 r_g is the geometrical transmission ratio, which is defined by:

$$r_g = \frac{R_p}{R_s} \quad (2.5)$$

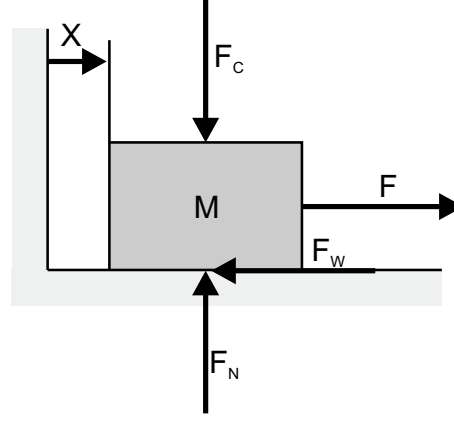


Figure 2.3: Moving block with friction

Definition 2 r_s is the transmission speed ratio, which is defined by:

$$r_s = \frac{\omega_s}{\omega_p} \quad (2.6)$$

In this equation $\omega_p = \frac{d\theta_p}{dt}$ is the rotational speed of the primary shaft and $\omega_s = \frac{d\theta_s}{dt}$ the rotational speed of the secondary shaft.

2.1.2. Models

As discussed in 2.1, for the calculation of the clamping force, the calculation of the belt tension, slip in the variator and traction between the belt and the pulley, several models have been developed. First, stationary models, that do not consider ratio changing effects, give insight into the tension force and compression force distributions and the required clamping forces needed for this equilibrium. Second, transient variator models that consider variator ratio changing, i.e. shifting, and can be used to predict the rate of change of the ratio.

The most detailed variator model accounts for the elastic deformation of belt and pulleys. This gives the most detailed analysis, but also the highest level of complexity.

All models use a model for friction. Several friction models are discussed in the next paragraph.

Friction

Friction plays a very significant role in modeling the behavior of a CVT. The friction force is the force counteracting a relative movement in a surface to surface contact, as shown in Figure 2.3. The relative motion of the surfaces is: $v = \dot{x}$. In this figure F_w is the friction force, F_c is a clamping force that hold the block on the surface, F is a pulling force driving the block and F_N is the contact normal force.

Friction is a subject of study for a very long time. Important works include that of Leonardo da Vinci [21], who found that the friction force was independent on the area of contact and Amontons [4], who found the relationship between the contact pressure and the friction force. Coulomb [20] found

that the friction force was independent on the relative velocity. The often used simple friction model, Coulomb friction, is described by:

$$|F_w| \leq \mu_c F_N, \text{ if } v = 0 \quad (2.7)$$

$$F_w = \mu_c F_N \text{sign}(v), \text{ if } v \neq 0 \quad (2.8)$$

where μ_c is the Coulomb friction coefficient.

A graphical representation is given in Figure 2.4(a). Coulomb friction does not take into account the decrease in friction when a stick-slip transition occurs. The Coulomb friction model causes difficulty in numerical simulations when the speed difference v becomes zero, because there is a change in the number of states in the system causing a discontinuity in some situations. In the case of zero sliding velocity the friction force is not defined by the friction coefficient, but must be determined otherwise. In numerical calculations the visco-plastic friction model, which will be discussed later, is often used with a very high viscous damping to avoid undetermined states.

To include static friction in the Coulomb friction model, a different friction coefficient ($\mu_s > \mu_c$) is taken at $v = 0$. This is shown in Figure 2.4(b). The model now becomes:

$$|F_w| \leq \mu_s F_N, \text{ if } v = 0 \quad (2.9)$$

$$F_w = \mu_c F_N \text{sign}(v), \text{ if } v \neq 0 \quad (2.10)$$

where μ_c is the friction coefficient for sliding motion (slip) and μ_s is the friction coefficient when there is no sliding motion (stick).

Viscous damping models the movement of an object through a viscous fluid. With viscous damping no stick exists. A graphical representation is given in Figure 2.4(c). Viscous damping is described by:

$$F_w = c_0 v \quad (2.11)$$

where c_0 is the viscous damping constant.

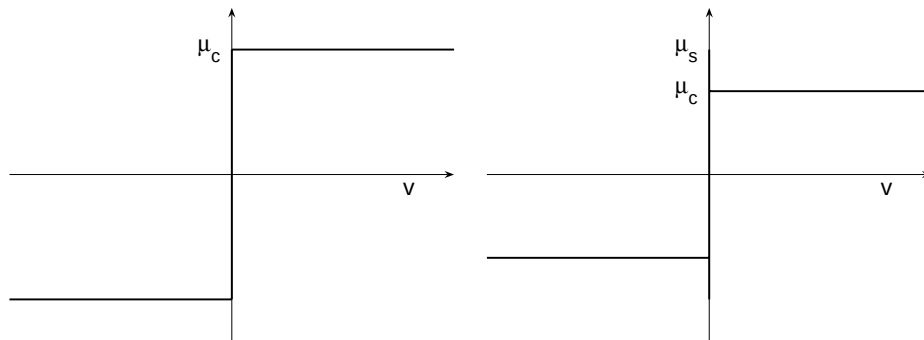
Viscous damping is not likely to occur in a variator, because a lubricated metal to metal contact is highly influenced by the normal force, which is not included in viscous damping.

A combination of both the Coulomb friction model and the viscous friction model called the visco-plastic friction model gives a more realistic behavior for lubricated contacts and because it is a continuous model, numerical problems are easier to solve. The friction can be calculated with:

$$F_w = \text{sign}(v) \min(c_0 |v|, \mu_c F_N) \quad (2.12)$$

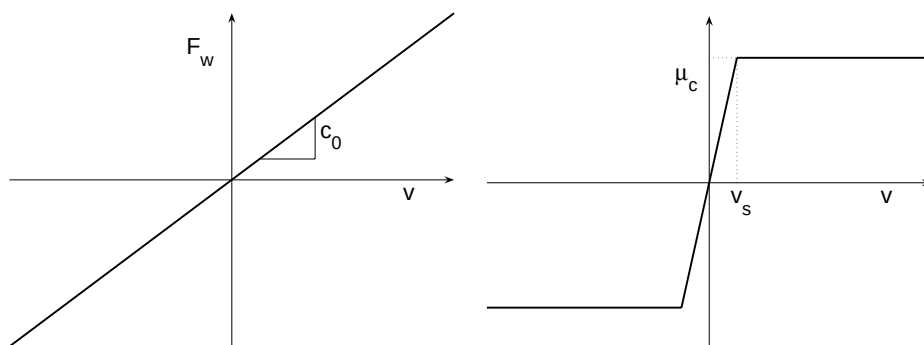
where c_0 is the viscous damping constant for low sliding velocities and μ_c is the friction coefficient for higher sliding velocities.

For lubricated steel-steel contacts the LuGre friction model was developed [57]. This model introduces an extra state that models the dynamics of the friction contact. If this extra state is omitted this



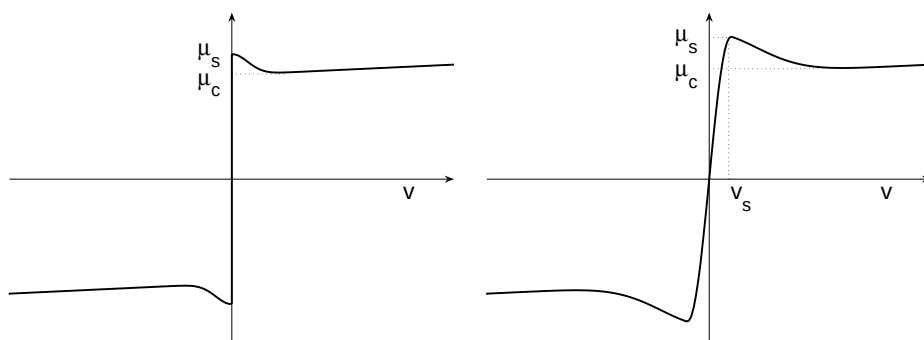
(a) Coulomb friction model

(b) Coulomb friction model with static friction



(c) Viscous friction model

(d) Visco-Plastic friction model



(e) Stribeck friction model

(f) Continuous friction model

Figure 2.4: Friction models

models still gives a good description of a lubricated contact, a good approximation of the Stribeck curve [80] (Figure 2.4(e)):

$$\begin{aligned} g &= a_0 + a_1 \exp\left(-\left(\frac{v}{v_1}\right)^2\right) \\ f &= c_0 v \\ F_w &= (\text{sign}(v)g + f)F_N, \text{ if } v \neq 0 \end{aligned} \quad (2.13)$$

$$F_w \leq (a_0 + a_1)F_N, \text{ if } v = 0 \quad (2.14)$$

In these equations the variables a_0 , a_1 , v_1 and c_0 are constants describing the Stribeck curve.

For numerical simulations models that are continuous at zero velocity are desirable. This model can be extended to be continuous at zero velocity by replacing the sign function with a continuous function. A smooth approximation of the sign function is found in the arctan function [87] [88]. The result is shown in Figure 2.4(f). This friction model is described by:

$$\begin{aligned} g &= a_0 + a_1 \exp\left(-\left(\frac{v}{v_1}\right)^2\right) \\ f &= c_0 v \\ F_w &= \left(\frac{2}{\pi} \arctan(\epsilon_w v)g + f\right)F_N \end{aligned} \quad (2.15)$$

In Figure 2.4 the different models are plotted.

2.2. Stationary Variator Modeling

Stationary variator models assume that the variator is in equilibrium and is not changing ratio. These models give insight into the forces in the belt and the normal forces acting on the pulleys. With the normal forces and a suitable friction model also the torque transmission can be calculated. The required clamping forces to transmit a certain amount of (engine) torque can also be derived. The first model that will be examined is the continuous belt model developed to describe the rubber V-belt variators. Next, a model for the metal pushing V-belt will be shown. These models do not consider pulley bending.

2.2.1. Continuous belt model

The continuous belt model considers the belt to have very little bending stiffness. Therefore the bending moments can be neglected. Furthermore, the pulley stiffness is considered to be much higher than the stiffness of the belt in longitudinal direction and is therefore also neglected. So, only stiffness of the belt in the longitudinal direction is taken into account. Furthermore, the inertial forces are omitted here. Several authors have discussed this topic. Among others Gerbert [31] and van Rooij [92] published detailed models describing a continuous belt.

The forces acting on a segment of the belt are shown in Figure 2.5. In this figure q_N is the distributed

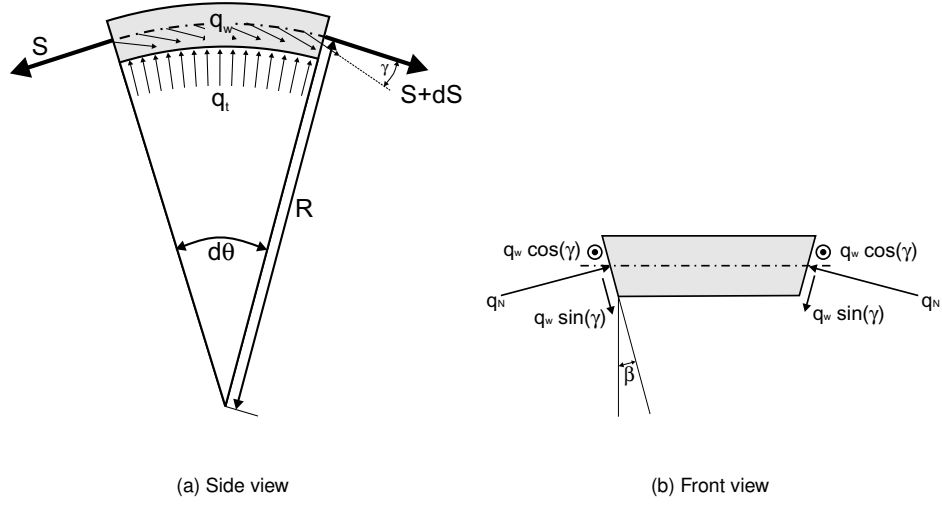


Figure 2.5: Forces acting on the belt according to the continuous belt model

normal force acting on the belt segment, S is the tension force in the belt, q_t is the radial component of the normal force distribution, q_w is the friction force in the belt pulley contact, γ is the angle between the radial direction and the direction of the friction force and β is the pulley groove angle. If the belt is in an equilibrium state, then the following equations hold:

$$\Sigma F_t = 0 \quad (2.16)$$

$$\Sigma F_r = 0 \quad (2.17)$$

In this equation F_t represent the tangential force components and F_r the radial force components. For the segment, shown in Figure 2.5, the following equilibrium equations are found:

$$2Rd\theta q_t - d\theta S - 2Rd\theta q_w \sin \gamma \cos \beta = 0 \quad (2.18)$$

$$dS + 2d\theta R q_w \cos \gamma = 0 \quad (2.19)$$

From these equations it follows that for the tension in the belt holds:

$$S = 2Rq_t - 2Rq_w \sin \gamma \cos \beta \quad (2.20)$$

$$\frac{dS}{d\theta} = -2Rq_w \cos \gamma \quad (2.21)$$

The normal force in tangential direction and the friction force are given by:

$$q_t = q_N \sin \beta \quad (2.22)$$

$$q_w = q_N \mu(v) \quad (2.23)$$

Using these equations together with Equations (2.20) and (2.21) the following differential equation can be obtained:

$$\frac{dS}{d\theta} = -\frac{S \mu \cos \gamma}{(\sin \beta - \mu \sin \gamma \cos \beta)} \quad (2.24)$$

To calculate a solution to this differential equation γ and μ need to be known for all values of θ .

If it is assumed that the belt runs in a circular path on the pulleys, i.e. no spiral running occurs, then γ is equal to zero. This greatly simplifies the differential equation, but information for the friction coefficient is needed to find a solution.

When a Coulomb friction model is used the tension force distribution the model follows the Eytelwein formula [29]. For an arbitrary friction model, the tension force distribution is given by:

$$\frac{dS}{d\theta} = \frac{-\mu(v)}{\sin(\beta)} S \quad (2.25)$$

To calculate the friction coefficient the relative motion between belt and pulley has to be known. From the equilibrium equations follow that when $\mu(v) = 0$, the belt tension does not change. However, if the belt tension is greater than zero the opposite is also true: if the belt tension does not change, then the friction coefficient must be zero. If the belt is not infinitely stiff, then a change of the tension in the belt should also change the strain in the belt and thereby changing the length of the belt, causing a relative motion between belt and pulley. So for any friction model must hold that $\mu = 0$ for $v = 0$. The strain of the belt can be calculated using Equation (2.26):

$$\epsilon = \frac{S}{AE} \quad (2.26)$$

and its derivative $d\epsilon$ with respect to the belt-tension:

$$\frac{d\epsilon}{d\theta} = \frac{dS}{d\theta} \frac{1}{AE} \quad (2.27)$$

With Equation (2.28) the relative motion (v) between belt and pulley can be calculated with the known belt strain (ϵ) and the belt strain at the belt entry point (ϵ_0):

$$v = \omega_p R_p (\epsilon - \epsilon_0) \quad (2.28)$$

If a Coulomb friction model is assumed, then the friction coefficient is constant when the relative motion of belt and pulley is not zero. If there is no relative motion of belt and pulley, the friction coefficient must be zero, because if $\mu \neq 0$, the tension gradient is non-zero and therefore the strain of the belt must change, which in turn causes a relative motion of belt and pulley. This means that there must exist an active and an inactive part in the belt-pulley contact when Coulomb friction is assumed.

Definition 3 *The active arc is the part of the wrapped angle where the tension in the belt changes.*

Definition 4 *The inactive arc is the part of the wrapped angle where the tension in the belt is constant.*

On the driving pulley the belt has to run equally fast or slower than the pulley. At the driven side, the belt has to run equally fast or faster than the pulley. In the part where the belt and pulley have

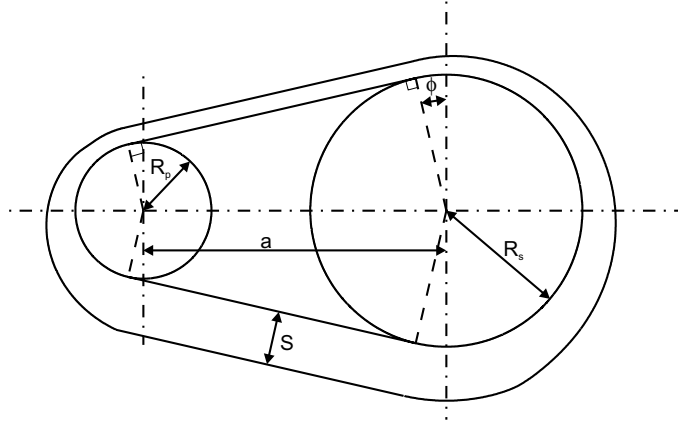


Figure 2.6: Tension in the belt

equal speed, the tension in the belt will be constant, because the friction coefficient will be zero. In the part where the speed is not equal, the tension in the belt will increase or decrease exponentially according to Equation (2.25).

In Figure 2.6 the tension in the belt is shown along the whole belt.

Pulley forces

The forces acting on the pulley are shown in Figure 2.7. To find the required clamping force for equilibrium, the forces acting on the pulley in horizontal direction have to be integrated along the wrapped angle of the belt along the pulleys according to:

$$F_{clamp}(\theta) = \int_0^\theta (q_N \cos \beta + q_w \sin \gamma \sin \beta) d\theta \quad (2.29)$$

F_p and F_s are the primary and secondary clamping force respectively and can be calculated with Equation (2.29). To find the torque transmitted by the belt, the friction forces acting in tangential direction have to be integrated along the wrapped angle of the belt on the pulley.

$$T_p = \int_0^{\theta_p} (q_w \cos \gamma) R_p d\theta_p \quad (2.30)$$

$$T_s = \int_0^{\theta_s} (q_w \cos \gamma) R_s d\theta_s \quad (2.31)$$

In these equations T_p is the torque on the primary side and T_s the torque on the secondary side. For ratio control the ratio between the primary and secondary clamping force for which the variator is in equilibrium is of interest. The rate of change of this clamping force ratio has a large influence on the stability of a ratio control system. Furthermore, it is used in feedforward control for clamping force calculation and ratio control.

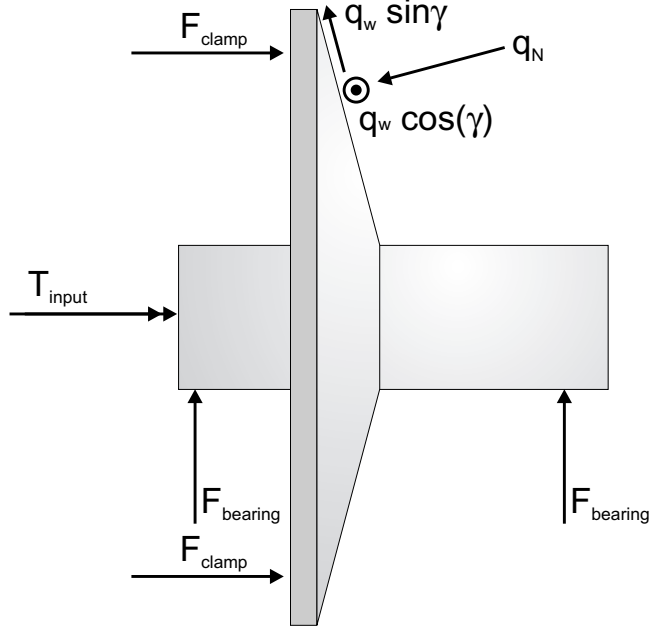


Figure 2.7: Forces acting on one pulley sheave

Definition 5 The clamping force ratio for which the variator does not shift ($\dot{r} = 0$), is defined as:

$$\Psi = \left. \frac{F_p}{F_s} \right|_{\dot{r}=0} \quad (2.32)$$

The torque that is transmitted from the primary side can be calculated using the belt-tension in the parts of the belt between the pulleys (S_1 and S_2) and the primary running radius (R_p):

$$T_p = (S_1 - S_2)R_p \quad (2.33)$$

and when a Coulomb friction model is used, with a Coulomb friction coefficient μ , this equation can be written as:

$$T_p = S_0(e^{\frac{\mu}{\sin \beta} \alpha} - 1)R_p \quad (2.34)$$

When on one of the pulleys the active arc equals the wrapped angle of the belt on the pulley, the maximum torque is reached that can be transmitted. In this model, this will always be on the pulley with the smallest wrapped angle, because if the friction coefficient is equal on both pulleys, then the rate of change of the tension in the belt will also be equal. The total amount of torque that can be transmitted depends then on the clamping force and the running radius and again for the Coulomb friction model. The maximum transmittable torque is given by:

$$T_{max} = S_0(e^{\frac{\mu}{\sin \beta} (\pi - |\phi|)} - 1)R_p \quad (2.35)$$

Definition 6 The ratio between the input torque and the maximum torque that can be transmitted at a certain given clamping force is called 'torque ratio' and is denoted by τ . This value can be

calculated with:

$$\tau = \frac{T_p}{T_{max}} \quad (2.36)$$

Parameter sensitivity

To investigate the sensitivity of the model for variations in the parameters, the calculation of the clamping force ratio Ψ is compared for low ($r_g \approx 0.44$), medium ($r_g \approx 1$) and overdrive ($r_g \approx 2.25$) for variations in the model parameters. The model parameters under investigation are: μ , AE , γ and the friction model. Ψ is calculated for τ between -1 and 1 .

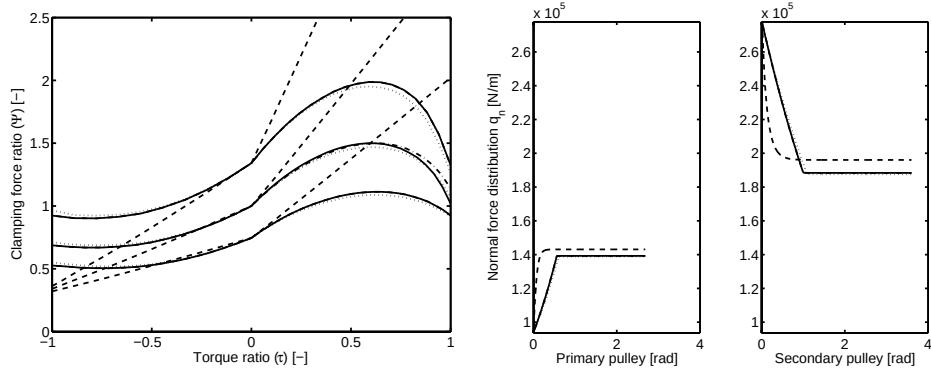
Friction models First, the effects of several friction models are compared. In Figure 2.8(a) Ψ is shown for several friction models. From the figure can be seen that except for the viscous friction model (the dashed line in this figure), the sensitivity of the value of Ψ for different friction models is small. Comparing the normal force distribution calculated with the continuous belt model as shown in Figure 2.8(b) shows the same result. The Coulomb friction model, the Visco-plastic model and the continuous Stribeck model give comparable results for the value of Ψ . The viscous friction model however does not compare well to the measured values as will be shown later.

The belt-tension is shown in Figure 2.8(c). The tension in the belt is a linear function of the normal force distribution (See Equations (2.29) and (2.23)). The different behavior of the viscous friction model can also be seen in the belt-tension and in the rate of change of the belt-tension along the wrapped angle of the pulley. This is shown in Figure 2.8(d).

All friction models have been tuned to have comparable friction coefficients. It can be clearly seen from the figure that viscous damping results in completely different behavior. This model gives very different results than actually measured in variators (See Figure 2.12). The viscous damping model is clearly not applicable for belt type variators.

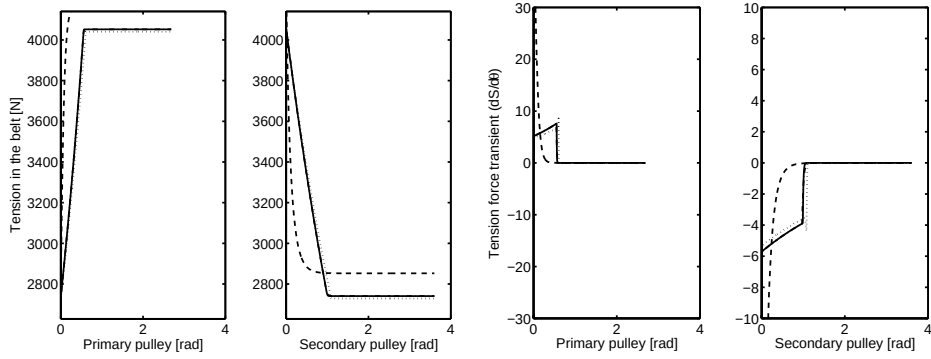
Friction coefficient The influence of the friction coefficient using the Coulomb friction model can be seen in Figure 2.9(a). From this figure can be seen that a higher value of the friction coefficient influences Ψ in such a way that the highest value becomes higher and the lowest value becomes lower. This is caused by the tension in the belt. If the friction coefficient becomes higher, the tension difference in the belt also increases due to the fact that more torque can be transferred with the clamping force remaining the same.

Parameter AE For parameter AE (the cross sectional area of the belt times Young's modulus) only the lubricated contact friction model is shown, because the Coulomb friction model has no sensitivity for this parameter. This is due to the fact that the Coulomb friction model is not sensitive to the speed of the sliding contact. The results are shown in Figures 2.10(a) and 2.10(b). Using the visco-plastic friction model or the continuous Stribeck curve the Ψ will change with changes of



(a) Ψ for variations in the friction model for Low (lower graphs), Medium (middle graphs) and Overdrive (upper graphs)

(b) Normal force distribution for variations in the friction model for ratio 2



(c) Tension in the belt for ratio 2

(d) Rate of change of the belt tension force $dS/d\theta$ for ratio 2

Figure 2.8: Sensitivity for the friction model of the continuous belt model, solid: Coulomb friction, dashed: viscous friction, dash-dot: Coulomb friction with micro-slip region, dotted: lubricated contact friction model

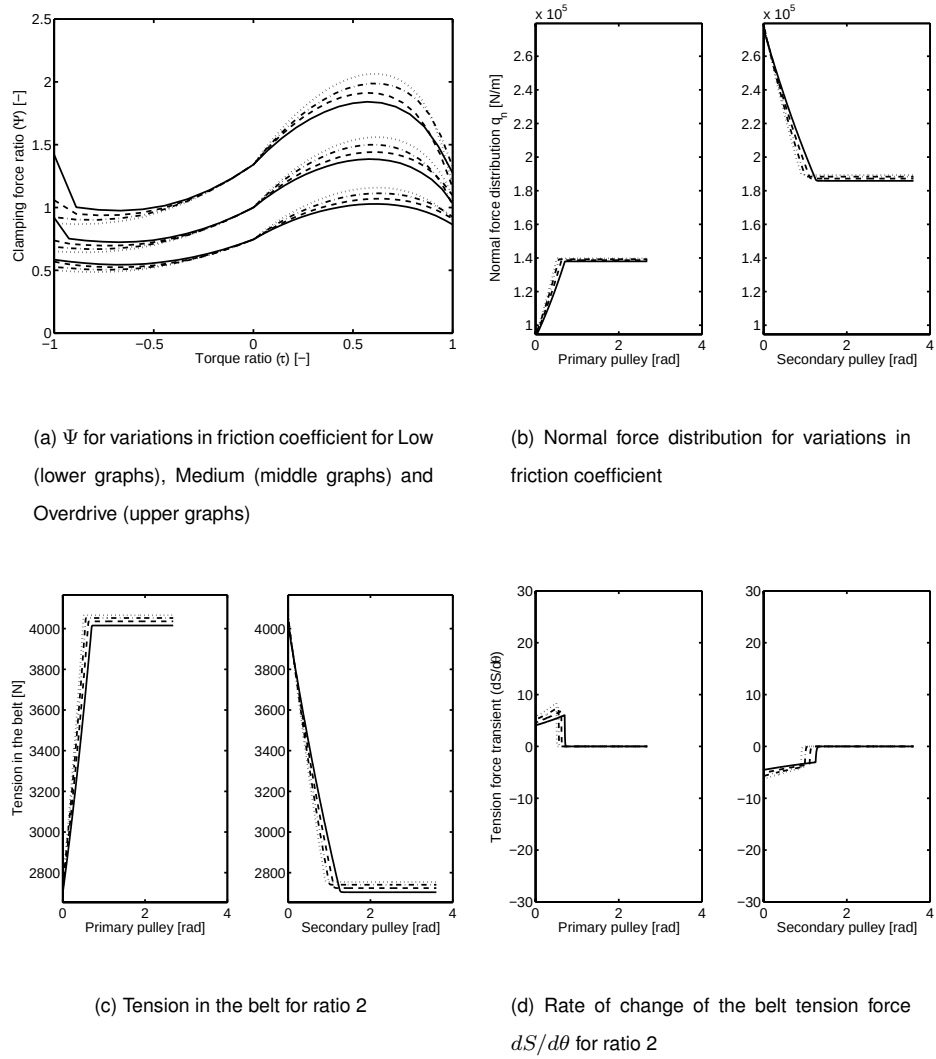


Figure 2.9: Sensitivity for friction coefficient of the continuous belt model shown for the visco-plastic friction model, solid: $\mu = 0.08$, dashed: $\mu = 0.09$, dash-dot: $\mu = 0.10$, dotted: $\mu = 0.11$.

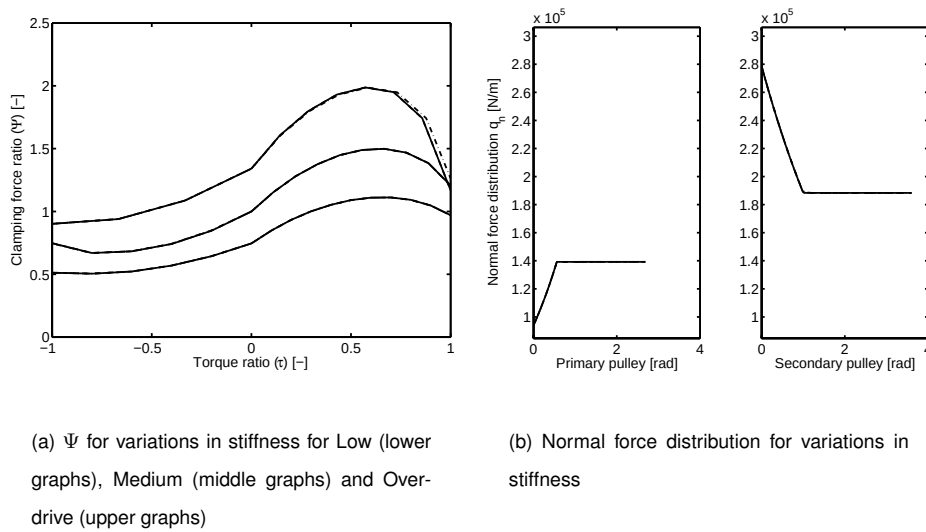


Figure 2.10: Sensitivity for stiffness of the continuous belt model

AE. If *AE* is high, the belt will slide along a larger portion of the pulley, because the sliding speed will remain small and therefore the friction coefficient will also remain small. If *AE* is lowered, the friction coefficient will increase until *AE* is low enough for full sliding friction to occur in some portion of the wrapped angle, while another part is still idle. If this happens, then the friction coefficient will decrease again with decreasing *AE*.

Slip angle Although the running radius is assumed constant in this model (zero pulley deformation) the effect of the radial friction can be approximated using γ , the belt-slip angle. If γ is taken constant over the whole wrapped angle of the belt on the pulley, the results for different values of γ are given in 2.11(a) and 2.11(b). The assumption of a constant value for γ is further evaluated in the section on the effects of pulley deformation. This way of tuning γ to fit the data implicitly assumes that the running radius on the pulley is continuously in- or decreasing. This is the case for spiral running.

Optimized parameters If the calculated Ψ is set against measured data, the difference can be quite large as can be seen in Figure 2.12(a). The initial value $\gamma = 0$ is apparently not very accurate. Therefore both the values of γ and μ are used to fit the model more accurately to the data. The friction model used is the Coulomb friction model, because the sensitivity of the model to the friction model is not very high. Using the Coulomb friction model also the parameter *AE* is not important for the parameter fit. The result is shown in Figure 2.12(b). The model is fitted more closely to the positive torque values, because this is the area in which the variator will operate in more frequently in automotive applications. The model matches the data more closely with the optimized values. The optimized values are:

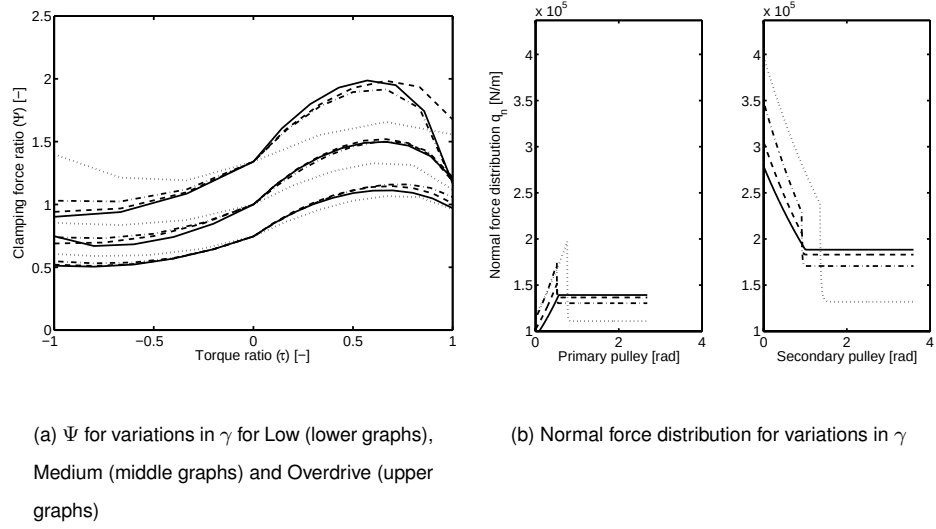


Figure 2.11: Sensitivity for γ of the continuous belt model

Parameter	Value
μ	0.1 [-]
γ	0.8 [rad]

The high value of γ that is needed to accurately track the measured values of Ψ is probably due to omission of the bands-blocks interaction in this model. This value is therefore not representative of a 'real' spiral running, but rather the compensation for the omission of some physical properties in the model.

Conclusions continuous belt model

From the results in this section follows that the continuous belt model can be very accurate for positive values of T_p , although the model does not take compression forces into account. The value Ψ which is an important property for ratio control, as will be shown later in Section 2.3, can be accurately simulated using the continuous belt model.

The friction model that is used in the calculation does not have a large influence on the calculated result, except for the viscous friction model. The viscous friction model apparently does not accurately describe the friction in the variator. Coulomb friction describes the variator friction in sufficient detail, therefore this model is used in simulations.

Furthermore, the stiffness of the belt is of little consequence to the belt tension distribution. The spiral running however, is an important factor in the calculation of the belt tension. This can be seen in the sensitivity to the angle γ , the slip angle of the belt on the pulley. Also the friction coefficient is an important factor.

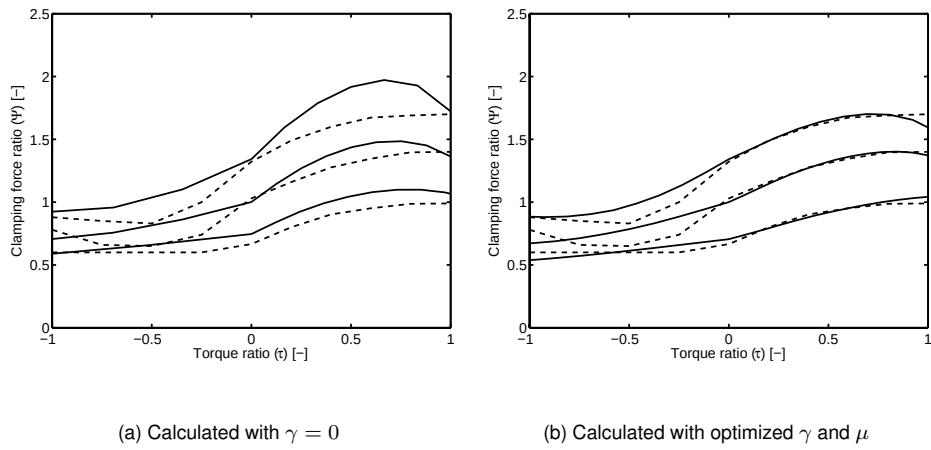


Figure 2.12: The calculated Ψ (solid) versus the measured data (dashed) for Low (lower graphs), Medium (middle graphs) and Overdrive (upper graphs)

2.2.2. Pushbelt model

The metal pushbelt as produced by Van Doorne's Transmissie has some special properties not captured by the continuous belt model. The pushbelt consists of blocks and bands as discussed earlier. The blocks can move freely along the bands. Between the blocks a small gap exists depending on the tension force in the bands.

Numerous works have been contributed to the modeling of the metal pushing V-belt variator. Gerbert [31] assumes the tension in the belt to be constant, because the stiffness of the bands is lower than the stiffness of the blocks. Also Kim [42] assumes the tension in the bands to be constant, but he assumes the friction between bands and blocks to be negligible. Others like Sun [82] assume the friction to be small with a value of around 0.01. In this section the following assumptions are made:

- the bands are assumed to be a continuous belt with negligible bending stiffness,
- friction exists between bands and blocks, therefore the tension in the bands will not be constant,
- in the belt there is always one part where compression is zero, this means that there is some play between the blocks in the belt,
- the compression force between the blocks can not be higher than the tension in the bands at any point in the pushbelt, the pushbelt would otherwise buckle,
- inertia forces can be neglected, the centrifugal forces are not taken into account,
- the friction coefficients are constant (Coulomb friction model), the Stribeck effect is not taken into account.

In a pushbelt the bands are packed tightly together, forming a belt. This belt has a very low bending stiffness due to the fact that the individual bands are very thin. Therefore, the bending stiffness can be neglected. Also, because of the tight packaging, the belt can be viewed to be continuous. In reality the bands move relative to each other as well, but this effect is neglected here.

The blocks and the bands are not fixed, i.e. they can move relative to each other. The contact surface is lubricated, lowering the friction, but high normal forces make this friction relevant. Furthermore, it is assumed that there is a (small) amount of play between the blocks in the belt. Therefore, there must be a point in the belt where the compression between the blocks is zero, since there is a gap between the blocks.

If the compression force between the blocks would be higher than the tensile force in the belt. If this was the case, a net compression force would result. The result of this would be buckling of the belt, due to the lack of bending stiffness in the belt.

The inertial forces are neglected here. In reality these forces are present and can be relevant for

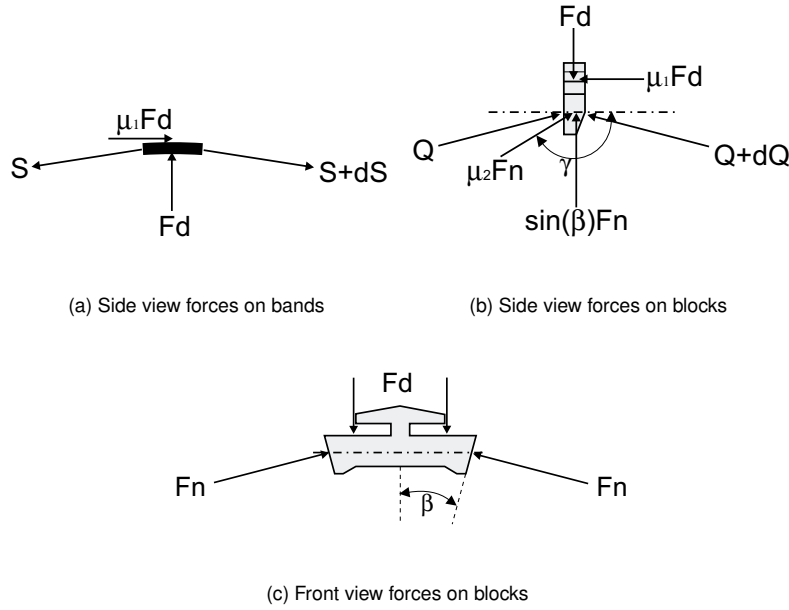


Figure 2.13: Forces acting on the belt according to the pushbelt model

high speed operation of the variator. For the comparison between the models and the model parameter sensitivity study the inertial forces are not of interest.

Because of the very small influence of the friction model on the outcome of the continuous belt model, only the Coulomb friction model is used in this section.

2.2.3. Forces in the pushbelt

Forces acting on the bands and blocks of a metal V-belt are shown in Figure 2.13. The blocks now form a layer between the belt and the pulley. There can be no tension between the blocks, but they can transmit torque when compressed, hence the name pushbelt. Because there is no pre-strain in the belt, there is some play between the blocks when not compressed. This is necessary for the belt to be able to bend.

The forces acting on the blocks and the bands have to be in equilibrium in stationary conditions:

$$dQ = 2\mu_2 \cos \gamma dF_n - \mu_1 dF_d \quad (2.37)$$

$$dS = -\mu_1 dF_d \quad (2.38)$$

$$dF_d = d\theta S \quad (2.39)$$

$$dF_d = d\theta Q + 2 \sin \beta dF_n + 2 \sin \gamma \mu_2 dF_n \quad (2.40)$$

The equilibrium Equations (2.37), (2.38), (2.39) and (2.40) can be combined to a set of differential equations given in 2.41 through 2.43.

$$\frac{dS}{d\theta} = -\mu_1 S \quad (2.41)$$

$$\frac{dQ}{d\theta} = -\frac{\mu_2 \cos \gamma}{2(\sin \beta + \mu_2 \sin \gamma)} Q + \left[\frac{\mu_2 \cos \gamma}{2(\sin \beta + \mu_2 \sin \gamma)} - \mu_1 \right] S \quad (2.42)$$

$$\frac{dF_n}{d\theta} = \frac{S - Q}{2(\sin \beta + \mu_2 \sin \gamma)} \quad (2.43)$$

In line with many publications a Coulomb friction model is assumed here. This gives the possibility to find an analytical solution to the differential equations. The solution then still depends on the boundary conditions. Integrating Equation (2.41)-(2.43) gives:

$$S = S_0 e^{-\mu_1 \theta} \quad (2.44)$$

$$Q = Q_0 e^{-\frac{\mu_2 \cos \gamma \theta}{2(\sin \beta + \mu_2 \sin \gamma)}} + \int \left(\left[\frac{\mu_2 \cos \gamma}{2(\sin \beta + \mu_2 \sin \gamma)} - \mu_1 \right] S \right) d\theta \quad (2.45)$$

$$Q = Q_0 e^{-\frac{\mu_2 \cos \gamma \theta}{2(\sin \beta + \mu_2 \sin \gamma)}} + \left[\frac{\mu_2 \cos \gamma}{2\mu_1(\sin \beta + \mu_2 \sin \gamma)} - 1 \right] S_0 [e^{-\mu_1 \theta} - 1] \quad (2.46)$$

In the more general form the differential equations can be used to find a numerical solution.

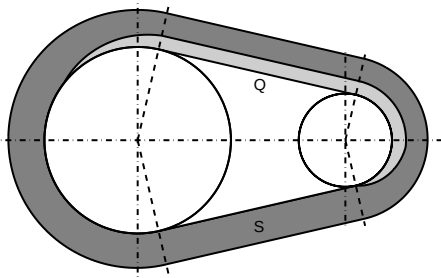
Results

If the model described in the previous paragraph is used to calculate the forces in the belt numerically, then the boundary conditions need to be known. In the case of the pushbelt model there exist four different regimes, shown in Figure 2.14, each having their own set of boundary conditions. The four different regimes are:

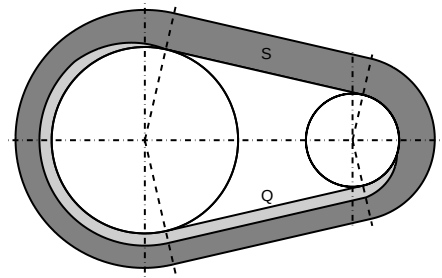
- $r_g \leq 1, T_p \geq R_p (S_1 - S_0)$
- $r_g \leq 1, T_p < R_p (S_1 - S_0)$
- $r_g > 1, T_p \geq R_p (S_1 - S_0)$
- $r_g > 1, T_p < R_p (S_1 - S_0)$

To determine, for a given operating point of the variator, in which regime the variator is, it has to be determined whether the bands slip on the blocks on the primary or on the secondary side. Secondly, it has to be determined on which side the compression is higher than zero. The direction of the friction force between the bands and the blocks is determined by the speed difference between the blocks and the bands. The speed difference depends on the transmission ratio.

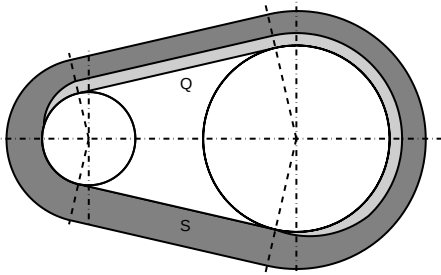
In low, ratios lower than one, when the belt has a larger running radius on the secondary pulley, the bands will slip on the primary side. Because the bands will run faster than the blocks, the tension will decrease on the primary side, aiding the transmission of torque. The torque at which the compression force changes sides will therefore be lower than zero. In ratios greater than one the bands will slip on the secondary side. In contrast to the first situation, the bands will counteract the torque transmission. The torque at which the compression side will change will therefore be greater than zero.



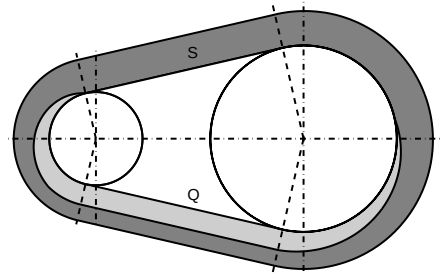
(a) $r_g > 1$, positive compression ($Q_0 > 0$)



(b) $r_g > 1$, negative compression ($Q_1 > 0$)



(c) $r_g \leq 1$, positive compression ($Q_0 > 0$)



(d) $r_g \leq 1$, negative compression ($Q_1 > 0$)

Figure 2.14: Tension and compression force distribution

The boundary conditions for the differential equations is given by the torque and its direction and the transmission ratio.

Low In low ratio ($r_g < 1$), where the belt runs at a smaller radius on the primary pulley and on a larger radius on the secondary pulley, the bands will slip on the blocks on the primary pulley. Because the relative difference between the running radius of the bands and the running radius of the blocks is larger on the primary pulley than on the secondary pulley, the bands will move slower than the blocks on the primary pulley. This effect causes a positive torque to be transmitted when no compression force is present. The corresponding tension and compression force distribution are shown in Figure 2.14(c) and 2.14(d) for $T > -T_0$ and $T \leq -T_0$ respectively, with $T_0 = R_p(S_1 - S_0)$.

Overdrive In overdrive ratio ($r_g > 1$) this situation occurs on the secondary pulley. This causes a negative torque for the situation of no compression force. In medium ratio this effect does not occur, because the difference between the bands and the blocks is zero on both sides (no slip occurs). The corresponding tension and compression force distribution are shown in Figure 2.14(a) and 2.14(b) for $T \geq T_0$ and $T < T_0$ respectively.

Parameter sensitivity

The influence of model parameters on the calculation of the Ψ are determined for the pushbelt model. For the parameter γ the results are shown in Figure 2.15. From this figure can be seen that the Ψ is lowered with an increasing value of γ . For negative values of γ the value of Ψ increases.

In Figure 2.16 shows the parameter sensitivity for μ_2 . For an increasing value of μ_2 , the friction coefficient between blocks and pulley, the value of Ψ increases for positive torque and decreases for negative torque. The inverse is true for decreasing values of μ_2 .

The parameter sensitivity of the friction coefficient between blocks and bands, μ_1 , is shown in Figure 2.17. Only slight change is observed for changes in μ_1 . For increasing values of μ_1 the ratios above medium (ratio > 1) show a decrease in Ψ and the results for low ratios (ratio < 1) show an increase in the value of Ψ . In medium no change is seen.

Parameter sensitivity for the clamping force is zero as is shown in Figure 2.18. The torque increases proportionally with the clamping force and therefore the resulting change in Ψ vs torque ratio equals zero.

If the model is tuned to measured data, the model parameters are:

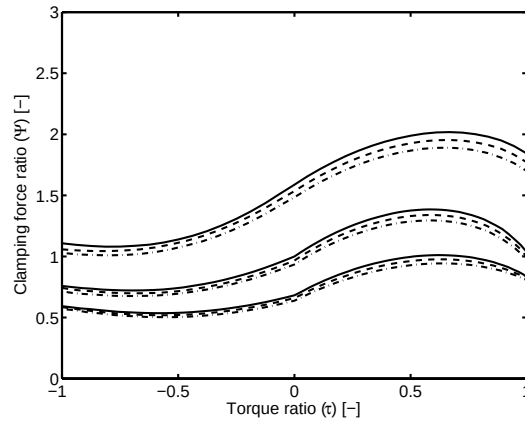


Figure 2.15: Ψ curve sensitivity for γ , solid: $\gamma = 0$, dashed: $\gamma = 0.04$ and dash-dot: $\gamma = 0.08$ (upper graph: $r_g = 2.25$ (OD), middle graph: $r_g = 1$ (MED) and lower graph: $r_g = 0.45$ (LOW))

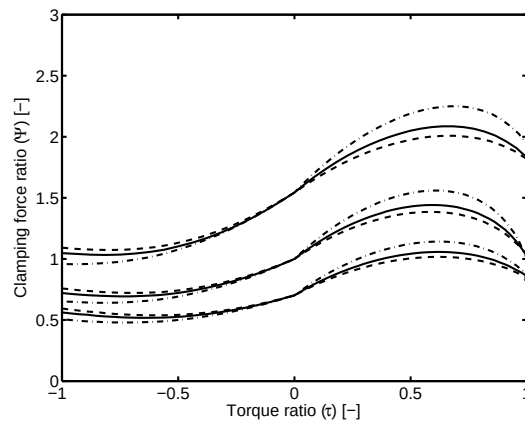


Figure 2.16: Ψ curve sensitivity for μ_2 , with solid: $\mu_2 = 0.09$, dashed: $\mu_2 = 0.08$ and dash-dot: $\mu_2 = 0.11$ (upper graph: $r_g = 2.25$ (OD), middle graph: $r_g = 1$ (MED) and lower graph: $r_g = 0.45$ (LOW))

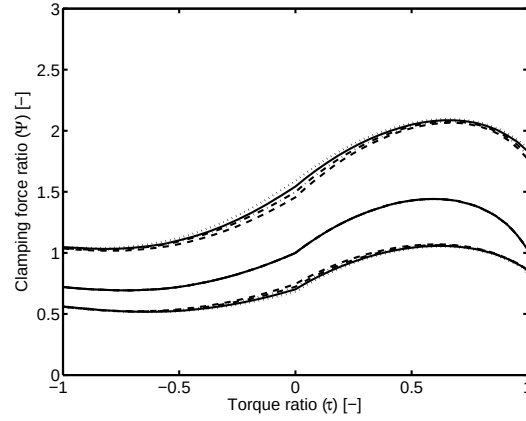


Figure 2.17: Ψ curve sensitivity for μ_1 , with solid: $\mu_1 = 0.02$, dashed: $\mu_1 = 0.0$, dash-dot: $\mu_1 = 0.01$ and dotted: $\mu_1 = 0.03$ (upper graph: $r_g = 2.25$ (OD), middle graph: $r_g = 1$ (MED) and lower graph: $r_g = 0.45$ (LOW))

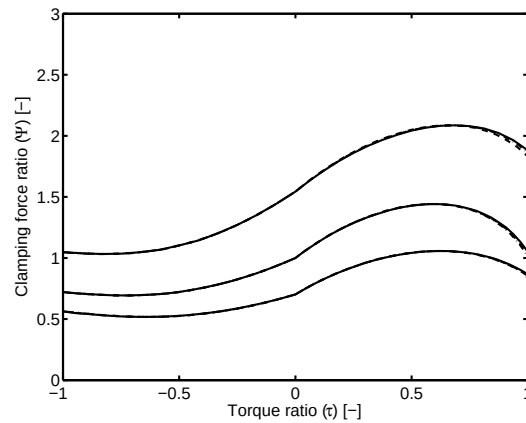


Figure 2.18: Ψ curve sensitivity for the clamping force, with solid: $F_s = 15 \text{ kN}$, dashed: $F_s = 20 \text{ kN}$ and dash-dot: $F_s = 25 \text{ kN}$ (upper graph: $r_g = 2.25$ (OD), middle graph: $r_g = 1$ (MED) and lower graph: $r_g = 0.45$ (LOW))

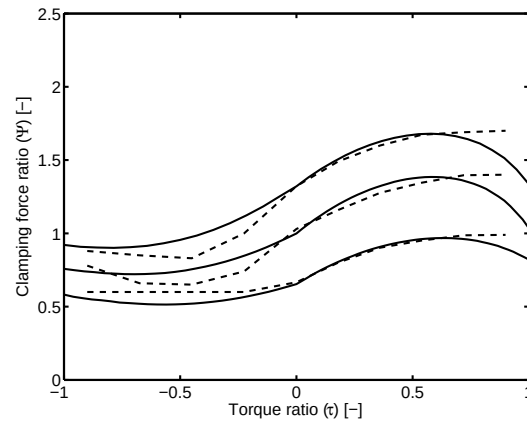


Figure 2.19: Ψ curve comparison with measured data (dashed curves) and calculated data (solid curves). (upper graph: $r_g = 2.25$ (OD), middle graph: $r_g = 1$ (MED) and lower graph: $r_g = 0.45$ (LOW))

Parameter	Value
μ_2	0.08
μ_1	0.03
γ	$r = 0.5: 0.05, r = 1: 0.00, r = 2.5: 0.22$

The value of μ_1 and μ_2 are not sensitive to the ratio. This finding is intuitive, because the tribological circumstances are equal in all ratios. The same is not true for the value of γ . This value however is very high in overdrive. Possibly some unmodeled effects are represented in this value. The result is shown in Figure 2.19. For positive torque the model matches very closely to the measured data, but for negative torque large differences can be seen.

2.2.4. Conclusion stationary model

The calculation of the clamping force ratio Ψ can be done with either the continuous belt model or the pushbelt model. Although the pushbelt model is more complex and takes the complex kinematics of the pushbelt into account, the results are not more accurate when compared to measured data.

For calculating the clamping force ratio for a pushbelt variator it is sufficient to use the continuous belt model. However, the continuous belt model does not take into account the discontinuous behavior of the pushbelt that occurs when the regime of the variator changes, i.e. when the compression side changes or the ratio changes through $r_g = 1$. If this behavior needs to be simulated, then the pushbelt model must be used.

2.3. Variator Transient Model

The transient behavior of a V-belt type variator, i.e. the shifting behavior, is discussed in this section. To shift a V-belt type variator either a creeping process caused by elastic deformations of pulley and belt or a sliding process has to occur. The shifting by elastic deformation is commonly referred to as creep-mode shifting. The sliding process is called slip-mode shifting. Therefore we define two types of slip:

Definition 7 *When stable, creep like, slipping occurs, i.e. the friction coefficient increases with increasing slip, it is said that micro-slip occurs.*

Definition 8 *When unstable slipping occurs, i.e. the friction coefficient decreases with increasing slip, it is said that macro-slip occurs.*

2.3.1. Creep-mode shifting

Creep-mode shifting is the main shift mode for pushbelt type variators. This shiftmode is characterized by the way the variator changes ratio. In creep-mode shifting the ratio change is caused by elastic deformation of the belt and pulleys. No macro-slip occurs. This type of shifting is modeled by several authors. Ide [36] developed a model for creep mode shifting that states that shifting is proportional to the input speed and the difference between the primary clamping force and the primary clamping force needed for equilibrium. Shafai [71] however suggests that the shifting process can be described as a moving mass with viscous damping. Several authors suggest that the shifting behavior of the CVT is linked to deformation of belt and pulleys [18] [17] [79].

In creep mode shiftspeed is dependent on the actuation forces, rotating speed of the pulleys and pulley and belt deformation. Pulley and belt deformation is the means by which shifting is possible. More flexible pulleys give faster shifting. Pulley deformation is also dependent on the running radius of the belt on the pulley.

If the variator is symmetrical, upshifting should be the opposite of downshifting. Because the stiffness of the pulley-belt system is dependent on the ratio of the variator, a ratio dependence can be expected in the ratio changing behavior. Furthermore a clamping force dependency follows from the increased deformation of the pulleys and the belt with a higher clamping force. Finally, with higher rotational speed faster ratio changing can be observed.

2.3.2. Slip-mode shifting

Slip-mode shifting is the characterization of the ratio changing behavior of a variator when macro-slip is occurring. This type of shifting is much faster than creep-mode shifting, because the sliding motion of the belt on the pulley reduces the friction in radial direction and the primary and secondary speeds are completely decoupled, i.e. the primary and secondary inertias do not have to be accelerated.

While shifting when slipping is very fast, ratio changing which causes slip to increase instead of decrease can cause serious damage to the variator since the absorbed energy in the pulley-belt contact will increase.

In slip-mode, shifting is not limited by the elastic deformations in the belt and/or pulley. Slip-mode shifting is therefore not dependent on the rotational speed of the variator and the speed of shifting is limited only by the actuation system. Ide has studied this type of shifting behavior of a pushbelt type variator [37].

2.3.3. Models

Over time several models have been proposed to describe the transient behavior of a V-belt type variator. A short overview of often used models is given. Also an analysis using dimensional analysis of the shifting behavior is given based on the work of Carbone in [17].

Dimensional Analysis

The rate of change of the ratio r_g is expressed as a function of: the clamping forces F_p and F_s , the input and output torques T_p and T_s , the primary and secondary running radii R_p and R_s , the length of the belt, the axial pulley distance, the primary and secondary shaft speeds, pulley and belt stiffnesses k_p and k_s , beltspeed and the pulley groove angle. This can be written as:

$$\dot{r}_g = f(F_p, F_s, T_p, T_s, R_p, R_s, L, a, \omega_p, \omega_s, k_p, k_s, v_b, \beta) \quad (2.47)$$

This function can be rewritten to a dimensionless form using as fundamental units the quantities R_p , F_s and ω_p using Buckingham's pi-theorem [14] [16] [15]. Equation (2.47) now can be written as:

$$\frac{\dot{r}_g}{\omega_p} = F\left(\frac{F_p}{F_s}, \frac{T_p}{F_s R_p}, \frac{T_s}{F_s R_p}, \frac{R_s}{R_p}, \frac{L}{R_p}, \frac{a}{R_p}, \frac{\omega_s}{\omega_p}, \frac{k_p R_p}{F_s}, \frac{k_s R_p}{F_s}, \frac{v_{belt}}{R_p \omega_p}, \beta\right) \quad (2.48)$$

This equation can be further simplified by using the fact that β , L and a are constants and that $\omega_s/\omega_p = r_s$, $T_p/(F_s R_p) = c\tau$, $T_p/T_s = r_g$ and $R_p/R_s = r_g$. Furthermore, if k_p and k_s are approximately equal k_s can be omitted. This is the case if the pulley stiffness is (nearly) equal of both pulleys. This will be the case if the variator is symmetrical. Equation (2.48) can then be rewritten as:

$$\frac{\dot{r}_g}{\omega_p} = F\left(\frac{F_p}{F_s}, \tau, r_g, r_s, \frac{k_p R_p}{F_s}, \frac{v_{belt}}{R_p \omega_p}\right) \quad (2.49)$$

For creepmode shifting, where no macro-slip occurs, the following assumptions can be made: $v_b/(R_p \omega_p) \approx 1$ and $r_s \approx r_g$. The shifting process therefore has to be a function of the input shaft speed, the clamping force ratio, the torque ratio, the transmission ratio and a factor determined by the pulley stiffness.

Ide

Ide found that the shifts speed of the variator is a function of the speed ratio r_s and is linearly dependent on the input speed [36]. He also found a linear relationship with the difference between

the primary clamping force and the primary clamping force for which the variator does not shift. Furthermore he found that this shift speed further depended on the ratio of the variator.

The resulting model is:

$$\dot{r}_g = \omega_p \kappa_{ide}(r_g) F_s \left(\frac{F_p}{F_s} - \Psi \right) \quad (2.50)$$

In this equation $\kappa_{ide}(r_g)$ is a function of r_g , which must be obtained using experimental data.

Shafai

Shafai models the variator as being a damped single mass system [71]:

$$m_r \ddot{x}_p + c_r \dot{x}_p = F_s \left(\frac{F_p}{F_s} - \Psi \right) \quad (2.51)$$

The acceleration term is very small compared to the damping term, therefore the acceleration terms may be neglected ($|m_r \ddot{x}_p| \ll |c_r \dot{x}_p|$). If the acceleration term is neglected the following differential equation remains:

$$\dot{x}_p = \frac{1}{c_r} F_s \left(\frac{F_p}{F_s} - \Psi \right) \quad (2.52)$$

In this equation c_r is a damping coefficient, which must be determined using experimental data. If this equation is rewritten in terms of the geometric ratio r_g substitution of the time derivative of Equation (2.3) for x_p , the following results:

$$\dot{r}_g = \frac{1 - r_g}{c_r R_s \tan \beta} F_s \left(\frac{F_p}{F_s} - \Psi \right) \quad (2.53)$$

CMM

The Carbone Mangialardi Mantriota (CMM) model was introduced recently. Carbone [17] reasons that the shift speed of the variator must be almost linear with the logarithm of the clamping force quotient F_p/F_s . Not only did this give a better approximation of the results they had from other, more complex models, but the reasoning that the shifting process must be symmetric around $r_s = 1$ gives rise to this suggestion.

According to the CMM model, the transient behavior of a V-belt type variator can be described by:

$$\dot{r}_g = \kappa_{cmm}(r_g, [F_s]) \omega_p \left(\ln \frac{F_p}{F_s} - \ln \Psi \right) \quad (2.54)$$

In this equation $\kappa_{cmm}(r_g, [F_s])$ is a function of r_g and for higher accuracy also F_s , which can be obtained from experiments or from theory.

2.3.4. Measurements

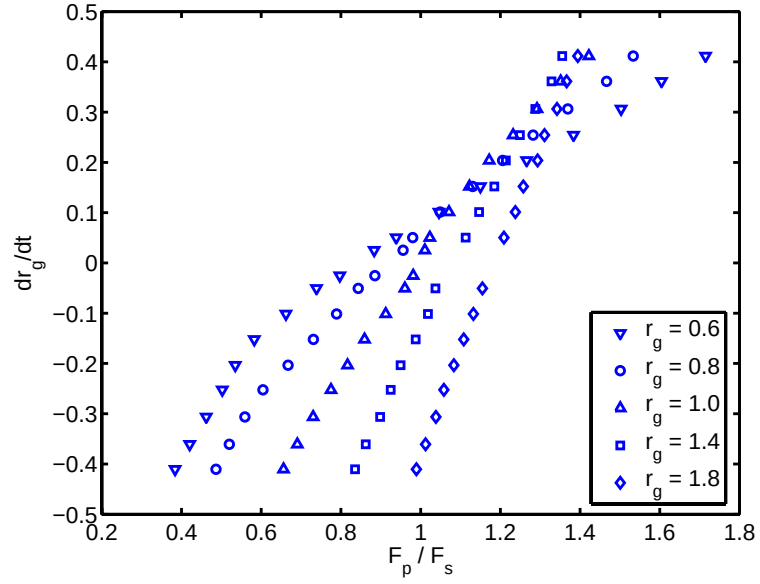
Measurements have been done to validate or invalidate the models mentioned in the previous paragraph. These measurements have been done on a beltbox testrig. These measurements will be compared with the given models and the results will be compared.

Analysis

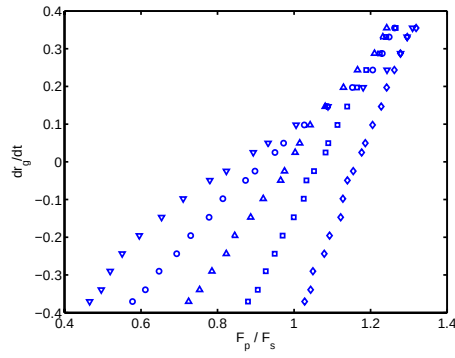
In Figure 2.20(a) measurements are shown of $\dot{r}_g$ against the clamping force ratio F_p/F_s . In this figure can be seen that indeed for low shifts speeds $\dot{r}_g$ is more or less linear with F_p/F_s . However, for higher shifting speeds, especially for lower ratio's a clear deviation from Ide's transient behavior theory is present. If these results are compared to the results shown in Figure 2.20(b), then it can be seen that increasing the clamping force increases the shifts speed. The relation however is not linear. With a doubling of the clamping force the shifts speed increases much less than a factor two. In Figure 2.20(c) the results are shown from measurements with a doubled input shaft speed. If these results are compared to the results from Figure 2.20(a), then it can be seen that doubling the input shaft speed also doubles the shifts speed. If these measurements are compared to the models shown earlier, then it can be seen that the model from Shafai does not take into account the dependency on the input shaft speed and assumes a linear relation with the secondary clamping force. Both can be seen to be untrue.

For the model presented by Ide follows that the input shaft speed is taken into account in a linear way as follows from the measurements. The clamping force however is also present in a linear fashion in the Ide model. This does not compare well to the measurements.

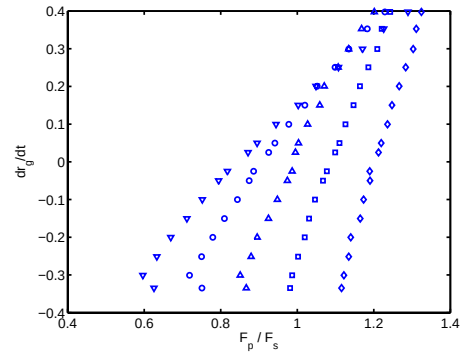
For the CMM model, the results should be compared to the logarithm of the clamping force ratio. These results are shown in Figure 2.21. These are the same measurements as shown earlier in a logarithmic scale. As can be seen in this figure, the logarithm of the clamping force ratio gives an almost linear relation to the shifts speed. Furthermore, the clamping force is not a factor in the CMM model. In the CMM model however, there is a constant factor for the pulley deformation. If this factor is related to the clamping force, the CMM model can very accurately describe the rate of change of speed ratio. In Figure 2.21(a), 2.21(b) and 2.21(c) the model results from the CMM model are compared to the measured values of $\dot{r}_g$. In these graphs the clamping force effect on shifts speed is taken into account in the CMM model.



(a) Measured shifting response $\dot{r}_g$ set against the clamping force ratio F_p/F_s , with $F_s = 20kN$ and $\omega_p = 100rad/s$.

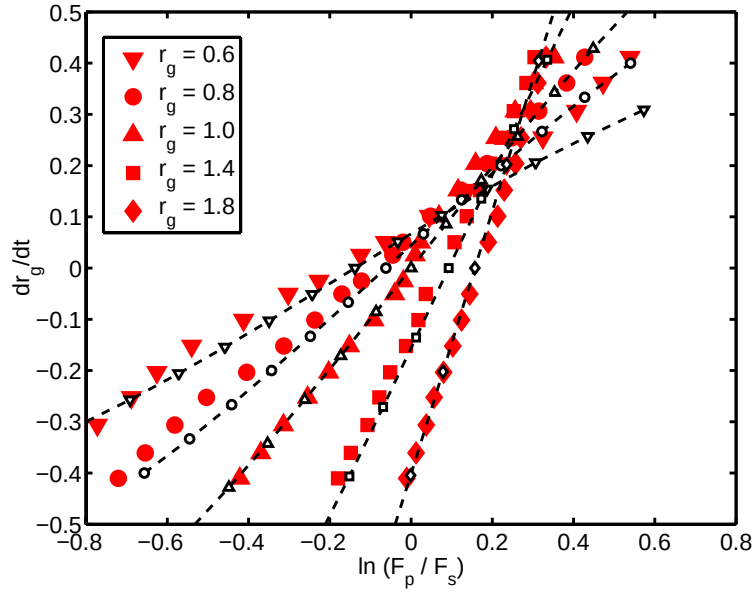


(b) Measured shifting response $\dot{r}_g$ set against the clamping force ratio F_p/F_s , with $F_s = 30kN$ and $\omega_p = 100rad/s$.

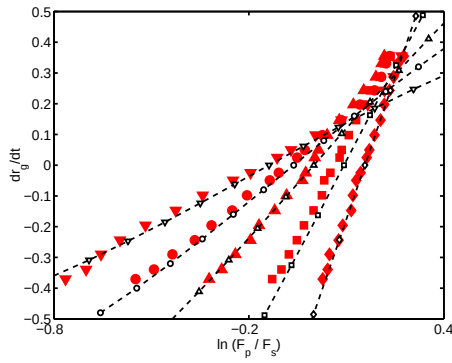


(c) Measured shifting response $\dot{r}_g$ set against the clamping force ratio F_p/F_s , with $F_s = 20kN$ and $\omega_p = 200rad/s$.

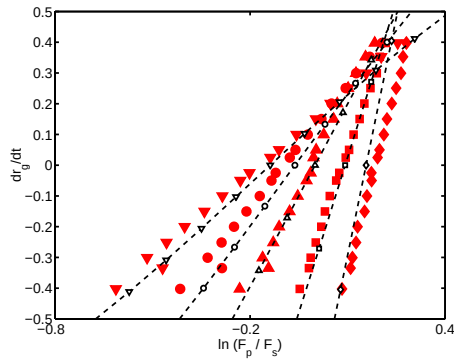
Figure 2.20: Measurements of the shifting response



(a) Measured shifting response $\dot{r}_g$ set against the clamping force ratio $\ln(F_p/F_s)$, with $F_s = 20kN$ and $\omega_p = 100rad/s$.



(b) Measured shifting response $\dot{r}_g$ set against the clamping force ratio $\ln(F_p/F_s)$, with $F_s = 30kN$ and $\omega_p = 100rad/s$.



(c) Measured shifting response $\dot{r}_g$ set against the clamping force ratio $\ln(F_p/F_s)$, with $F_s = 20kN$ and $\omega_p = 200rad/s$.

Figure 2.21: Measurements of the shifting response

2.3.5. Comparison

If the CMM model is compared to both the Ide model and the model from Shafai, the difference can be illustrated with the Taylor expansion of $\ln(F_p/F_s) - \ln(\Psi)$:

$$\ln \frac{F_p}{F_s} - \ln \Psi = \frac{1}{\Psi} \left[\frac{F_p}{F_s} - \Psi \right] - \frac{1}{2} \frac{1}{\Psi^2} \left[\frac{F_p}{F_s} - \Psi \right]^2 + \dots \quad (2.55)$$

Looking at this Taylor expansion can be seen that the Ide model and the Shafai model are a first order approximation of the logarithm of the clamping force ratio used in the CMM model. Also it can be seen that for values of Ψ smaller than one the difference can become large. From the measurements indeed the large differences can be found for low ratios were the value of Ψ is smaller. With the overstated influence of the clamping force in both Ide's and Shafai's model, it must be concluded that overall the CMM model gives the most accurate results compared to our measurements.

Slip in the variator

Slip in the variator is important for several reasons. First, slip can cause serious damage to the belt and the pulleys. For this reason it is important to know the conditions that lead to the situations where damage occurs. Second, to optimize the performance of the variator, both to maximize the torque capacity and to minimize the clamping force, detailed information on the dynamics of slip and the relation between slip and traction is necessary. Why slip is especially valuable for efficiency optimization of the variator will become apparent in the next chapter, on variator efficiency.

Models for the slipping behavior of the variator are studied by Klaassen [44] and Bonsen [11], proposing a dynamical model for slip in a CVT. Kim [42], [10] and Asayama [5] propose a model for the traction behavior of a pushbelt type variator. Van Drogen and van der Laan [89] study the relation between slip and damage in the variator experimentally.

In this chapter slip in the variator is studied. First, the relation between slip and traction is discussed, using a kinematical pushbelt model. Measurements of the traction in a pushbelt variator will be shown.

Second, a dynamic model to describe slip in the variator is derived. The dynamics of slip are important when designing a slip control system. Furthermore, the model is used to determine the stability criteria for the pushbelt variator.

Third, several slip estimation methods will be discussed. Correct estimation of the slip in a variator is essential to be able to use the variator at its full capability, since its characteristics are closely related to this variable.

3.1. Slip and Traction

The V-belt type variator utilizes friction to transmit power from the driving pulley to the belt and from the belt to the driven pulley. In this chapter it is assumed that the primary pulley is the driving pulley. The effective friction coefficient, i.e. the average friction coefficient, along the wrapped arc of the belt on the pulley is called traction.

Definition 9 The dimensionless traction coefficient μ_{eff} is defined here by:

$$\mu_{eff} = \frac{T_s \cos \beta}{2 \min(F_s, F_p) R_s} = \frac{T_p \cos \beta}{2 \min(F_s, F_p) R_p} \quad (3.1)$$

where T_p and T_s are the torque on the primary and secondary shaft of the variator, β is the pulley angle, F_p and F_s are the primary and secondary clamping forces and R_p and R_s are the running radii of the belt on the pulley on the primary and secondary side respectively. $T_s/R_s = T_p/R_p$, because the forces acting on the belt on either pulley must be equal.

The traction coefficient μ_{eff} is also referred to as simply μ .

In this definition μ_{eff} is the transmitted output torque divided by the contact normal force between the belt and the pulley with the lowest clamping force. According to the models presented in Chapter 2, the pulley with the smallest wrapped angle will also have the smallest clamping force when traction is maximal.

Measurements show [25] that the traction coefficient depends on the (micro-)slip between the belt and the pulleys. Since the traction coefficient defined according to 3.1 is a global variable, a global measure for the slip is required to characterize the relation between slip and traction. A possible global measure for the slip is the so-called speed loss v_s in the variator, defined by:

$$v_s = \omega_p R_p - \omega_s R_s \quad (3.2)$$

Later in this section it will be shown that the traction does depend only weakly on the input speed $\omega_p R_p$. Therefore a relative slip is defined as the speed loss divided by the input speed $\omega_p R_p$.

Definition 10 Relative slip is a dimensionless variable describing the relative motion between the pulleys and the belt.

$$\nu = \frac{v_s}{\omega_p R_p} = 1 - \frac{\omega_s R_s}{\omega_p R_p} = 1 - \frac{r_s}{r_g} \quad (3.3)$$

Several authors have used this definition of slip [47] [10] [94] [89]. In the sequel, the term slip means relative slip, unless stated otherwise. Besides, the term traction is often used as an abbreviation for traction coefficient.

3.1.1. Traction curve

The traction curve is a diagram of the relation between slip (3.3) and traction (3.1). In Figure 3.1 measured traction curves of a typical variator are shown for three different ratios. The notable differences between the traction curves will be discussed in Paragraph 3.1.2.

As can be seen in Figures 3.1, 3.2 and 3.3, traction is zero at zero slip. For increasing values of the slip, the traction increases until a certain maximum value is reached and then is constant or slightly decreasing. For small values of slip the traction is nearly linear with slip. This part of the curve with viscous behavior, is called the micro-slip area. For larger values of slip, when traction decreases

with increased slip, it is said that the variator operates in the macro-slip area.

Figure 3.2 shows three traction curves for different input speeds. The traction curves are almost identical in the micro-slip area. In the macro-slip area, there are distinct differences. Speed has no influence in the micro-slip area, because the slip velocity is very small and the amount of slip is determined predominantly by the wrapped angle. In macro-slip, the friction coefficient decreases with increasing slipspeeds, causing a lower traction coefficient. This is caused by the Stribeck effect [80].

In Figure 3.3 the traction curve is shown for different clamping forces. The micro-slip region slightly moves to the right. As will be shown in paragraph 3.1.2 this is caused by the lengthening of the belt and other elastic deformations under the higher load.

3.1.2. Play in the belt

It is assumed that all deformations of blocks and pulleys can be neglected. Furthermore, it is assumed that $r_g < 1$, so that $R_p < R_s$. Then, if slip occurs, this must occur at the primary pulley (See Chapter 2). Let δt be the total play between the blocks in the belt, let the primary pulley be the driving pulley and let the compression between the blocks have a positive contribution to the transfer of power from the primary to the secondary pulley.

The speed of the blocks at the exit of the primary pulley is denoted by v_e . This speed is constant between the primary and secondary pulley, on the secondary pulley and between the secondary and primary pulley. Because there is no slip between the secondary pulley and the belt, it follows that $v_e = R_s \omega_s$. From the secondary pulley to the primary pulley and in the inactive arc on the primary pulley there is no compression between the blocks. Obviously, play between the blocks can only occur in this part of the belt. Here it is assumed that the total play in the belt is distributed evenly between the blocks in the inactive part of the belt on the primary pulley (see Figure 3.5). Experiments by Kobayashi [47] confirm this assumption.

The total gap between the block elements in the belt, δ_t , can be estimated by adding an initial gap δ_o to the increase in belt length (ΔL) due to the internal stresses in the bands:

$$\delta_t = \delta_o + \Delta L \quad (3.4)$$

Hence, the mean gap δ_m is given by:

$$\delta_m = \frac{\delta t d_e}{\alpha_p R_p} \quad (3.5)$$

where d_e is the thickness of a block and α_p is the extend of the primary inactive arc.

The speed at which the blocks enter the pulley is $\omega_p R_p$. Because the primary pulley is the driving pulley $\omega_p R_p > v_e$. Therefore, the mean gap δ_m will be:

$$\delta_m = (\omega_p R_p - v_e) T \quad (3.6)$$

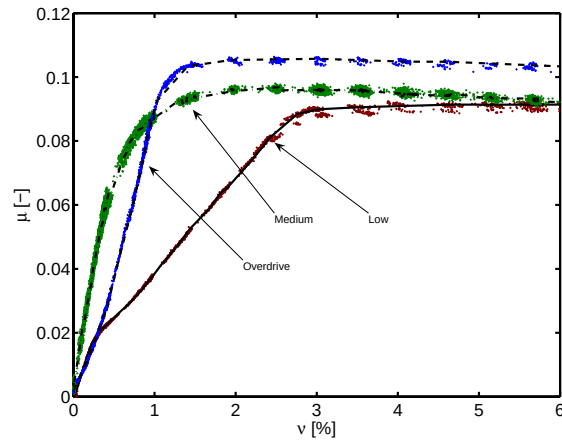


Figure 3.1: Traction as a function of slip measured at low, medium and overdrive ($r_g = 0.45$ (solid), $r_g = 1$ (dash-dot) and $r_g = 2.25$ (dashed))

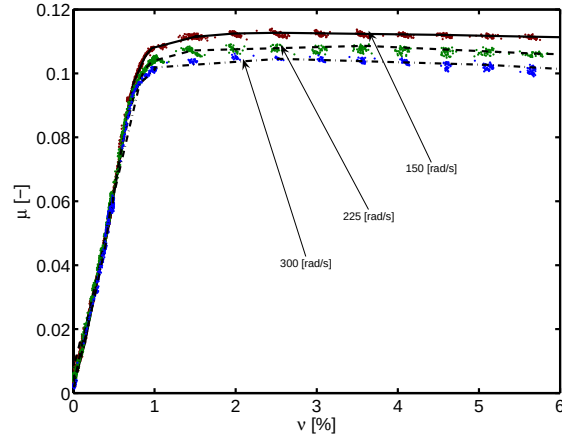


Figure 3.2: Traction as a function of slip measured at overdrive for different input speeds ω_p (150 rad/s (solid), 225 rad/s (dashed) and 300 rad/s (dash-dot))

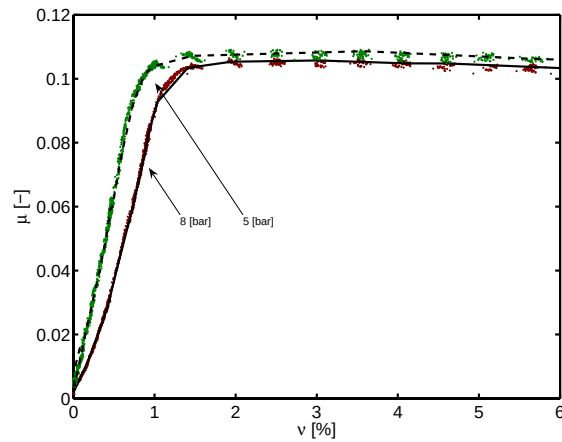


Figure 3.3: Traction as a function of slip measured at overdrive ratio for different clamping forces (5 bar $\sim$ 5 kN (dashed) and 8 bar $\sim$ 8 kN (solid))

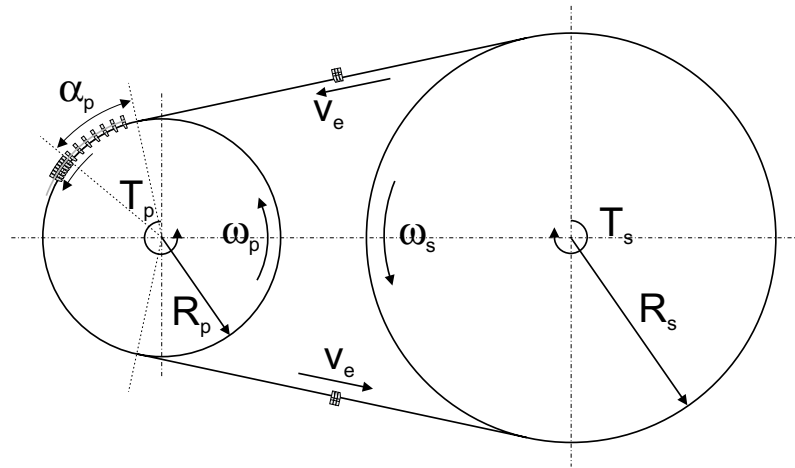


Figure 3.4: Pushbelt variator kinematics

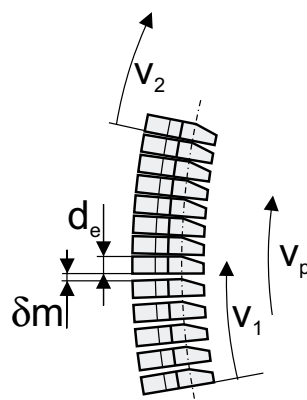


Figure 3.5: Gaps in the belt

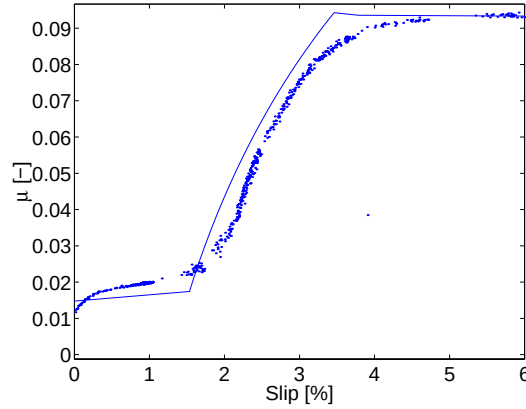


Figure 3.6: Traction curves with increased gap (1.8 mm), (solid: model and dots: measured data)

Here T is the time that passes between the entry of two sequential blocks. Using $v_e = \omega_s R_s$ and Equation (3.3) it follows that:

$$\nu = \frac{\delta_t}{\alpha_p R_p} \quad (3.7)$$

The inactive arc should always contain at least one block element, otherwise (macro-)slip will occur. It must be concluded that the presence of a gap δt must result in slip relative to the active arc on the driving pulley. Furthermore, the decrease of the friction coefficient with increasing slip speed has to be taken into account as shown by Kobayashi [47]. For this purpose the Stribeck friction model may be used [58].

With Equation (3.7) slip and traction can be determined from measured data as will be shown in Section 3.3. The tension and compression force distribution needed to calculate the lengthening of the belt are found using the continuous belt model or the pushbelt model described in Chapter 2. Also, we can calculate the idle arc from this model. From the idle arc, the length of the belt and the initial gap we can calculate an estimate for slip in the belt for a given load.

In Figure 3.6 a comparison is shown for the calculated traction curve and measured data. The measured data is acquired with an adapted belt. This belt had an increased gap, with one block element of thickness $d_e = 1.8mm$ removed. The influence of play is quite obvious. The point where the slope of the traction curve changes, nearly discontinuously, coincides with the point where the compression part changes to the other side. In the other traction curves this discontinuity is very small, but can be seen in Figure 3.1 for low. For overdrive this discontinuity occurs for reversed torque.

3.1.3. Results

The traction curve can be accurately described for micro-slip using this model. Measurements shown in Figure 3.6 show that for a situation where the gap is enlarged by removing a block, the model still accurately describes the traction curve.

3.2. Dynamic Variator Slip Model

The dynamics of slip as defined in Definition 10 is of interest for the design of a control system with slip as the design variable. First a nonlinear model will be derived for the dynamics of the variator. Then this model will be converted to a nonlinear model for slip in the variator. Finally a linearization of this model is given, which will be the input for the control design process.

Slip is a function of four variables: ω_p , ω_s , R_p and R_s . With R_p and R_s mutually dependent by the geometric configuration given by Equation (2.1), the geometric ratio is used as the dynamic variable as defined in Definition 1.

The variator consists of two shafts each with one rotational degree of freedom and a variable transmission ratio, which also introduces one degree of freedom. All other parts of the transmission, engine or driveline are modeled as a varying input torque and a varying output torque.

Figure 3.7 gives a schematic representation of the test setup, which consists of a driving motor, a variator and a load motor. The dynamical behavior of this system can be described by three

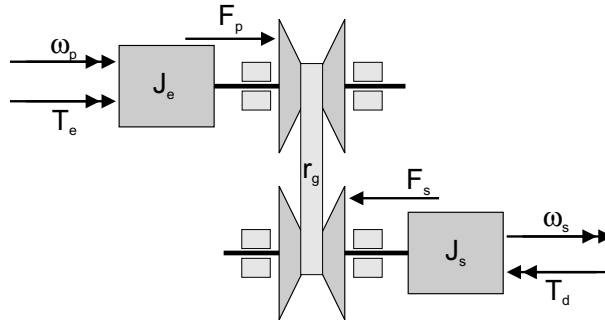


Figure 3.7: Scheme of the test setup

differential equations, *i.e.* one for the primary side (with moment of inertia J_e and driving torque T_e), one for the secondary side (with moment of inertia J_s and load torque T_d) and one for the variator. In Figure 3.8 the free body diagrams of the primary and secondary side are shown. The equations of motion for these part are given by:

$$J_e \dot{\omega}_p = T_e - T_p - T_{p,loss} \quad (3.8)$$

$$J_s \dot{\omega}_s = T_s - T_{s,loss} - T_d \quad (3.9)$$

where T_p and T_s are the torques, exerted by the belt on the primary and secondary pulley whereas $T_{p,loss}$ and $T_{s,loss}$ are the loss torques at the primary and secondary side. These loss torques are due to, amongst others, friction in the bearings of the shafts. It is assumed here that they are small enough to neglect them. Rewriting of Equation (3.1) results in:

$$T_p = \frac{2}{\cos(\beta)} \mu_{eff} \min(F_p, F_s) R_p(r_g) \quad (3.10)$$

for the primary torque T_p . If all inertia effects of the belt are neglected then, as shown in Chapter 2, the primary and the secondary torque are related by $T_s = r_g T_p$.

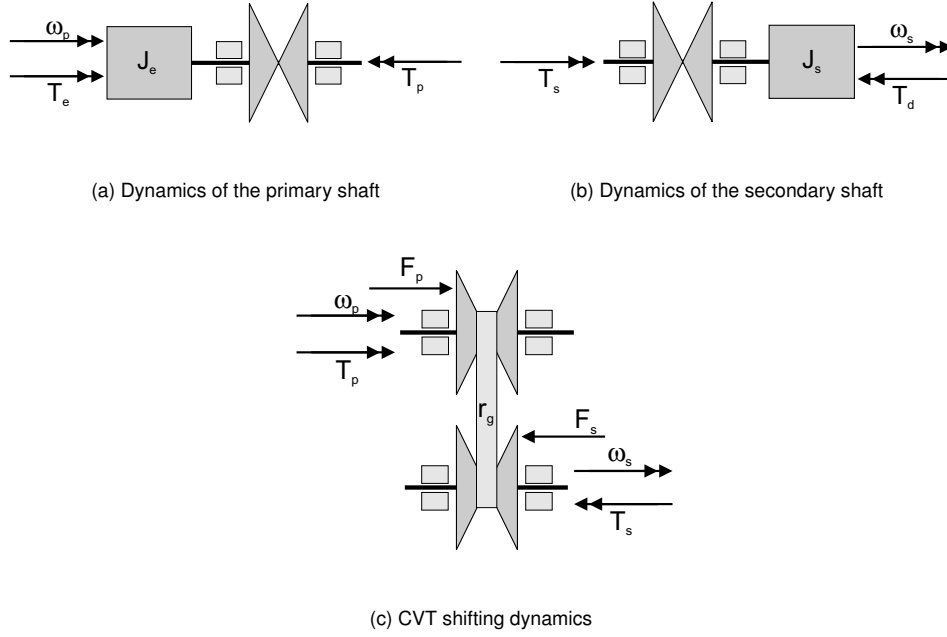


Figure 3.8: Free body diagrams of the primary and secondary side

The third differential equation for the system in Figure 3.7 is the relation for the rate of change of the geometric ratio r_g . Here the CMM model from Section 2.3 is adopted, so $\dot{r}_g$ is given by:

$$\dot{r}_g = \kappa(\nu, r_g) \omega_p \cdot \left[\ln\left(\frac{F_p}{F_s}\right) - \ln(\Psi(\nu, r_g)) \right] \quad (3.11)$$

The system equations now become:

$$\begin{bmatrix} \dot{\omega}_p \\ \dot{\omega}_s \\ \dot{r}_g \end{bmatrix} = \begin{bmatrix} \frac{T_e - T_p}{J_e} \\ \frac{T_s - T_d}{J_s} \\ \kappa \omega_p \left\{ \ln\left(\frac{F_p}{F_s}\right) - \ln(\Psi(\nu, r_g)) \right\} \end{bmatrix} \quad (3.12)$$

with state $\underline{x}$, input $\underline{u}$ and disturbance $\underline{w}$, defined by:

$$\underline{x}^T = [\omega_p \ \omega_s \ r_g] \quad (3.13)$$

$$\underline{u}^T = [F_p \ F_s] \quad (3.14)$$

$$\underline{w}^T = [T_e \ T_d], \quad (3.15)$$

the system equations can also be written in state space form, yielding:

$$\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u}) + \underline{L}\underline{w} \quad (3.16)$$

From Equation (3.3) it is seen that $\omega_s = (1 - \nu)r_g\omega_p$ and using this relation ω_s can be eliminated from the system equations, which is useful when studying the dynamics of slip. This results in:

$$\begin{bmatrix} J_e & 0 & 0 \\ (1 - \nu)r_g J_s & -\omega_p r_g J_s & (1 - \nu)\omega_p J_s \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\omega}_p \\ \dot{\nu} \\ \dot{r}_g \end{bmatrix} = \begin{bmatrix} T_e - T_p \\ T_s - T_d \\ \kappa \omega_p \left\{ \ln\left(\frac{F_p}{F_s}\right) - \ln(\Psi(\nu, r_g)) \right\} \end{bmatrix} \quad (3.17)$$

with state $\underline{x}$, input $\underline{u}$ and disturbance $\underline{w}$, defined by:

$$\underline{x}^T = [\omega_p \ \nu \ r_g] \quad (3.18)$$

$$\underline{u}^T = [F_p \ F_s] \quad (3.19)$$

$$\underline{w}^T = [T_e \ T_d], \quad (3.20)$$

the system equations can also be written in state space form, yielding:

$$\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u}) + \underline{L}(\underline{x})\underline{w} \quad (3.21)$$

For subsequent analyses a linearization of the model around stationary working points is needed. In each stationary working point $\dot{\underline{x}} = \underline{0}$, so $\dot{\omega}_p = 0$, $\dot{\nu} = 0$ and $\dot{r}_g = 0$. From Equations (3.10), (2.5) and (3.11) it then follows that¹:

$$T_{e,0} = T_{p,0} = 2\mu(\nu_0, r_{g,0}) \min(F_{p,0}, F_{s,0}) R_{p,0} / \cos(\beta) \quad (3.22)$$

$$T_{d,0} = T_{s,0} = r_{g,0} T_{p,0} = r_{g,0} T_{e,0} \quad (3.23)$$

$$F_{p,0} = \Psi(\nu_0, r_{g,0}) F_{s,0} \quad (3.24)$$

For a unique specification of the stationary point it is necessary to prescribe the primary rotational speed $\omega_{p,0}$, the geometric ratio $r_{g,0}$, the driving torque $T_{e,0}$ and the clamping force $F_{p,0}$ or $F_{s,0}$. The slip ν_0 then can be determined from Equation (3.22). In state space terms the stationary working point is characterized by the state $\underline{x}_0 = [\omega_{p,0} \ \nu_0 \ r_{g,0}]^T$, the input $\underline{u}_0 = [\Psi(\nu_0, r_{g,0}) F_{s,0} \ F_{s,0}]^T$ and the disturbances $\underline{w}_0 = [T_{e,0} \ r_{g,0} T_{e,0}]^T$.

Linearization requires that the partial derivatives of $R_p = R_p(r_g)$, $\mu = \mu(\nu, r_g)$, $\Psi = \Psi(\nu, r_g)$ and $\kappa = \kappa(\nu, r_g)$ are known. The derivative of R_p with respect to r_g can readily be determined from the geometric relations in Chapter 2. With respect to the experimentally determined functions μ , Ψ and κ it is assumed that they are continuous and at least once differentiable with continuous partial derivatives. Linearization of the system equations around a specified stationary working point then results in:

$$\delta \dot{\underline{x}} = \underline{A}_0 \delta \underline{x} + \underline{B}_0 \delta \underline{u} + \underline{L}_0 \delta \underline{w} \quad (3.25)$$

where $\delta \underline{x} = \underline{x} - \delta \underline{x}_0$, $\delta \underline{u} = \underline{u} - \delta \underline{u}_0$ and $\delta \underline{w} = \underline{w} - \delta \underline{w}_0$. Explicit relations for the entries of the state matrix $\underline{A}_0$, the input matrix $\underline{B}_0$ and the disturbance matrix $\underline{L}_0$ are given in Appendix B.

¹Quantities that refer to a stationary point are labelled with a subindex ₀

3.3. Ratio and Slip Estimation

This section describes some methods to estimate the geometric ratio r_g and relative slip ν . The geometric ratio and slip have to be estimated, because due to, for instance, pulley deformation, temperature influences and belt deformation, they cannot be measured directly. The investigated methods are:

- Pulley position measurement,
- Running radius measurement,
- Torque measurement,
- Normal load modulation,
- Engine load modulation.

Methods relying on the measurement of a geometric variable like pulley position or radial belt position, of belt speed measurement or of torque measurement are used to estimate the geometric ratio. Several authors have used pulley position measurement, (e.g. Kobayashi [47]) and running radius measurement (e.g. Nishizawa [54]).

Differential speed measurement as used in some ABS systems try to distinguish between ratio changes and slip by the speed at which the change occurs. Modulation methods are used to estimate the slope of the traction curve. A normal load modulation method was proposed by Faust [30].

After the overview of possible slip estimation methods a selection will be made based on this overview and some measurements.

3.3.1. Estimation vs Measurement

Slip as defined in Definition 10 cannot be used, because the geometric transmission ratio cannot be measured directly. The running radius of the belt on the pulley is in reality not constant along the wrapped angle of the belt around the pulley and therefore not uniquely defined. Therefore the ratio $r_g = R_p/R_s$ does not exist, since R_p and R_s are not uniquely defined.

In order to be able to estimate slip we need to define a measure for the geometric ratio and the relative slip between belt and pulley.

Definition 11 *The zero load ratio is defined by:*

$$r_{s0} = r_s, \text{ when } T_s = 0 \quad (3.26)$$

The zero load ratio r_{s0} can be seen as an estimate for the geometric ratio $\hat{r}_g$:

$$\hat{r}_g = r_{s0} \quad (3.27)$$

Proposition 1 *The estimation of slip can be calculated with:*

$$\hat{\nu} = 1 - \frac{\omega_s}{\omega_p \hat{r}_g} \quad (3.28)$$

Where $\hat{\nu}$ is the estimation of relative slip between belt and pulley and $\hat{r}_g$ the estimated geometric ratio [47].

For the control of slip in a variator it is not necessary that the estimated value of slip is actually very close to the 'real' value of slip. What is needed is a relation between the estimated slip and the variator efficiency that is known a priori.

3.3.2. Position measurement

If it is assumed that the change in pulley deformation due to changes in torque is negligible, a pulley position or running radius measurement can be used to determine an estimate $\hat{r}_g$ for r_g .

To accurately estimate r_g with a pulley position or running radius measurement, the position measurement has to be compensated for temperature changes and for changing clamping force levels. When the temperature rises, the transmission housing and all its parts will expand and the measured pulley position will change causing the estimated ratio $\hat{r}_g$ to change due to the fact that the sensor is mounted on the housing.

Clamping forces will have a large influence on the deformation of the pulleys, the belt and the housing. Therefore also this variable has to be taken into account when estimating r_g from the pulley position measurement.

The geometric ratio can be estimated using:

$$\hat{r}_g = f(x_{p,s}, T, F_s) \quad (3.29)$$

where f is a lookup table obtained by measurements of the zero-load ratio r_{s0} for different settings of the pulley position x_p or x_s or the running radius R_p or R_s , the temperature T and the clamping force F_s .

In Figure 3.9 a possible sensor position is shown for a running radius sensor. This sensor measures the running radius at one location on the wrapped angle of the belt. Figure 3.10 shows a possible configuration for the pulley position sensor [95].

3.3.3. Input/Output torque

The geometric ratio of the variator can also be approximated by the ratio of the input and output torque of the variator. If the belt is cut in between the two pulleys, the forces on both sides must be equal. The torque on the shafts is now given by the resultant of both belt ends:

$$T_p = (F_2 - F_1)R_p \quad (3.30)$$

$$T_s = (F_2 - F_1)R_s \quad (3.31)$$

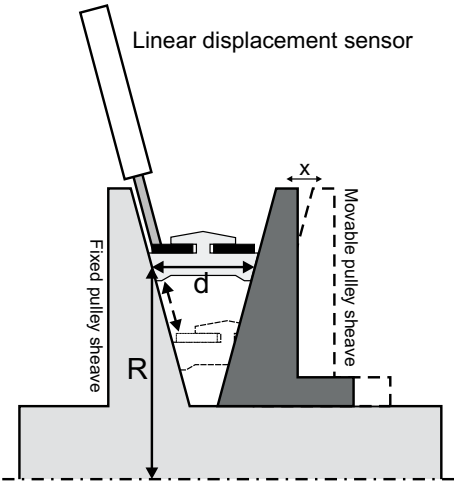


Figure 3.9: Running radius sensor

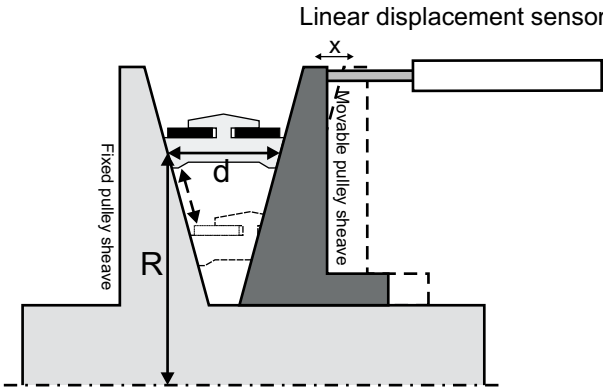


Figure 3.10: Sensor setup for pulley position measurement

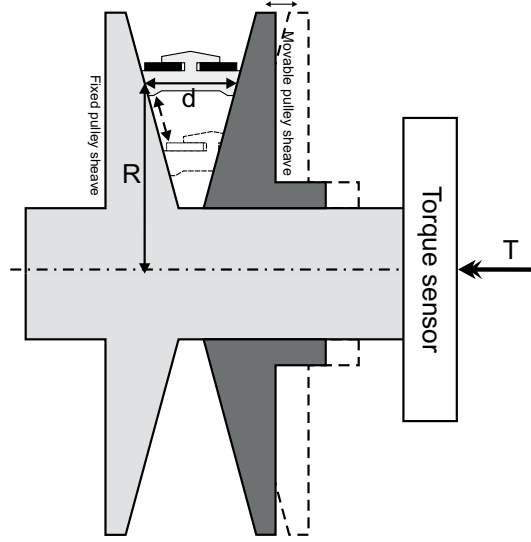


Figure 3.11: Torque sensing method

Therefore, the ratio T_p over T_s is the same as R_p over R_s , the geometric ratio. The geometric ratio r_g can therefore be estimated by:

$$\hat{r}_g = \frac{T_p}{T_s} \quad (3.32)$$

However, the torques are never measured directly after the variator, but rather on the shaft. Therefore, the measured torque will include losses due to slip, bearings etc. Since no real variator is completely lossless, the torque loss in the variator should be known to correctly use the torque ratio as an estimate for the geometric ratio:

$$\hat{r}_g = \frac{T_p + T_{loss}}{T_s} \quad (3.33)$$

here T_{loss} is the torque loss in the variator. This loss is caused by the losses in the belt and the losses in the bearings. This method can only yield good accuracy for higher torques and is therefore not suited for slip estimation in the lower torque regions.

Another disadvantage of this method is that torque measurement is expensive and error-prone and therefore not very interesting in automotive transmissions.

3.3.4. Beltspeed

By measuring the beltspeed in addition to the pulley speeds, an estimate can be determined of slip between belt and pulley [90]. If there is no slip and no other deformations occur, then $r_s = r_g =$

$$\frac{\omega_s}{\omega_p} = \frac{R_p}{R_s}.$$

It is assumed that if slip occurs, the belt slips only on one pulley. The models presented in Chapter 2 suggest that slip will always occur on the pulley with the smallest running radius. However, these models are simplifications of reality. Therefore it is not known a priori on which pulley this slip will

occur, especially around $r_g = 1$ this is unsure. Two situations must therefore be considered:

$$\hat{v}_{belt}^s = \omega_s \hat{R}_s \quad (3.34)$$

$$\hat{v}_{belt}^p = \omega_p \hat{R}_p \quad (3.35)$$

The running radius can be estimated by stating that if the slip occurs on one side, the running radius on the other side is equal to the beltspeed divided by the rotational speed of that pulley:

$$\hat{R}_s = v_{belt} / \omega_s \quad (3.36)$$

$$\hat{R}_p = v_{belt} / \omega_p \quad (3.37)$$

With the geometric relation for the belt, given in Equation (2.1) the running radius on the other pulley can be calculated.

$$\nu_p = 1 - \frac{v_{belt}}{\hat{v}_{belt}^s} \quad (3.38)$$

$$\nu_s = 1 - \frac{\hat{v}_{belt}^p}{v_{belt}} \quad (3.39)$$

If it is assumed that slip between belt and pulley takes place at the secondary side, then we can calculate the slip. This is a safe assumption, because if slip exists at the primary side, the estimated slip value will be much higher than the real value. The comparison is shown in Figure 3.12. Of course if slip exists only on the primary side this over estimates the slip by as much as 400%. Therefore it should be known on which side the slip occurs.

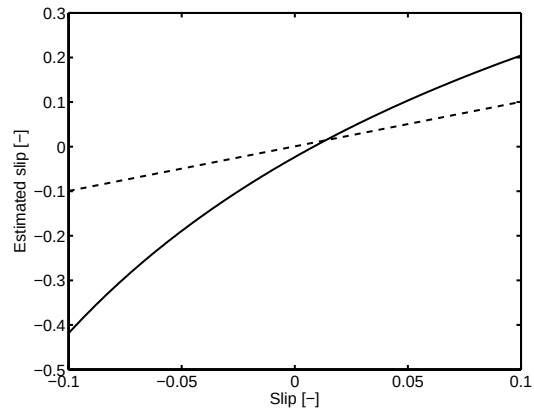
There are several methods for measuring the speed of the belt. One way is to measure the rotational speed of a wheel that runs on the belt. Another way is to measure the time between the passing of two blocks. There are a few options to do this. One is to measure the passing by of the airgap between the blocks using an inductive sensor on the outside of the belt when it runs on one of the pulleys. The difficulty here is that the running radius is not constant, therefore the sensor must follow the belt. Another option is to measure the passing by of the airgap between the blocks on the inside in the straight part of the belt in between of the pulleys. If a chain is used, also the passing of the pins on the side of the chain can be measured. In both of these cases the sensor must follow the belt closely.

If a belt is specially constructed, with a few magnetic blocks, or with alternating differently shaped blocks, then a hall-effect sensor can be used to measure the speed of the belt without the need to follow the belt closely.

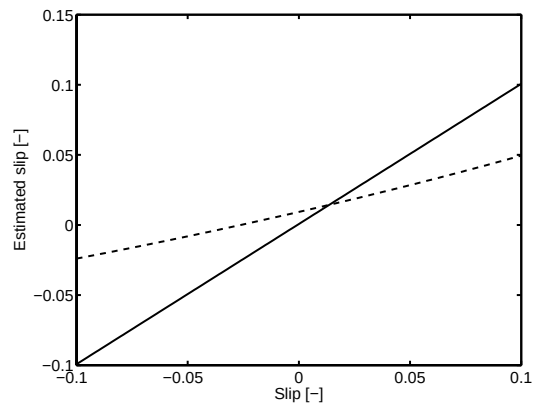
A sensor that measures the block pass frequency as described above must follow the belt closely. Therefore, a mechanism must be used that is stiff in the running direction of the belt, but flexible in all other directions. In Figure 3.16(a) such a device is shown for the inductive sensor.

3.3.5. Modulation methods

The frequency response functions of the variator will vary with the slope of the traction curve [66] [56]. In a single operating point of the variator the slope of the traction curve can be viewed upon



(a) Estimated slip with slip on secondary shaft (dashed: assumed secondary side slipping, solid: assumed primary side slipping)



(b) Estimated slip with slip on primary shaft (dashed: assumed secondary side slipping, solid: assumed primary side slipping)

Figure 3.12: Estimated slip

as being a viscous damping constant. This value influences the response of the output speed to changes in the input speed. This form of slip estimation was introduced for CVT applications by Faust et al. [30].

This method can be used to design an estimator for the slope of the traction curve. This slope is different for micro- and macro-slip, therefore it is possible with this method to determine whether the variator is in the micro- or in the macro-slip area.

For the modeling we look back at the slip model from Chapter 3. Assuming that the ratio remains unchanged the variable states are given by: $x = [\omega_p, \omega_s]^T$.

Normal load modulation

By linearizing in a certain working point a linear state space description is obtained. First, the effective friction coefficient is taken piecewise linear. The sections of the curve where the traction is increasing with increasing slip is the so called micro-slip area. The part of the traction curve where the traction is constant or slightly decreasing with slip, is called the macro-slip area. Like in Section 3.3, the traction coefficient can be approximated by:

$$\mu_{lin}(\nu, r_g) = M_\nu \nu + M_r r_g + M_0 \quad (3.40)$$

The state-space input is: $u = [F_s]$. The dynamic slip model, Equation (3.12), is linearized in point $x_0 = [\omega_{p0}, \omega_{s0}]^T$.

$$\begin{aligned} \dot{x} = & \begin{bmatrix} -\frac{2M_\nu \omega_{s0} F_s R_p}{\omega_{p0}^2 r_g \cos(\beta) J_e} & \frac{2M_\nu F_s R_p}{\omega_{p0} r_g \cos(\beta) J_e} \\ \frac{2M_\nu \omega_{s0} F_s R_s}{\omega_{p0}^2 r_g \cos(\beta) J_s} & -\frac{2M_\nu F_s R_s}{\omega_{p0} r_g \cos(\beta) J_s} \end{bmatrix} x + \dots \\ & \dots + \begin{bmatrix} -\frac{(2M_\nu \nu + 2M_0) R_p}{\cos(\beta) J_e} \\ \frac{(2M_\nu \nu + 2M_0) R_s}{\cos(\beta) J_s} \end{bmatrix} u \\ y = & \mathbf{I}x + \mathbf{0}u \end{aligned} \quad (3.41)$$

$$y = \mathbf{I}x + \mathbf{0}u \quad (3.42)$$

As can be seen from the bode plot of this system shown in Figure 3.13, the frequency responses of the system with macro-slip are significantly different from the bode plot of the system in micro-slip.

Engine load modulation

Basically engine load modulation is the same method as the Normal Load Modulation, except that the normal load is not modulated, but the driving torque. Because in Automotive applications this happens naturally with IC engines when the explosion in the cilinder occurs, no additional actuation is needed. Also the high frequency of the torque peaks from the engine give higher resolution than the normal load modulation technique. This type of observer is also under investigation by Toyota [98].

The input is now chosen as: $u = [T_e]$. The dynamic slip model, Equation (3.12), is linearized in

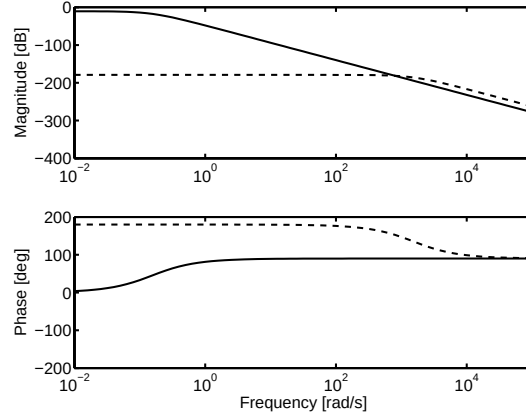


Figure 3.13: Transfer function ω_p/F_s for different slip conditions (dashed macro-slip, solid micro-slip)

point $x_0 = [\omega_{p0}, \omega_{s0}]^T$. The resulting system is:

$$\dot{x} = \begin{bmatrix} -\frac{2M_\nu \omega_{s0} F_s R_p}{\omega_{p0}^2 r_g \cos(\beta) J_e} & \frac{2M_\nu F_s R_p}{\omega_{p0} r_g \cos(\beta) J_e} \\ \frac{2M_\nu \omega_{s0} F_s R_s}{\omega_{p0}^2 r_g \cos(\beta) J_s} & -\frac{2M_\nu F_s R_s}{\omega_{p0} r_g \cos(\beta) J_s} \end{bmatrix} x + \dots$$

$$\dots + \begin{bmatrix} \frac{1}{J_e} \\ 0 \end{bmatrix} u$$

(3.43)

$$y = \mathbf{I}x + \mathbf{0}u$$

(3.44)

The bode plots for the transfer function y_1/u (ω_p/T_e) is shown in Figure 3.14. As for the normal load modulation, also the modulation of the input torque can be used to determine the traction conditions of the variator.

There is a distinct relation between the slope of the tractioncurve and this transfer function.

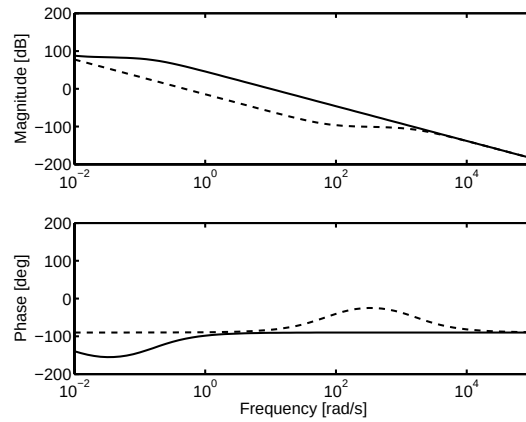


Figure 3.14: Transfer function ω_p/T_e for different slip conditions (dashed: macro-slip, solid: micro-slip) [66] [56]

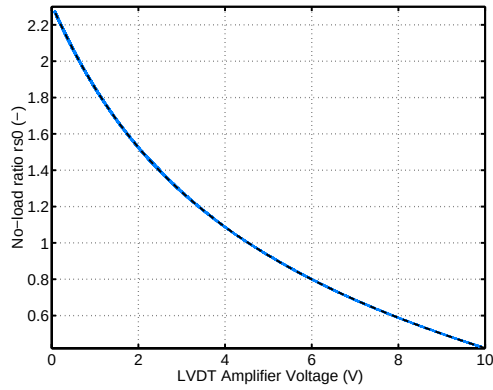
3.3.6. Results

The pulley position measurement is used in this research for the estimation of the geometric ratio and slip. In Figure 3.15 experimental results are shown using a LVDT sensor to measure the pulley position on the primary side. In this figure is also shown a polynomial fit of the measured sensor output vs. the measured zero-load ratio. Also shown is the drift of the sensor output with respect to the temperature. This temperature dependence is quite large and should therefore be compensated for. As can be seen from the graph, a linear fit gives a good estimation of this drift. For the pressure, not shown in the graphs, also holds that its influence is not negligible and should be compensated for. If these compensations are made good accuracy can be achieved for the slip estimation.

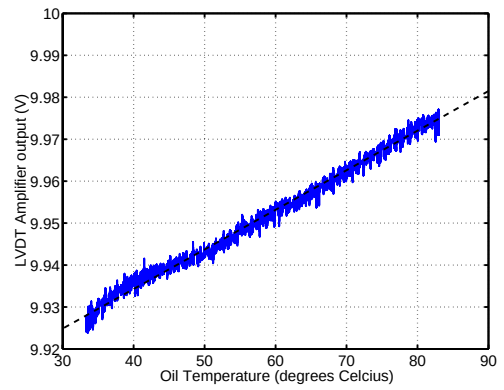
In Figure 3.16 the measurement setup and measurements of the beltspeed are shown using an inductive sensor placed close to the belt in between the pulleys. This type of measurement is less sensitive to temperature and pressure, because if the beltspeed changes, so does the input or output shaft speed. However, the pulley on which slip occurs is uncertain. This has to be investigated further to make the beltspeed measurement method a reliable estimator for the slip between belt and pulley.

The other techniques like the modulation technique are less accurate, due to the sensitivity for noise in the case of the modulation technique and due to the unknown side on which slip occurs for the beltspeed measurement. Also very high resolution measurements are required to make a reliable estimation.

The torque measurement is only accurate for high torques and therefore not very interesting for estimating the geometric ratio for slip control applications. Slip control is especially interesting for low input torques, as will be explained in Chapter 4.

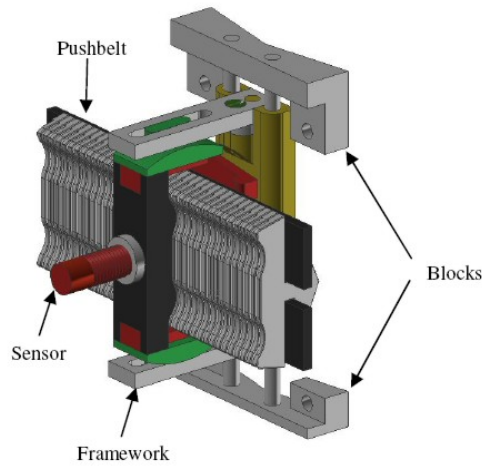


(a) Measurements vs Polynomial fit (dashed line) of the pulley position measurement output (LVDT) vs the measured speed ratio at zero load ($T_s = 0$).

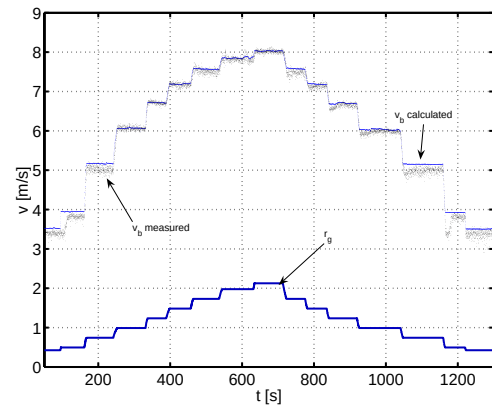


(b) Measurements vs Linear fit (dashed line) of the temperature drift at a constant speed ratio at zero load ($T_s = 0$).

Figure 3.15: Pulley position measurement and calibration



(a) Image of the beltspeed sensor.



(b) Measurement of the beltspeed using an inductive sensor. Shown are the transmission ratio, the calculated beltspeed (no slip) and the measured belt speed. The output torque is zero ($T_s = 0$).

Figure 3.16: Beltspeed measurement

Variator system losses

The efficiency of a variator is dependent on many factors. In this chapter all losses of the components in a hydraulically actuated variator, including the actuation system, are assessed on their contribution to the total energy loss. A mathematical model for the efficiency of a pushbelt type variator is derived and the parameters are estimated. This model can be used to determine the optimal operating points of a variator.

Earlier work on the efficiency of pushbelt variators include that of Micklem [53], who investigated power losses in the variator as a function of pressure and speed, Akehurst [1] [2] [3], who showed some detailed models of the loss mechanisms in the variator and Sue [81] who used Finite Element Analysis to determine the influence of radial slip in the efficiency of the variator. In this chapter the efficiency as a function of slip is investigated. As will be shown by measurements, the efficiency can be seen as a function of a torque loss and slip. This model was earlier presented by Veenhuizen [94], but here more detailed measurements are presented. Furthermore, the effects of a hydraulic actuation system will be discussed. The actuation system has a large influence on the overall efficiency and cannot be neglected.

The effect of efficiency optimization of the variator is discussed at the end of this chapter. The optimum operating point can be used as a setpoint in a variator control system.

4.1. Losses

Losses in a variator and its actuation system can be attributed to different components. A hydraulic actuation system needs a pump to generate hydraulic power. Other types of actuation systems also require auxiliary power to operate. This power is not used to drive the car and is therefore considered as a loss. Other losses occur in the bearings in the variator. These bearings are under considerable load because of the clamping forces. These losses are considered 'torque losses'.

The variator itself also contributes to the power loss. Since it is a friction based drive there is always some speed loss in the frictional contact. Also, internal friction in the belt or chain accounts for a

decrease in efficiency. These losses can be considered as a torque loss in the variator.

4.1.1. Efficiency of the variator

For power flow from primary to secondary pulley, the efficiency η of a variator is defined by:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{T_s \omega_s}{T_p \omega_p} \quad (4.1)$$

If T_{loss} is denoted as all torque losses discussed earlier, and ω_{loss} as the speed loss in the variator, then using the torque loss and the speed loss the output torque and output speed can be calculated by:

$$T_s = \frac{T_p}{r_g} - T_{loss} \quad (4.2)$$

$$\omega_s = \omega_p r_g - \omega_{loss} \quad (4.3)$$

The efficiency of the transmission now becomes:

$$\eta = \left(1 - \frac{r_g T_{loss}}{T_p}\right) (1 - \nu) \quad (4.4)$$

4.1.2. Losses in the hydraulic actuation system

The power P_h consumed by a hydraulic pump without leakage as given in Equation (4.5) is therefore linear with both pressure p_l and speed ω_h of the pump shaft. The pressure is approximately linearly dependent on the clamping forces. Furthermore, the hydraulic pumps used always have a certain amount of leakage.

The pumping losses, P_h , are:

$$P_h = \omega_h c_{fl} p_l + T_{fr} \quad (4.5)$$

where c_{fl} is the flow per rotation of the pump and T_{fr} is the friction torque of the hydraulic pump. Lowering the pressure decreases the hydraulic loss in the transmission. In most production CVTs the pump runs at a speed proportional to the engine speed. Because the CVT has to function properly at idle speed, the pumped volume is higher than necessary in most driving situations.

4.1.3. Losses in the variator

To calculate the bearing losses SKF [73] supplies models and specifications. The frictional moment of the bearings depends on the type of the bearing, the load, the speed of the axle and the type and quantity of the lubricant used.

The bearings have a power loss that depends on the input speed and the load. Because the load increases with input torque and therefore with input power and the speed also is proportional to the input power, these have a constant influence on the efficiency with varying loads or speeds, but there is also a constant friction part that accounts for very low transmission efficiencies at low input

power.

In a pushbelt speed and torque losses occur. These are caused by the bands running at a different radius than the blocks. This difference in running radius causes a slip between the blocks and the bands in ratios other than medium ($r_g \neq 1$). Also friction between the blocks account for some losses. This has been discussed earlier in Section 2.2.2. Also chains and other types of belts have internal friction when bending and stretching. These losses are also modeled in the parameter torque loss.

The torque loss T_{loss} is not negligible, even not if the transmitted power is very small. Hence, the variator efficiency is very low if the input torque T_p is small.

Speed loss is mostly caused by slip in the variator. The speed loss consists of the slip between the belt and the pulley and the slip between blocks and bands. This slip is discussed in Chapter 3

4.2. Measurements

Efficiency measurements are shown in Figure 4.1. The testrig on which these measurements have been performed is described in detail in Chapter 6. From the figure can be seen that the optimal efficiency for different transmission ratios is found at different values of the slip.

The sensitivity of the efficiency for clamping force and speed are shown in Figures 4.2 and 4.3 respectively. Increasing speed has an adverse effect on efficiency as can be expected as does increasing clamping force. If the clamping force is increased both the forces in the bearings and the losses in the belt increase. However, as can be seen from Figure 4.3, the maximum efficiency for the higher pressure setting increases. This effect is caused by the higher torque that can be transmitted. The higher torque more than offsets the higher losses. When torque is constant however, the efficiency will still decrease with higher clamping force.

Efficiency is optimal for slip speeds of around 1%, but the optimal value varies with clamping force and ratio. Efficiency in medium ($r_g = 1$), is much higher due to the absence of slip in the belt. Also lower clamping forces give higher efficiency as can be expected, as does lower speed.

4.3. Torque loss

To obtain the torque loss from the measurements Equation (4.4) can be rewritten to:

$$T_{loss} = \left[1 - \frac{\eta}{1 - \nu} \right] T_p \quad (4.6)$$

T_{loss} is calculated for the efficiency measurements shown in Figure 4.1, the results are shown in Figure 4.4. With the torque loss estimation from the figure the efficiency can be estimated when the primary torque is known. This torque however is itself a function of slip, clamping force and ratio.

To compare the efficiency measurements to the torque loss model first a model for the traction has

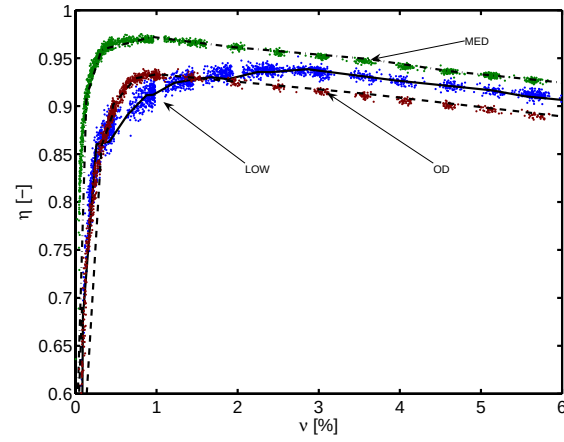


Figure 4.1: Efficiency as a function of slip measured at $r_g = 0.43$: solid line, $r_g = 1$: dash-dotted line and $r_g = 2.25$: dashed line, with $F_s = 8 \text{ kN}$ and $\omega_p = 225 \text{ rad/s}$

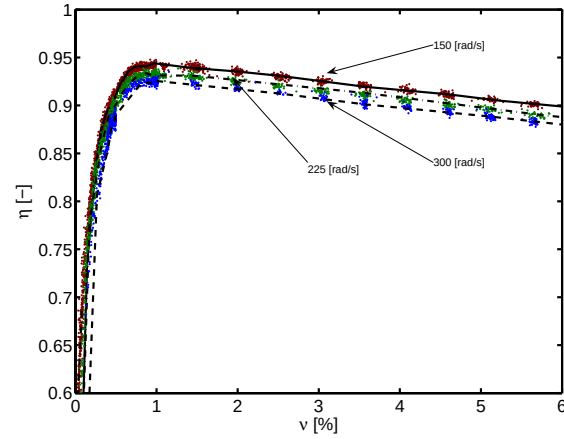


Figure 4.2: Efficiency as a function of slip measured at $r_g = 2.25$ and $F_s = 8 \text{ kN}$ for $\omega_p = 150 \text{ rad/s}$: solid line, $\omega_p = 225 \text{ rad/s}$: dash-dotted line and $\omega_p = 300 \text{ rad/s}$: dashed line

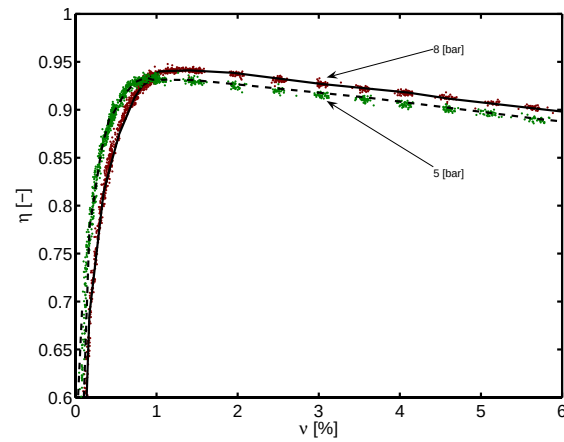


Figure 4.3: Efficiency as a function of slip measured at $r_g = 2.25$ and $\omega_p = 225 \text{ rad/s}$ for $F_s = 8 \text{ kN}$: solid line and $F_s = 5 \text{ kN}$: dashed line

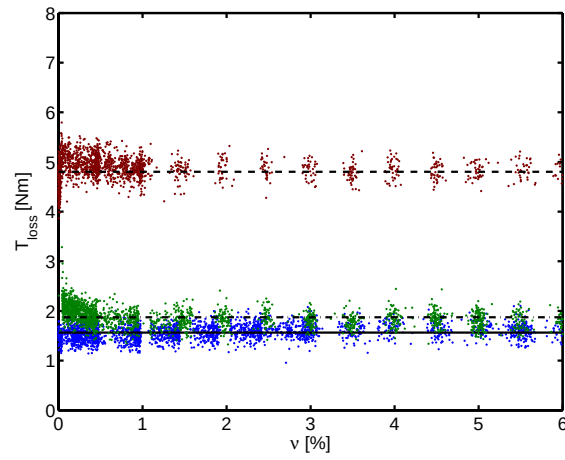


Figure 4.4: Torque losses in the variator as a function of slip, solid: low ratio, dash-dot: medium ratio and dashed: overdrive ratio ($F_s = 8 \text{ kN}$, $\omega_p = 225 \text{ rad/s}$).

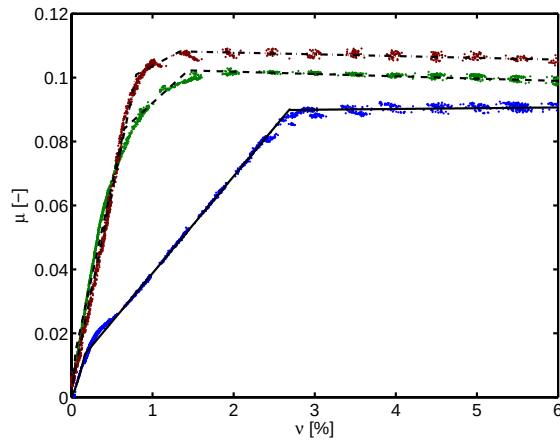


Figure 4.5: Traction curves as a function of slip for different ratios, solid: low ratio, dash-dot: medium ratio and dashed: overdrive ratio ($F_s = 8 \text{ kN}$, $\omega_p = 225 \text{ rad/s}$).

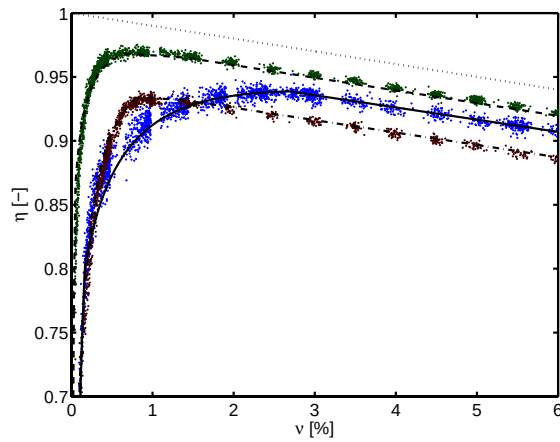


Figure 4.6: Measurements of the efficiency as a function of slip for different ratios and the model fit, solid: low ratio, dash-dot: medium ratio and dashed: overdrive ratio ($F_s = 8 \text{ kN}$, $\omega_p = 225 \text{ rad/s}$).

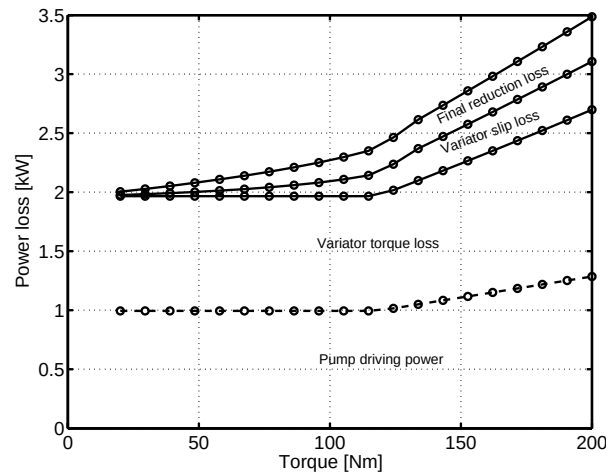


Figure 4.7: Breakdown of the losses in a CK2 transmission.

to be known. The traction coefficient is modeled piecewise linear with slip. This approximation is then used to calculate the torque through the variator depending on slip and ratio. The traction-curve measurements and the linearized model are shown in Figure 4.5. In this graph three different slopes are used to approximate the traction curve. More pieces give better accuracy.

4.4. Results

With this information the torque loss can be estimated by comparing the model to the efficiency measurements. With the appropriate torque loss the efficiency model is evaluated using the given functions in Equation (4.4). The result is plotted in Figure 4.6. With the dotted line the speed loss due to slip is shown. Efficiency can not be higher than that line due to the slip loss.

From Figure 4.6 can be seen that the model gives a very good prediction of the variator efficiency using a constant torque loss for a given transmission ratio, input speed and clamping force. Of interest is further that the efficiency is much higher for medium ratio, than it is for low and overdrive ratios. Maximum efficiency decreases with increasing clamping force or increasing input shaft speed as can be expected.

4.5. Efficiency improvement potential

To estimate the efficiency improvement potential in a CVT that is currently in production, the model is applied to a Jatco CK2 transmission [40]. In Figure 4.7 the different loss components are shown for this transmission. From the figure it can be seen that the pump loss and the variator torque loss are the major components in the CVT losses. By using energy optimized clamping force control both the variator torque loss and the pump loss could be reduced. The variator torque loss is optimized by the controller and the pump loss is reduced as a side effect due to the reduction in clamping force and therefore in hydraulic pressure. In the lower torque region the variator torque

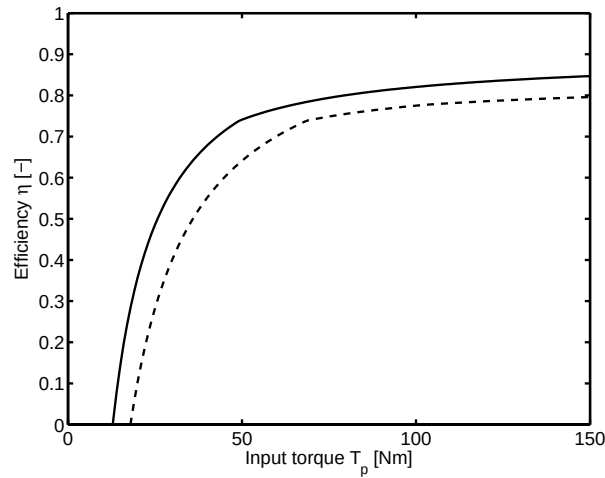


Figure 4.8: Comparison of efficiency vs torque input. Solid line: optimal control, and dashed line: standard TCM control ($r = 0.64$, $\omega_p = 300 \text{ rad/s}$).

loss and the pump loss are constant, because in this region the clamping force is equal to the minimum clamping force and therefore not dependent on the load.

In Figure 4.8 a comparison is shown between a CK2 controlled in the most efficient operating point, i.e. the clamping force for which the losses are minimized given the torque input, ratio and speed, and a CK2 controlled by the Transmission Control Module (TCM) from Jatco. The TCM uses higher clamping forces than necessary for transmitting the required torque. This causes the efficiency to drop considerably compared to an optimally controlled CK2, especially in the lower ratios. In high ratios (overdrive) the efficiency gain from optimal clamping force control in a CK2 is limited, because there is a lower limit to the clamping force. Therefore, it is not possible to lower the clamping forces substantially in the overdrive ratios, limiting the possible efficiency gains.

From Figure 4.8 it can be seen that considerable efficiency gains can be obtained using optimal control, from 5% for high torques to more than 20% in the low torque regions. Although this figure is lower for higher transmission ratios, there is still some gain possible. The efficiency gains in higher ratios are lower, because the clamping forces are generally lower and therefore the losses are lower, also when using the TCM control system.

The efficiency can be improved much in the lower torque regions, because the control system will use the lowest possible clamping force where the TCM will use clamping forces which are much higher [22].

If slip in the variator is used as the control variable, an optimal clamping force controller could be constructed, because slip in the variator is a very good predictor for the optimal efficiency point of the variator. The possibilities of control of slip in the variator are discussed. Fuel consumption can be compared on the basis of the total fuel consumption on the New European Drive Cycle (NEDC) (see Appendix D). In Figure 4.9 the simulation result for the NEDC cycle is shown for a

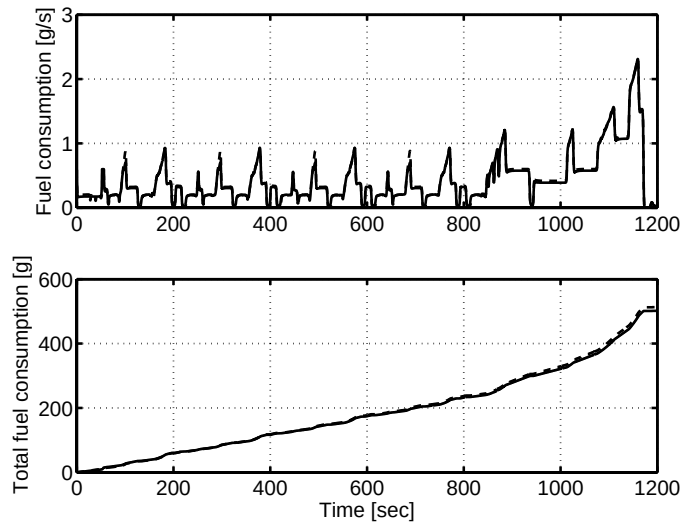


Figure 4.9: The results for the NEDC with slip control. The solid line is the result for an optimized slip control with a low minimum pressure and the dashed line is the result for a CK2 with slip control.

CK2 transmission in a Nissan Primera. Using the engine characteristic, transmission losses and control system used in Chapter 7, an estimation is made for the fuel economy of the Nissan Primera with slip controlled CVT. The total is 516.5 g which means that on the NEDC the fuel consumption was 6.6 l/100km. Compared to the data from Nissan (7.0 l/100km on the highway cycle and 8.8 l/100km in the combined cycle [55]). This is a large improvement. However, due to the limited accuracy of the simulation probably the total gain will be less, but still significant.

4.6. Conclusions variator efficiency

To fully understand the efficiency potential of a CVT all losses in the variator were included in the analysis. In Chapter 2 only the variator torque loss was examined. Also the losses of an actuation system have to be taken into account when optimizing the efficiency of the CVT. If a hydraulic system is used to actuate the CVT, also the losses caused by this system will be dependent on the clamping force and therefore on the type of control that is used. A torque loss map is shown in Figure 4.10.

From Figure 4.4 it was clear that the torque loss in the variator is independent from the value of slip. The torque losses depend on the ratio r_g , the input shaft speed ω_p and the clamping force F_s . Efficiency measurements of an hydraulically actuated CVT were already shown in Figure 4.6. These measurements only show the variator efficiency, the pump losses have not been taken into account. Finally, the efficiency including the pump losses is shown in Figure 4.11. Not only are the maximum efficiencies lower, but also the level of slip at the maximum is higher. Using the information in this figure the optimal operation points of this CVT can be found.

If these figures are compared to the traction curve measurements shown in Figure 4.5, it can be

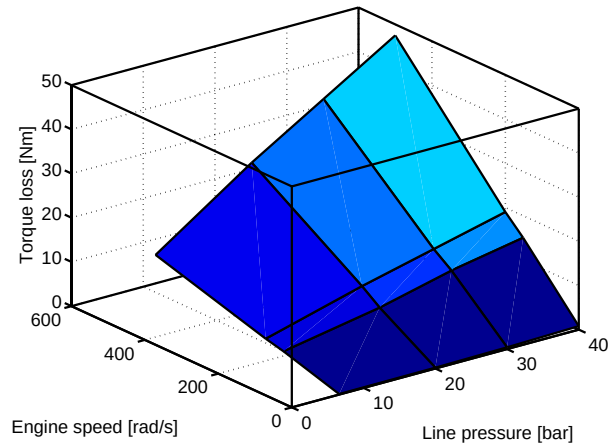


Figure 4.10: Torque loss map of the hydraulic pump unit in a CK2 transmission

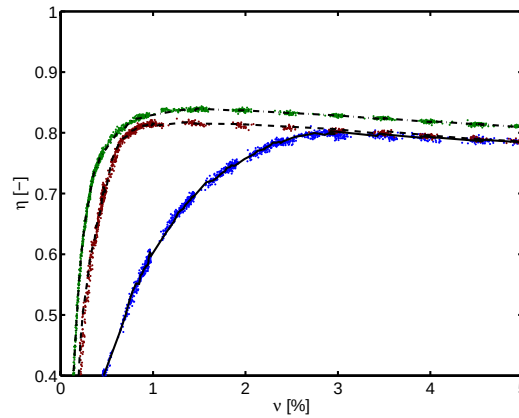


Figure 4.11: Efficiency vs Slip in a CK2 transmission with pumping losses

seen that without the losses in the hydraulic pump the maximum efficiency is reached before the maximum traction coefficient is reached. However, when the pump losses are included, maximum efficiency is reached near the same slip level where the maximum of the traction coefficient occurs. The simulation shows that it is very likely that a large potential for fuel economy improvement exists for this transmission.

Variator slip can be used to obtain optimal clamping force control, because variator slip is a good predictor for the efficiency of the variator. This topic will be further discussed in Chapter 5.

CVT Control

Control of slip is very common in automotive applications like traction control and anti lock braking systems. However, slip control has not attracted much attention in the field of CVT systems until now. Van Doorne's Transmissie started investigating slip in a variator, in order to improve the efficiency of the CVT. The problem of very high torques in geared-neutral transmissions around the geared neutral point can be solved with slip control as well[93]. Until now the main problem for implementation was that insufficient knowledge was available on measurement of slip in the variator, which was discussed in Section 3.3, and the behavior of the variator under slip conditions, which was discussed in Sections 3.1 and 3.2. Furthermore, until recently it was not known how much slip the variator could withstand. Findings by van Drogen et al. [89] suggest that a metal pushing V-belt can sustain a significant amount of slip for a substantial period of time.

The relation between slip and efficiency of the CVT was shown in Chapter 4. Here it was argued that control of slip in the variator indeed offers appealing opportunities for efficiency optimization. The relation between slip and traction shown in Chapter 3 is important for the understanding of the slipping behavior of the variator.

In this chapter the control problem of a variator is discussed. First, a short overview of the current control methods is given. Then, the most important properties of a clamping force control system are discussed. Finally, some simulations are made to assess the possibilities of slip control in a CVT.

In the next chapters several realizations of slip control will be discussed based on the slip dynamics model and the transient variator model from Chapter 3. The first implementation of a slip controlled variator is discussed in Chapter 6. Then in Chapter 7 an implementation will be shown in a CVT. In Chapter 8, an implementation will be shown in a Nissan Primera.

5.1. Control problem

As was shown in Chapter 4, the variator should be controlled at or very close to the maximum traction point to achieve optimal efficiency in the case of a hydraulically actuated CVT as was shown in Figure 4.11. However, controlling the variator at or near the maximum traction coefficient is not trivial due to instability in that point.

The control problem of slip in a variator is similar to that of ABS systems, because the traction curve of a variator is similar to the traction of a tire on the road. In the case of ABS several methods have been used. Hybrid control methods like found in Automotive Control Systems [41] were the first to be implemented. More recently, continuous control methods have been examined to improve the performance and the comfort. For example the gain-scheduled LQR method [63] or the gain-scheduled PI method [77] and optimal braking using the slip ratio [23]. Nonlinear control techniques like sliding mode control [74] [49] [85] have also been examined for ABS systems. A special case of sliding mode controllers, extremum seeking controllers, have been used for finding the maximum traction point in ABS systems [26].

When operating the variator near the maximum traction point the stability of the open-loop system is marginalized. The variator is stable when the traction curve has a positive slope as was shown in Section 3.2. This is no longer the case at maximum traction. The non-controlled system will become vulnerable to torque disturbances, because a small torque increase can put the variator in the unstable area of the traction curve, which could result in very high slip values in the open loop system.

There are uncertainties in the estimate of the maximum traction point. The traction coefficient will not be constant over the lifetime of the variator. The changes due to the dependence of ratio, pressure and speed can be compensated. The change over time however, is not exactly known.

5.2. Classic Clamping-force Control

Classic clamping force control in CVTs uses only feedback of the line-pressure, i.e. the secondary cylinder pressure in most cases. The required clamping forces are determined offline, resulting effectively in open-loop slip control. A lookup table is used to store the required clamping force depending on the ratio and the engine torque. Because open-loop control is used, a margin is needed in the clamping force to prevent disturbances to cause slip that could potentially cause damage to the variator. This margin is called the safety-margin. This method is described well by Vroemen [96].

In a CVT the belt is clamped by different forces on the primary and secondary axes. Ratio is held by controlling the balance of these forces and slip is prevented by prescribing the absolute level of the lowest of the two clamping forces. However, in most production CVT systems the clamping force is controlled only with the secondary clamping force and the ratio is controlled by changing

the primary clamping force. The reason for this is that the clamping force ratio is greater than one ($\Psi > 1$) for most operating points where slip might occur, as seen in Chapter 2. The clamping force on the primary side is therefore mostly higher than on the secondary side.

A commonly used clamping force strategy is a so-called safety strategy. This method is designed to keep the maximum torque capacity of the variator above the actual input torque at all times. Safety with respect to V-belt type variators is defined as the torque margin between the current input torque and the maximum transmittable input torque. This margin can be absolute, a safety torque, or relative to the expected input torque, a safety factor. If an absolute margin is taken then the minimally needed torque capacity of the variator needed to achieve this margin can be calculated with:

$$T_{min} = T_{sf} + |T_{input}| \quad (5.1)$$

Here T_{sf} is the safety torque. If the safety margin is taken relative to the input torque, then the minimally needed torque capacity of the variator is:

$$T_{min} = S_f |T_{input}| \quad (5.2)$$

where S_f is the safety factor. The safety needed depends on the uncertainty in the estimated input and output torque and the expected input level of the disturbances.

Using Definition (9) the minimally needed clamping force can now be calculated with:

$$F_{min} = T_{min} \frac{R_p \cos \beta}{2\mu_{max}} \quad (5.3)$$

If torque is very low, a minimum safety torque (T_{sf}) of around 30 Nm to 50 Nm is used to prevent slip in near zero-load conditions in the event of sudden shocks caused by for example road bumps. Increasing the safety margin decreases variator efficiency as discussed in Chapter 4, therefore the margin that should be kept depends on the desired robustness for torque load disturbances of the variator. A typical safety vs. efficiency relation is shown in Figure 5.1.

Engine torque is estimated using the throttle input signal from the driver, the actual engine speed, the state of the torque converter and an engine torque map. With this data a reasonable estimate can be made for the engine torque. However, since it is an estimate a certain margin has to be kept to be sure that modeling errors have no adverse effect. Also, sudden loads from the road can give higher torques. Therefore a good margin is needed to be able to cope with these uncertainties. The actual clamping forces applied to the pulleys (with $\dot{r} = 0$) can be calculated from Ψ . The primary and secondary clamping forces are given by:

$$F_p = \max(F_{min}, \Psi F_{min}) \quad (5.4)$$

$$F_s = \max(F_{min}, \frac{F_{min}}{\Psi}) \quad (5.5)$$

Over the lifespan of a variator the tribological properties will change. The maximum traction coefficient could decrease due to changes of the surface of the belt and the pulley and the traction fluid.

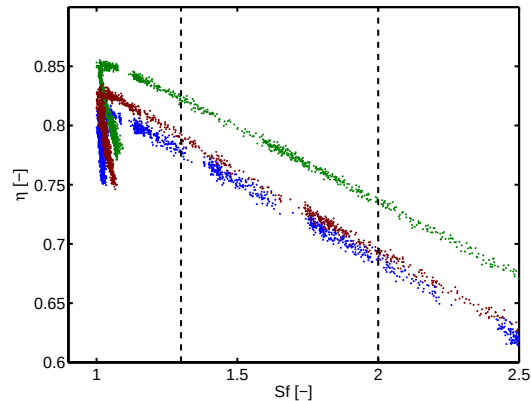


Figure 5.1: Safety factor vs. Efficiency given for three different transmission ratios.

An additional margin is needed to cope with these effects. Even if the driveline torque is measured, a safety-factor substantially higher than 1 is still necessary.

5.3. Ratio control

The speed ratio of a variator is controlled by changing the balance of the clamping forces. For each ratio there is a clamping force ratio for which the speed ratio of the transmission is constant as explained in Chapter 2. If the clamping force ratio is changed with respect to this equilibrium point, then the ratio of the transmission will change. A model for this was given in Section 2.3.

In modern CVTs the ratio is electronically regulated. The setpoint for the speed ratio of the transmission is determined by a predetermined algorithm. More on this issue can be found in Appendix E. Mostly, a speed ratio map is used that is a compromise between fuel economy and driveability.

In this chapter and the following chapters the ratio control system will often remain unmentioned. Although ratio control is used to keep the transmission in the correct position. No special attention has been paid to this issue. The ratio control algorithm is assumed to be adequate.

The influence of ratio control on slip in the variator is not assumed to be negligible, but to keep the initial control implementation simple, its effects are not taken into account. In Appendix A a proposal for a control method is given that does take ratio changing into account.

5.4. Slip Control

One way of coping with the uncertainties of the driveline torque, the traction coefficient and reducing the safety-factor at the same time is slip control. With slip control the danger of unstable slip is reduced using feedback control, while at the same time the clamping force is reduced, increasing the efficiency.

The first step to slip control is estimation of slip in the variator. This topic was discussed in Chapter 3. In this and the following chapters the pulley position measurement from Chapter 3 will be

used for estimation of slip. This method is chosen, because of the good accuracy and simplicity of implementation.

The second step is the design of the controller. Two designs will be proposed. First, a slip control algorithm is implemented on the beltbox testrig and second, a controller implemented in a production CVT. In Appendix A, a nonlinear design is proposed that decouples and linearizes ratio changing and slip. Stability, performance and robustness will be discussed.

Stability

First, the uncontrolled case is considered. To obtain a measure for the stability of the system the eigenvalues are calculated. The system is stable in an operation point if the eigenvalues for that point all lie in the left half of the complex plane.

For quasi-static situations ($\dot{\omega}_p = 0$ and $\dot{r}_g = 0$), no losses and no disturbances, the eigenvalues can be calculated from the linearized system matrix $\mathbf{A}_0$ from Section 3.2. Like in Section 3.3, Equation (3.40), the effective friction coefficient is taken piecewise linear. Defining the state space as $x = [\nu]$ and $u = [F_s]$ and $w = [T_e, T_d]^T$ the system can be linearized around a certain operating point $x_0 = \nu_0$, resulting in a linear system given by:

$$\delta \dot{x} = \mathbf{A}_0 \delta x + \mathbf{B}_0 \delta u + \mathbf{L}_0 \delta w \quad (5.6)$$

where $\delta x = x - x_0$, $\delta u = u - u_0$ and $\delta w = w - w_0$. The eigenvalue of the system is given by matrix $\mathbf{A}_0$. The linearized matrix $\mathbf{A}_0$ can now be derived (assuming $\nu_0 \ll 1$ and neglecting higher order terms):

$$\lambda = \mathbf{A}_0 = \frac{1}{\omega_{p0}} \left[\frac{-2R_{s0}F_0M_\nu}{J_{t0} \cos \beta} \right] \text{ with } J_{t0} = \left(\frac{r_{g0}}{J_e} + \frac{1}{J_s r_{g0}} \right)^{-1} \quad (5.7)$$

The eigenvalue is negative if $M_\nu > 0$, because F_s , J_e and J_s are always greater than zero. Therefore, the system is stable for positive values of M_ν and unstable for negative values of M_ν .

For micro-slip (Definition 7) the system will be open-loop stable. If macro-slip occurs (Definition 8), the slope of the traction curve will be zero or negative, resulting in an unstable open-loop system.

The control system should stabilize the slip in the variator in macro-slip.

Now the controlled case is considered. With a simple proportional controller with gain K , the eigenvalue can be found to be:

$$\lambda = \left[\frac{2(K_p M_0 + F_0 M_\nu)((\nu_0 - 1)r_{g0}^2 J_s - J_e)}{R_p(r_{g0})r_{g0} \cos(\beta) J_e \omega_{p0} J_s} \right] \quad (5.8)$$

To achieve stability the gain of the controller has to satisfy the following on the entire traction curve:

$$KM_0 > -F_0 M_\nu \quad (5.9)$$

Furthermore, stability is influenced by the bandwidth of the actuation system (i.e. the 'plant capacity'). In Figure 5.2 the frequency response function of the hydraulic actuation system in a CK2 transmission is shown. From this figure can be seen that the gain of the actuation system already

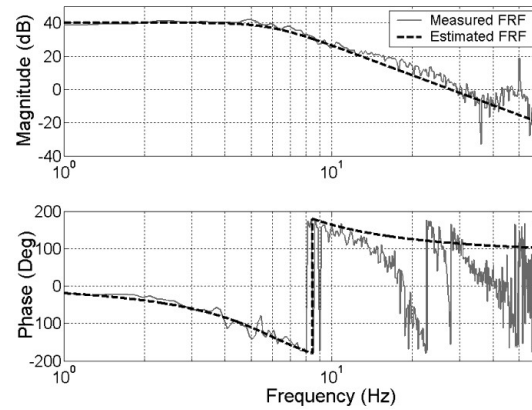


Figure 5.2: Frequency Response Function of the hydraulic actuation system of the CK2 transmission and the fitted third order low pass filter

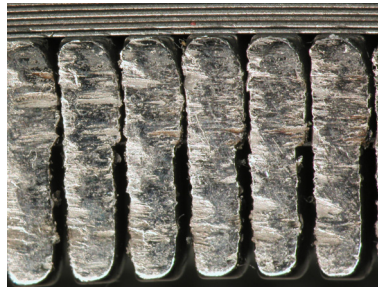


Figure 5.3: Damaged belt resulting from one of the experiments.

decreases starting at 5 Hz and the 180° phase lag is reached at 8 Hz . This will clearly limit the achievable bandwidth. This could cause problems if the disturbances acting on the system, mainly torque peaks, have a higher bandwidth than the actuation system. The effects of such disturbances cannot be suppressed and could potentially cause high levels of slip.

Performance

High slip values can seriously damage the belt and the pulleys. An example of a damaged belt is shown in Figure 5.3. Performance of the slip control system has to be good enough to keep the peak levels of slip within allowable levels. The slip in the variator must not exceed these levels in any situation, e.g. when the gas pedal is pressed suddenly, when moving from slippery surface to rough, or when a pothole is hit.

The F-v failure diagram, as shown in Figure 5.4 [68] [89], gives the relation between the element normal force and the absolute slip speed that indicates the boundary between adhesive wear and mild wear. The product of these two variables gives a measure of the power absorbed in the element-pulley contact for a certain friction coefficient ($P = Fv$). The limit of operation depends on the temperature, lubrication, input speed and other factors and must be determined experimentally as shown by van Drogen and van der Laan [89].

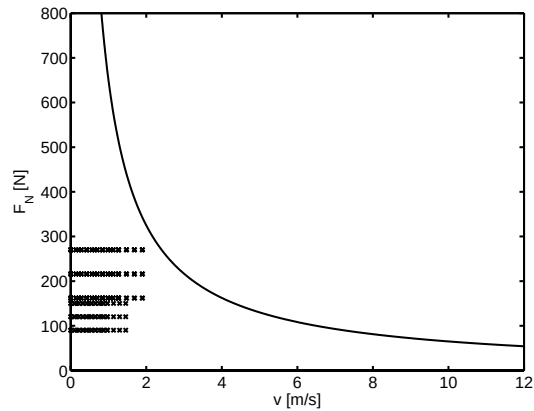


Figure 5.4: Example of a F-v failure diagram. The element normal force is given as a function of the absolute slip (real speed difference between the belt and the pulley).

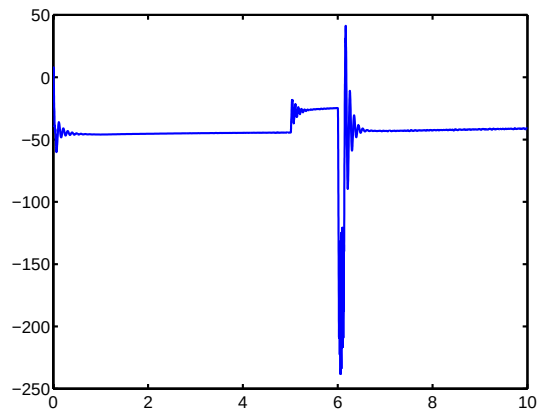


Figure 5.5: Torque peak from a road surface transition from an icy surface to a rough surface.

The variator can withstand some amount of slip given that the energy absorbed by the pulley-belt contact can be dissipated in the belt and pulley. And the normal force - slip speed combination should be below the adhesive wear boundary line. Here, ATF is used to cool the belt and pulley.

If slip in the variator is stabilized by a feedback controller, then the performance of this controller must be sufficient to keep the slip below the critical line from the F-v diagram (Figure 5.4). The difficulty is to do this for unknown disturbances caused by the road-wheel contact. These disturbances include torque peaks from crossing speed-bumps, suddenly changing tire friction due to road surface changes (ie going from slippery to rough surface) and μ -split situations. Disturbance rejection should be strong enough to keep the slip in the variator below the critical slip limit for all disturbances from the drivetrain. In Figure 5.5 an example is given of a surface transition torque peak.

Robustness

The control system will be designed using the slip dynamics model derived in Chapter 2. Modeling inherently introduces errors with respect to reality. These errors can be divided into two categories, the first one is unmodeled dynamics, the second one is parametric errors.

Unmodeled dynamics are caused by simplifications of the physical properties of the system or unknown properties of the system. If a linearized model is used, the nonlinear parts of the system are left out and are thus cause for unmodeled dynamics. Parametric errors occur when parameters of the model are not identified properly. This can be caused by limited accuracy of the estimation, or by varying parameters.

The control design should be robust for these modeling uncertainties and parameter errors. By demanding a maximum sensitivity of the closed loop system, robustness can be achieved for model imperfections, unknown parameters and aging effects.

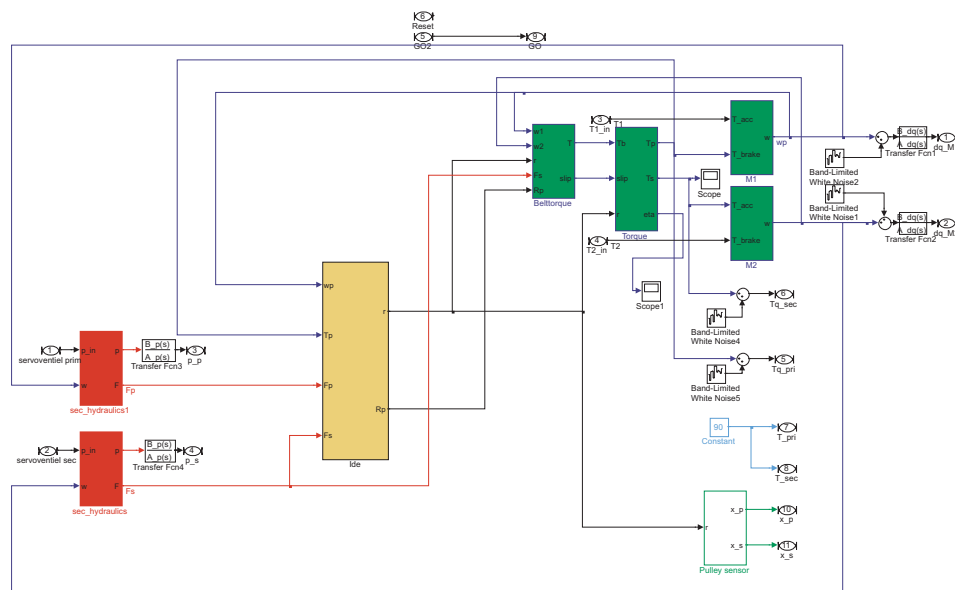
5.5. Simulation

The feasibility of slip control is tested using simulation. The simulation model, with a structure as shown in Figure 5.6(b), is a combination of the variator model, the traction model and the slip-dynamics model as discussed in Chapter 2 and Chapter 3. The Matlab[®] /Simulink[®] model is shown in Figure 5.6(a). Also the Delft Tyre model is used for road-load simulation [43] [69].

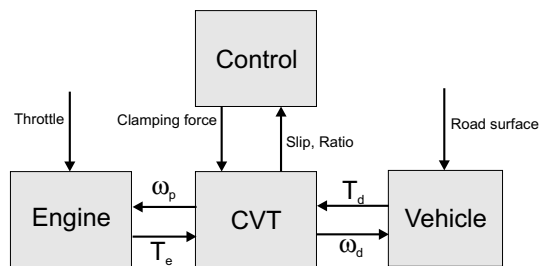
In Figure 5.7 the results are shown for the classic clamping force control system. A vehicle driving over a bump of 15 cm height is simulated. Plots for a safety-factor of 1.3, 1.5 and 2.0 are shown in this figure. In Figure 5.8 the same simulation is performed with a slip control strategy. The simulation is based on an input shaft torque of $T_p = 200 \text{ Nm}$, a CVT model based on the CK2 transmission and a vehicle that hits a 10cm bump at $v = 17 \text{ m/s}$. The torque peak that is generated is more than 150 Nm at the variator output shaft.

From these figures can be seen that with a normal safety strategy the slip is only limited using high safety factor, i.e. around 2.0. If a slip control strategy is applied, actuation bandwidth is essential. In Figure 5.8 several graphs are shown with varying bandwidth. From this figure can be seen that higher actuation bandwidth gives better results, reducing the slip peak to around 10%, comparable to a safety factor of 1.5. In the case of slip control an effective safety factor of 1.0 is used. This means that the clamping force is lower, resulting in less danger of damage for the same amount of slip for the variator.

In Figure 5.9 the result of torque peaks caused by sudden throttle actuation. In this graph the results for the different control strategies are shown in response to a 150 Nm engine torque increase. A comparison is shown for the situation of a slip controlled variator with and without feedforward. A feedforward is added based on a safety-factor strategy with a safety-factor of 1. This almost completely eliminating dangerous slip peaks, where in the case of only feedback slip control slip



(a) Matlab®/Simulink® model of the variator



(b) Simulation Structure

Figure 5.6: Simulation details

increases to around 10% due to the sudden change in engine torque.

5.6. Conclusions and recommendations variator control

In the first simulations the driven tires come completely off the ground and start spinning, when they touch the ground again the wheels are suddenly slowed at the friction limit of the tires, effectively the wheels are slipping. This is one of the most severe incidences that cause large torque peaks in the driveline. Also torque peaks that are caused by suddenly stepping on the throttle can cause large torque peaks. Although the torque peaks are not quite as high as the wheel slip situation, the duration can be much longer.

The simulations showed that a control system using a constant safety-factor is also vulnerable for high slip peaks caused by severe torque peaks. Also from the simulations can be seen that the bandwidth of the actuation system is a very important factor for slip control in a V-belt type variator. If the bandwidth is too low, torque peaks from the road cannot be compensated. The simulation of driver induced torque peaks shows that when a slow slip controller is used, this can cause serious problems with large amounts of slip. However, a feedforward control action can reduce this risk in this kind of predictable torque peaks.

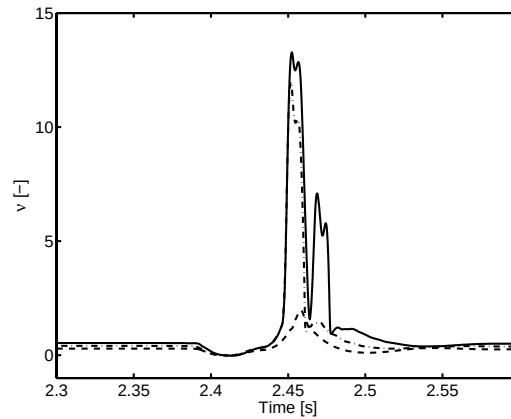


Figure 5.7: Simulation of vehicle driving over a 15 *cm* high obstacle using a safety-factor clamping force strategy (solid: $S_f = 1.3$, dash-dot: $S_f = 1.5$, dashed: $S_f = 2.0$).

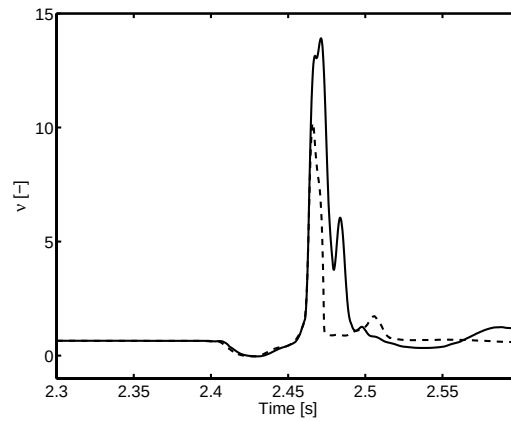


Figure 5.8: Simulation of vehicle driving over a 15 *cm* high obstacle using slip control (solid: 7 *Hz* bandwidth actuation, dashed: 20 *Hz* bandwidth actuation).

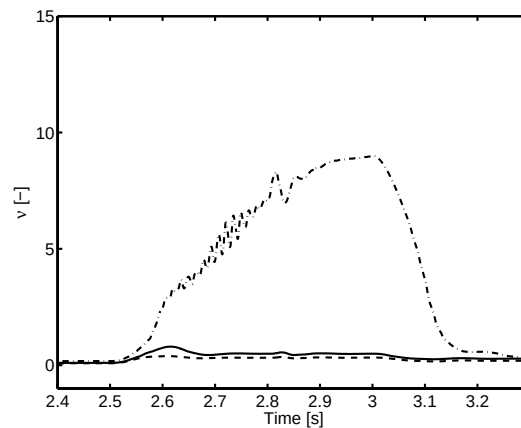


Figure 5.9: Simulation of a driver induced torque peak, engine torque going from 50 *Nm* to 200 *Nm* in 0.1 *s* for different control strategies (solid: safety-factor, dash-dot: slip control (7 *Hz*), dashed: slip control with feedforward (7 *Hz*)).

First generation slip controlled variator

Slip control is first implemented on the variator test rig [8], [11]. This test rig allows fast control of the clamping force and accurate measurement of the pulley position. Also the possibility exists to use fixed transmission ratio using so-called ratio rings. Because the ratio is fixed, the ratio changing of the variator can be neglected for the controller design. This setup is also used for the measurements of the traction curves and the efficiency shown in previous chapters.

This chapter will focus on the first implementation of slip control in a variator. First, the setup is described, then the control implementation is explained. Finally, the experimental results from the measurements on the test rig will be shown and some conclusions will be drawn.

6.1. Experimental setup

The experiments in this chapter are done on the test rig shown in Figure 6.1 and schematically represented in Figure 6.2. This test facility comprises two identical asynchronous electric motors with a maximum power: 78 kW , maximum torque of 298 Nm and a maximum speed of $525 \text{ rad/s} \simeq 5000 \text{ rpm}$. The pressure in the primary and secondary hydraulic cylinders of the variator can be controlled independently between 1 bar and 50 bar . The primary and secondary pressure are independently controlled using two Bosch-Rexroth solenoid valves. The hydraulic pressure directly controls the clamping force on the pulley.

The variator used in this setup is a Van Doorne's Transmissie beltbox type Alaska P811. The pulley shafts in this transmission have an axial distance of 155 mm and the length of the belt is 650 mm . The primary cilinder area, A_p , is 259.8 cm^2 and the secondary cilinder area, A_s , is 124.0 cm^2 . The pulley angle is 11° . This beltbox is lubricated and cooled using Automatic Transmission Fluid (ATF).

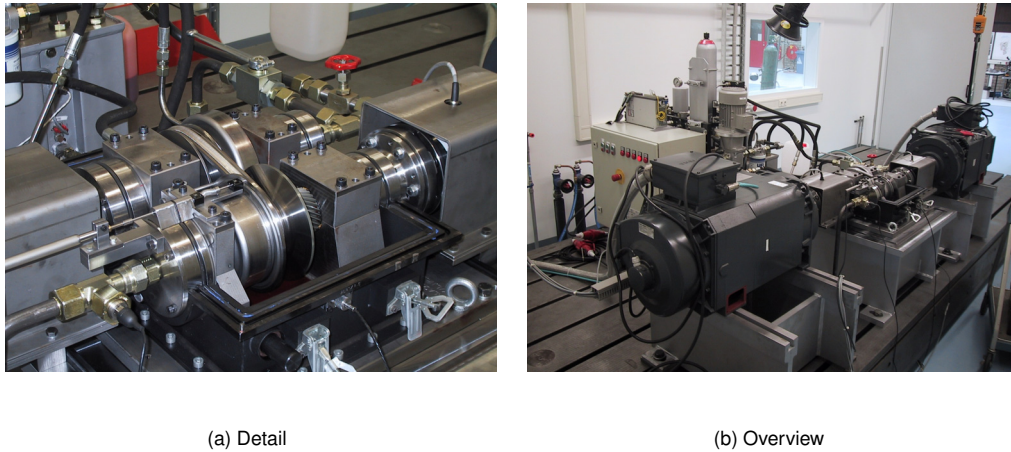


Figure 6.1: Test setup as used in the experiments

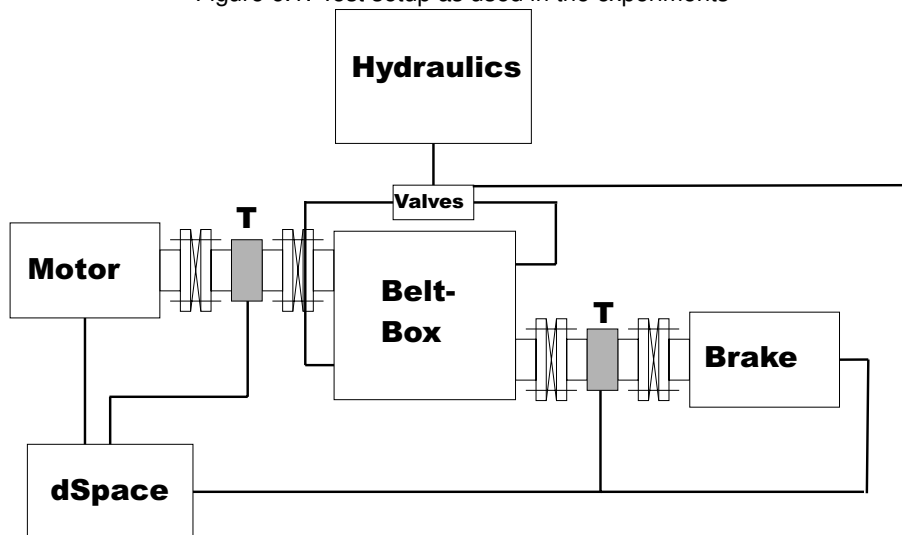


Figure 6.2: Layout of the test setup

The belt slip measurement system comprises three elements, two rotary encoders on both the input and output shaft, and a linear encoder on the secondary pulley position using a sliding contact. For the measurement of the pulley position contact is needed between the sensor and the rotating pulley. The environment in the transmission causes problems with contact-free methods and limits the space available to the sensor equipment. The pulley position is measured at the outer rim of the pulley. The signal of the two rotary encoders and the linear encoder is read by the dSpace[®] [27] encoder interface. The speed signal is derived using numerical differentiation of the position counter.

The oil sump temperature is regulated at 80 °C. Furthermore the torque measurement system consists of two HBM torque sensors.

For the data acquisition and controller implementation a dSpace[®] system is used in combination with dSpace[®] ControlDesk software. Matlab[®] /Simulink[®] [52], combined with the realtime tool-

Table 6.1: Measurement equipment for angular speeds and sheave position.

Measurement	Sensor	Resolution
Sec. pulley position	Heidenhain ST3078	$1 \mu m$
Incremental encoders	Heidenhain ERN 1381	$1/2048 rad$
Torques	HBM T20WN 200Nm	$0.2 Nm$

box, is used for the programming of the dSpace[®] system.

The clamping forces f_p and F_s are controlled by the hydraulic pressure. The pressure is measured at the hydraulic input side of the cylinders. The pressure inside the cylinders however, is also dependent on the shaft rotational speed. This is due to the centrifugal force acting on the oil inside of the cylinder. To maintain the required clamping forces, the pressure caused by the rotational speed of the shaft is deducted from the required pressure calculated by $p = F/A$ (the pressure is therefore calculated by: $p = F/A - f_c\omega_2$). The primary and secondary cylinder pressures are controlled using two independent servo-loops.

6.2. Control implementation

6.2.1. Introduction

The variator has two control inputs: F_p and F_s , the primary and secondary clamping force respectively, and two outputs: ν and x_p , the slip and the transmission ratio respectively. In Figure 6.3 the block diagram is shown of the control system.

The geometric ratio is controlled using the clamping force ratio as the control output and the position of the primary pulley as the control input. For the controller is chosen for an integral controller with low gain. During the tests the reference of the ratio is kept constant. This is done to prevent instability the ratio control loop and minimize the ratio control loop influence on the slip control loop. The slip is controlled using the secondary clamping force as the control output and the absolute slip as the control input. For this implementation is chosen for a PI controller for the slip control loop. A differentiating control action is omitted, because of the noise in the control input signal. In this setup slip is estimated using the primary and secondary shaft speeds and the primary pulley position measured by the linear encoder. The absolute value of the slip is taken, because the slip should be maintained for both positive and negative torques, causing positive and negative slip respectively.

6.2.2. System identification

Using measurements of the frequency response of the system, the characteristics of the system are identified. These measurements are done by adding noise to the output signal of the controller. With the measured input of the system and the measurement of the system output, the process

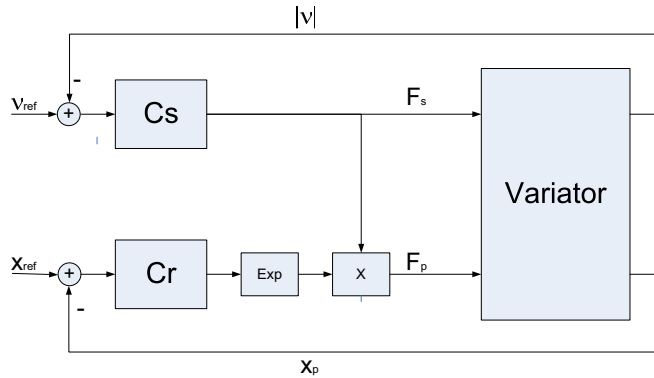


Figure 6.3: The block diagram of the control loop.

sensitivity function is obtained. The process sensitivity function is defined as:

$$S(s) = \frac{G(s)}{1 + C(s)G(s)} \quad (6.1)$$

with $C(s)$ equals Cs in Figure 6.3 and $G(s)$ equals combination of the actuator dynamics and the variator slip dynamics. Sensitivity of the controlled system was measured by adding band-limited noise to the controller output signal. By estimating the transfer function of the noise to the plant output signal an estimate for the process sensitivity is obtained. The plant characteristics can be deducted from the process sensitivity function with $G = \frac{S}{1-S}$. The measured sensitivity of the system for the clamping force control is given for 0.5% slip in Figure 6.4. The ratio of the CVT was near overdrive ($r_s \approx 2$). The primary speed was 50 rad/s . The sensitivity for 0.5% slip is low for very low frequencies and has a dip at 1.5 Hz . The process FRF is shown in Figure 6.5. The process at 0.5% has an eigenfrequency of 1.5 Hz . This frequency strongly depends on the inertia on the driveshaft (normally including the vehicle mass). From the first order model a first order response was expected. The measurement results support this. The slope of the process FRF goes to roughly -1 for high frequencies, with a phase shift of -90 degrees.

6.2.3. Controller tuning

Stability is achieved using a proportional feedback. Integral gain is added to increase the gain at low frequencies. Other than the proportional gain no other filters are used. The gains are given by:

Control action	Gain
Proportional	5000 N
Integral	8000 N/s

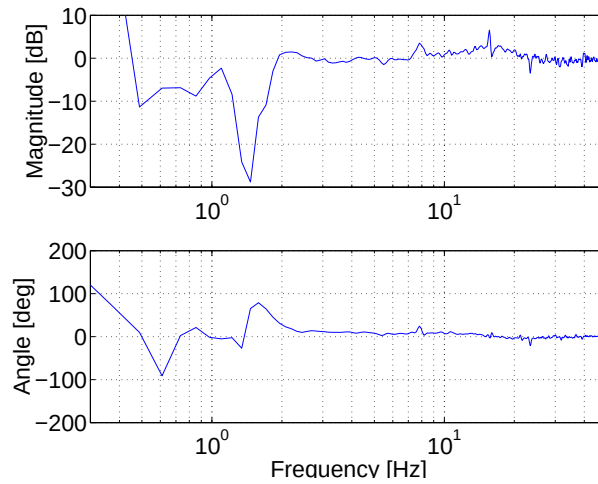


Figure 6.4: Sensitivity of slip controlled system at 0.5% slip

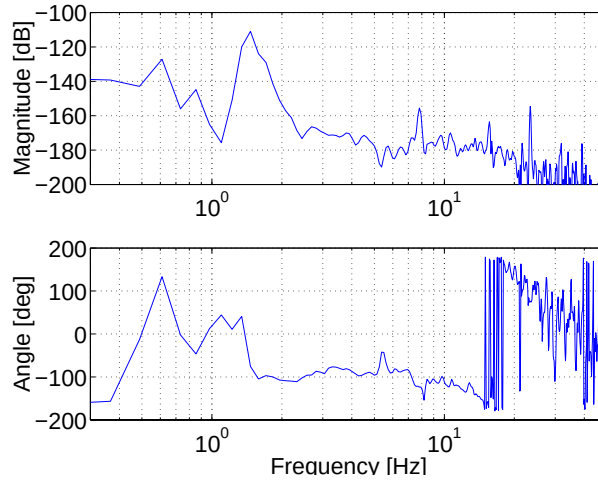


Figure 6.5: Frequency response of slip controlled plant at 0.5% slip

6.3. Experimental results

On the Beltbox testrig a slip controlled variator is tested for robustness in torque peaks from the driveline. In Figure 6.6 the results from this test are shown. It shows the response of the variator slip to several peaks in the torque at the CVT output side. The primary drive shaft of the variator is controlled at a constant speed. The primary and secondary pressures are controlled by the slip control algorithm.

From this figure it can be seen that peaks in the torque also cause peaks in the slip signal. This is unavoidable, because no prior knowledge of the disturbances is available and the controller can only react to the measured slip value. The slip control algorithm reacts to the increase in slip by raising the clamping pressure, thereby reducing slip. The amount of slip that is reached is therefore highly dependent on the bandwidth of the closed loop system. From the figure can be seen that slip increases only slightly and is reduced in a short period of time.

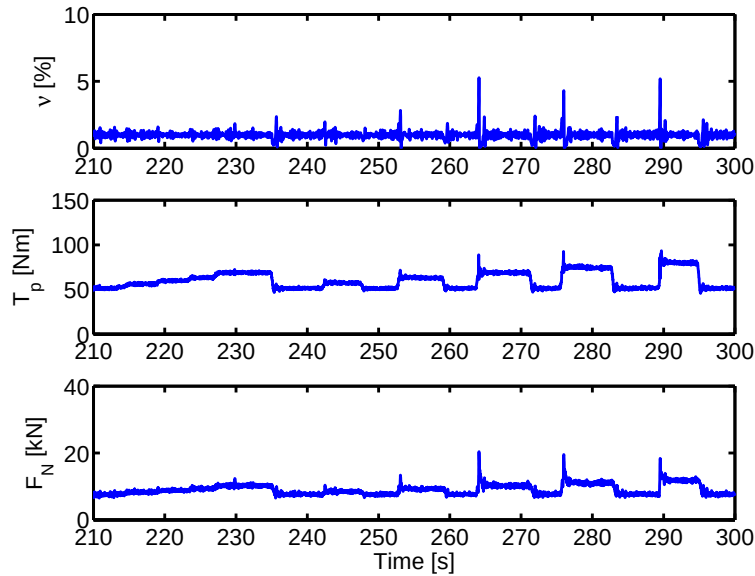


Figure 6.6: The slip and clamping force response of the variator to torque steps and peaks. The upper graph shows the slip as a function of time, the middle graph shows the input torque as a function of time and the lower graph shows the clamping force as a function of time.

In this measurement torque peaks of up to 50 Nm are shown and steps of up to 30 Nm . The resulting slip peaks are between 2% and 6%, which is in the safe area of the F-v diagram. The efficiency of the variator is tested on the beltbox testrig. The results have already been shown in Figure 4.6 in Chapter 4. The variator efficiency can be over 90% for higher torque levels. The pressure in the beltbox testrig is not constrained to a minimum, so maximum efficiency can be reached until very low torques. However, the bandwidth of the hydraulic system depends also on the nominal pressure level. For pressures lower than 3 bar the bandwidth was too low for stable control. Therefore lower torque levels will cause lower transmission efficiency.

6.4. Conclusions and recommendations first implementation

The first implementation of slip control is fairly rudimentary. However, it shows that slip control in a pushbelt type variator is feasible. The experimental results show that the variator can be controlled in a stable way in an operating point in the part of the traction curve that is decreasing. Furthermore it was shown that a slip controlled variator can be robust for disturbances in the form of torque peaks and steps.

In this first implementation it was found that the response of the hydraulic system at low pressure was slower than its response at high pressure. This was a limitation, because the system can no longer react to torque peaks in time to prevent high values of slip. Therefore it is recommended that a minimum pressure is maintained to avoid this problem.

The response of the slip in the variator to torque peaks and to changes in clamping force varies

6.4 Conclusions and recommendations first implementation

with the ratio and the rotational speed. Performance of the slip control system could be improved by using gain scheduling to make use of these variations. Load disturbances did not cause very large slip peaks and slip was regulated both in the macro-slip area and the micro-slip area. The system was tested with torque peaks of up to 50 Nm and steps of up to 30 Nm primary torque.

Gain scheduled PI control of slip in a CVT

After the successful implementation of slip control on the beltbox testrig, slip control on a full CVT transmission is attempted [9], [64], [65]. The transmission that is used in the CVT transmission testrig is a CK2 transmission from Jatco. This transmission is different with respect to the beltbox. The transmission contains several components that the beltbox does not: a torque converter, a DNR set, intermediate shaft and the differential gear. The main differences between the variators of these two are the hydraulic system, the size of the pulleys and the speed and position measurement system.

Implementation of slip control in a production model CVT is more difficult than its implementation on the beltbox testrig, mainly because of the slower hydraulic system response. The slower response severely limits the achievable bandwidth of slip control system. Also there is a minimum hydraulic line pressure. The ratio control system of the factory supplied transmission control unit is used and the ratio control itself will not be taken into account.

In this chapter first the details of the transmission testrig will be explained. Then the implementation of the control system will be reviewed. The approach from Panagopoulos [60] and Solyom [75], which they used for ABS control design, is used for controller synthesis. This method ensures stability of the closed loop system for a system with a cone bounded uncertainty, while maximizing disturbance rejection. The experimental results will be shown and finally, some conclusions and recommendations will be given for this chapter.

7.1. Transmission testrig

The reference transmission is tested on a testbed with the complete driveline of a Nissan Primera. The layout of this testbed is shown in Figure 7.1. This testbed is fitted with a 2.0 litre IC engine

from a Nissan Primera, the CK2 transmission, the driveshafts, a flywheel and both a hydraulic disc brake and an eddy current brake.

With the flywheel, the vehicle mass is simulated and with the eddy current brake the road-load or other loads can be simulated. The disc brake is used to simulate emergency braking.

Limitations of this testrig are the limited bandwidth of the braking system. Also because of the flywheel fixed to the axis no wheelslip situations can be simulated.

The testrig is equipped with a torque sensor between the engine and the transmission. This sensor uses strain gages for measurement of torsion. The signals are transferred using a wireless connection. The same system is used for the torque measurement on the driveshaft.

Engine speed, primary speed, secondary speed and driveshaft speed are measured using the built in sensors of the engine and the transmission respectively. These sensors use hall-effect sensors to measure the rotational speeds.

Furthermore, the pulley position on the primary side is measured using a Linear Variable Differential Transformer (LVDT), a capacitive linear displacement sensor.

A dSpace[®] system is used to control the CK2 transmission, the engine throttle and the brakes. Also the original Transmission Control Module (TCM) can be used for comparison.

The 4 cylinder 2.0litre IC engine has a maximum torque of 181 Nm at 5000 rpm and can deliver up to 104 kW of power at 6000 rpm . The maximum torque curve is shown in Figure 7.2.

The efficiency-map of the engine is shown in Figure 7.2. With this efficiency-map an optimal operation line (OOL) can be calculated. The OOL is also shown in this figure. The OOL can be used as a reference for the engine speed for a given output power level. With the optimal engine speed and output speed an optimal transmission ratio can be calculated.

7.2. Control implementation

On the above described transmission testrig, slip control is implemented. First, a linearized dynamic model for slip is used to analyse the system. Then, a controller is designed using a robust PI-controller synthesis method.

7.2.1. Slip model

The torque generated on both shafts of the variator can be conveniently described on the basis of Equation (3.10). Note that by using this description, torque losses are neglected. Although this limits model accuracy, the effect of torque loss is assumed small enough and not significant for the description of the variator dynamics. It is also assumed that speed ratio changes due to the axial motion of the variator sheaves are much smaller than those associated with slip. This assumption may impose limitations on the control strategy derived below, for those cases where fast ratio changes occur. This assumption allows the contribution of $\dot{r}_g$ to $\dot{\nu}$ to be neglected. The slip dynamics from Section 3.2 can now be simplified. The simplified model is derived using Equations (3.3)

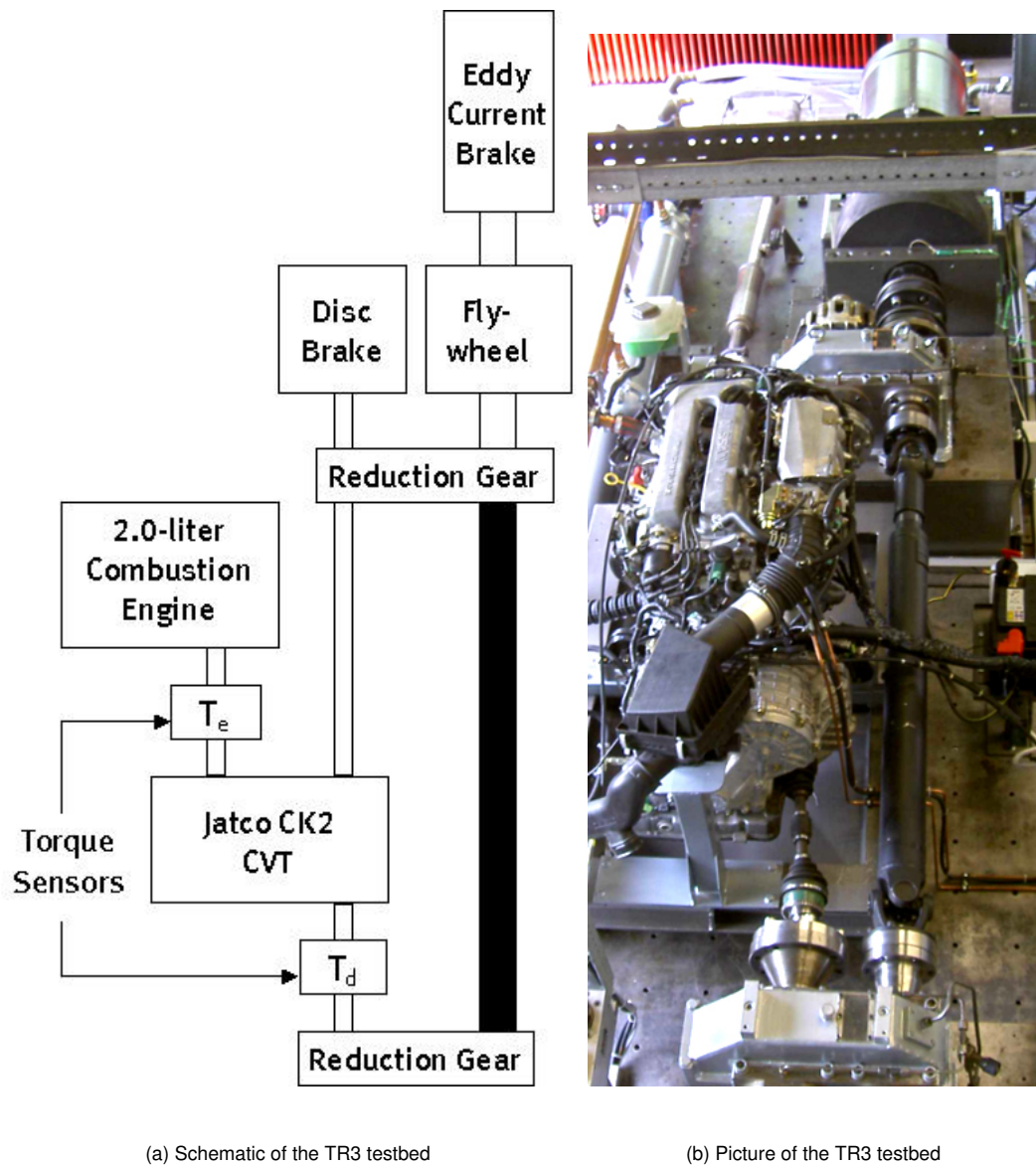


Figure 7.1: Layout of the TR3 testbed

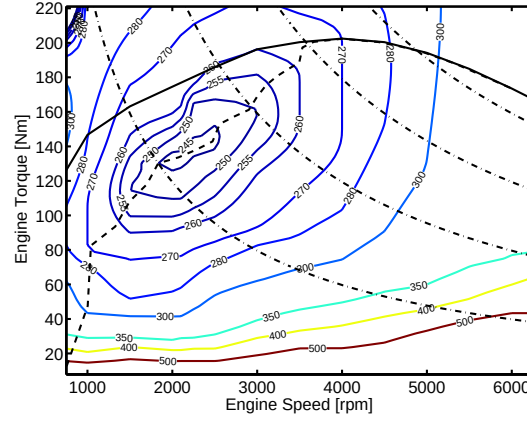


Figure 7.2: Efficiency map of the 2.0litre IC engine mounted on the TR3 testbed, also the OOL (dashed line) and the maximum torque curve (solid line) are shown.

and (2.6), resulting in:

$$\dot{\nu} = -\frac{\dot{r}_s}{r_g} \quad (7.1)$$

In this equation the rate of change of the speed ratio in terms of the system states is given by:

$$\dot{r}_s = \frac{\dot{\omega}_s \omega_p - \omega_s \dot{\omega}_p}{\omega_p^2}. \quad (7.2)$$

The dynamics of the variator as shown in Figure 3.7 were derived Chapter 3. If the ratio changing of the variator is considered to be much smaller than the dynamics of slip, then the model derived in Chapter 3 can be simplified. Substituting Equations (3.10), (7.2), (3.8) and (3.9) in Equation (7.1) leads to the following description of the slip dynamics:

$$\dot{\nu} = \frac{1}{\omega_p} \left(-\frac{2F_s R_s \mu(\nu)}{\cos(\beta) J_s r_g} + \frac{T_d}{J_s r_g} \right) + \frac{(1-\nu)}{\omega_p} \left(-\frac{2F_s R_s r_g \mu(\nu)}{\cos(\beta) J_e} + \frac{T_e}{J_e} \right) \quad (7.3)$$

This equation is non-linear in ν . By linearizing this equation at a certain operating point a linear state space representation is derived [44].

Linearized model

Like in Section 3.3, Equation (3.40), the effective friction coefficient is taken piecewise linear. Defining the state space as $x = [\nu]$ and $u = [F_s]$ and $w = [T_e, T_d]^T$ the system can be linearized around a certain operating point $x_0 = \nu_0$, resulting in the linear system:

$$\delta \dot{x} = \mathbf{A}_0 \delta x + \mathbf{B}_0 \delta u + \mathbf{L}_0 \delta w \quad (7.4)$$

where $\delta x = x - x_0$, $\delta u = u - u_0$ and $\delta w = w - w_0$. The linearized matrices A, B and L can now be derived (assuming $\nu_0 \ll 1$ and neglecting higher order terms):

$$\mathbf{A}_0 = \frac{1}{\omega_{p0}} \left[\frac{-2R_{s0} F_0 M_{\nu 0}}{J_{t0} \cos \beta} \right] \text{ with } J_{t0} = \left(\frac{r_{g0}}{J_e} + \frac{1}{J_s r_{g0}} \right)^{-1} \quad (7.5)$$

$$\mathbf{B}_0 = \frac{1}{\omega_{p0}} \left[\frac{-2R_{s0} \mu(\nu_0)}{J_{t0} \cos \beta} \right] \quad (7.6)$$

$$\mathbf{L}_0 = \frac{1}{\omega_{p0}} \begin{bmatrix} \frac{1}{J_e} \\ \frac{1}{J_s r_{g0}} \end{bmatrix}^T \quad (7.7)$$

The derived linearized system will be used for controller design. The system matrix $\mathbf{A}$ indicates that stability requires M_ν to be positive. This is only the case in the micro-slip region, which is the main reason for its common use in production CVT's. A control action is needed in order to stabilize the system in the macro-slip region.

The model has 3 inputs, but only the clamping force F_s can be controlled on implementation. In a vehicle application, the input torque T_e is controlled by the driver via the throttle pedal and the output torque T_d is determined by road conditions. Therefore they must be regarded as disturbances acting on the system. If an electronic throttle controller is used, then this would possibly eliminate the disturbance from the engine torque. This is not investigated further in this thesis.

System analysis

From the linearized system can be seen that the dynamics of slip depend on:

- the slope of the traction curve M_ν ,
- the input shaft speed ω_p ,
- the transmission ratio r_g ,
- the running radius R_s and
- the clamping force F_s .

From the system matrix $\mathbf{A}_0$ can be seen that the system has a stable pole when the slope of the traction coefficient M_ν is positive, which will move to the RHP for negative values of M_ν . The other values in the system matrix are always positive. This means that the controller should stabilize all possible negative values of M_ν .

The dynamics of slip are scaled by the input shaft speed as can be seen from Equation (7.3). A controller gain proportional with the input shaft speed would simplify the control design problem, because it eliminates ω_p from the equation. A gain scheduling approach with respect to ω_p can therefore be useful.

The dynamics further depend on the transmission ratio r_g and on R_s . R_s is in turn a function of r_g . Matrix $\mathbf{B}_0$ is negative for positive values of slip. This means that an increase in clamping force leads to a decrease in slip, which is intuitive. This can be seen as a phase shift of 180° in the bode plots.

The proposed variator control scheme is shown in Figure 7.3. The slip control system consists of a controller C , the actuator dynamics G_2 and the slip dynamics G_1 . In Figure 7.4 several (open-loop) bode plots are shown for different operating points of the linearized Plant. The first plots show the difference between the response of the system in micro- and macro-slip. In the left

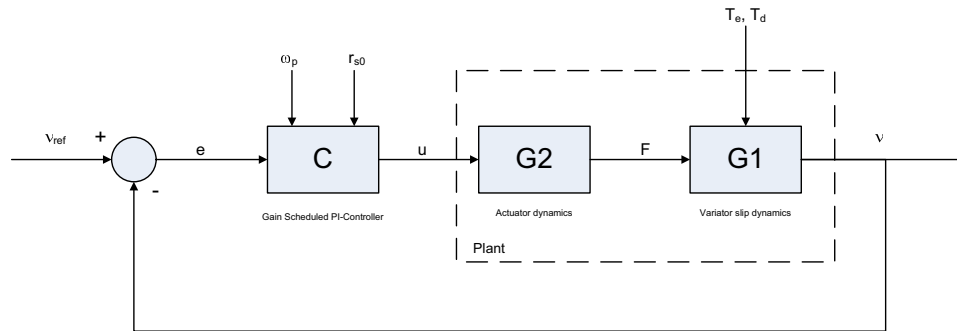


Figure 7.3: Proposed gain scheduling PI slip controller

column the responses for the slip dynamics are shown and in the right column the responses for the combined response of the slip dynamics and the actuator dynamics are shown. The second row shows the difference between Low and Overdrive ratios and the influence of the input speed ω_p on the response in macro-slip.

In Figure 7.4(a) the difference between the frequency response of the variator with respect to slip is shown for both micro- and macroslip. Contrary to microslip, macroslip is unstable, which can be seen from the phase of macroslip, which is 180° at low frequencies.

The transmission ratio of the variator changes the sensitivity of the system with respect to higher frequencies, as can be seen in Figure 7.4(c). However, the response of the system including actuation dynamics is rather similar. This is due to the lower cut-off frequency of the actuation system.

The input shaft speed ω_p changes the high frequency gain of the system as can be seen from Figure 7.4(d). This is caused by the larger speed difference that is needed to cause the same slip difference at higher speed.

Controller design

With the linearized model of the slip dynamics as described in Section 7.2 and the estimated transfer function of the actuation system (Figure 5.2) a slip controller can be designed. The variables that influence the slip dynamics have been discussed in the previous section. There is a large difference in the system response between the micro- and macro-slip region. For slip control design, attention is mainly focused on the macro-slip region. In this region, ratio and primary speed have the largest influence on the dynamics. To maximize performance in all operating points gain scheduling is used. The gain scheduling parameters are slip, ratio and input shaft speed.

A gain scheduled controller is designed by linearizing the slip dynamics in a number of operating points and by calculating the controller parameters for each operating point. As mentioned earlier, the slip controller requires good load disturbance attenuation and must be robust to deal with model

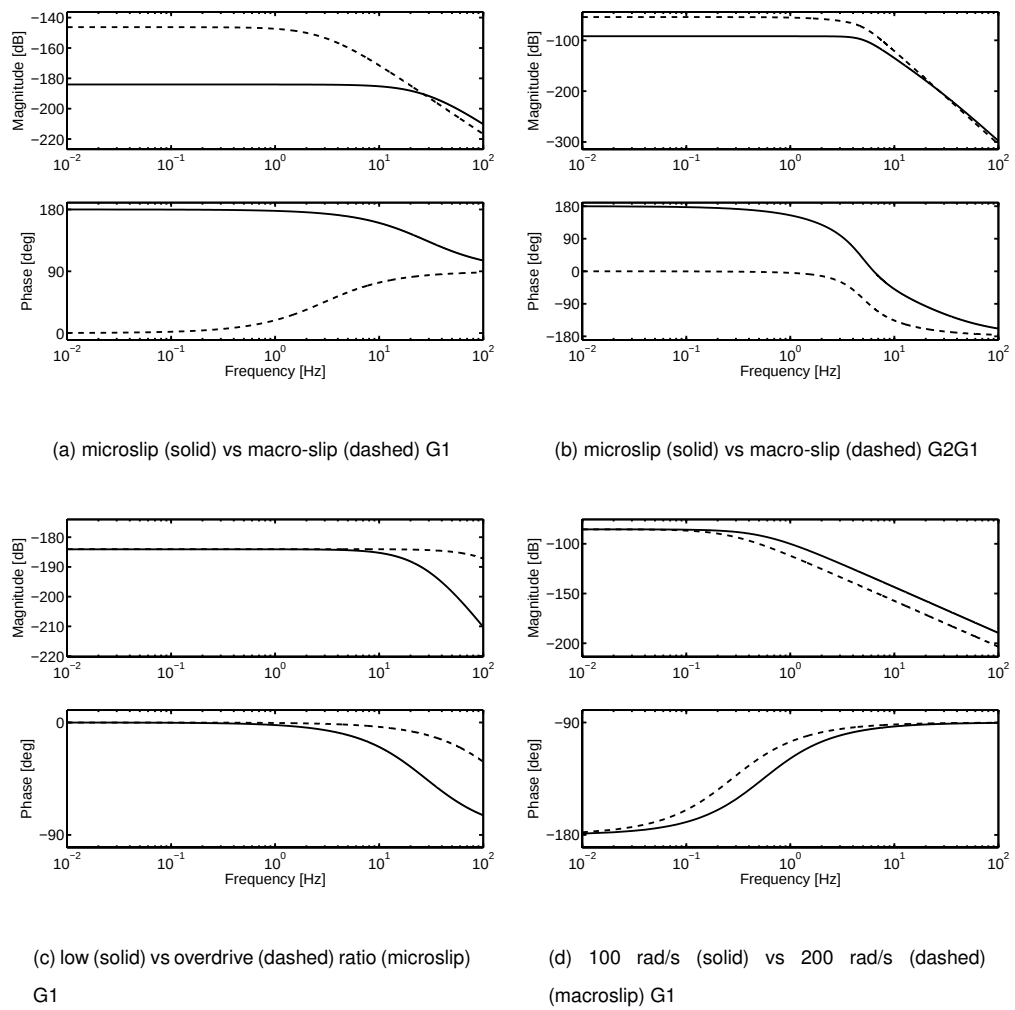


Figure 7.4: Bode plots of the linearized system

uncertainties.

Robust PI-controller synthesis method

To easily design controller parameters for multiple operating points, while meeting both design requirements, a synthesis method for robust PI(D)-controllers with optimal load disturbance response is used [75] [61]. The method is based on a constrained optimization problem that maximizes the integral gain of the PI(D)-controller while making sure that the maximum sensitivity, i.e. the modulus margin, is less than a specified value.

Using the maximum sensitivity as the main design parameter, a trade-off can be made between load disturbance response at low frequencies and robustness with respect to model uncertainties (modulus margin). The resulting controller parameters of this optimization process can be obtained graphically for a PI-controller. In Appendix C this method is explained in more detail.

Using this synthesis method for different ratios in the micro- and macro-slip region, the gain-scheduling scheme presented in Table 7.1 is obtained.

Table 7.1: CONTROLLER PARAMETERS FOR THREE OPERATING POINTS

	Micro-slip region		Macro-slip region	
Ratio	P-gain	I-gain	P-gain (@ 100 rad/s)	I-gain (@ 100 rad/s)
0.43	1.7	30	0.435	1.61
1	1.9	53	0.294	1.087
2.25	3.6	110	0.21	0.772

The differences between the micro- and macro-slip region mentioned earlier, result in very different values for the controller parameters. This is because the system dynamics drastically change at the transition from the micro- to the macro-slip region. The system matrix A in Equation (7.4) almost becomes zero in the macro-slip region. This means that a part of the system dynamics disappears, resulting in important changes in the systems gain.

Another reason is that in the macro-slip region the gain becomes scalable by the primary speed. This can be seen in Equation (7.6), considering the fact that system matrix A is practically zero. Therefore the gains in Table 7.1 for the macro-slip region are scaled by the primary speed (in rad/s) in the controller. Based on ratio, slip, and primary speed, the proper controller parameters are used. Between the operating points shown in table I interpolation will be used. To ensure the stability of the controller between these operating points, several measures were taken. In the micro-slip region load disturbance response is not very important since slip will not cause any damage in this region. However, many model uncertainties are present, because the slip dynamics

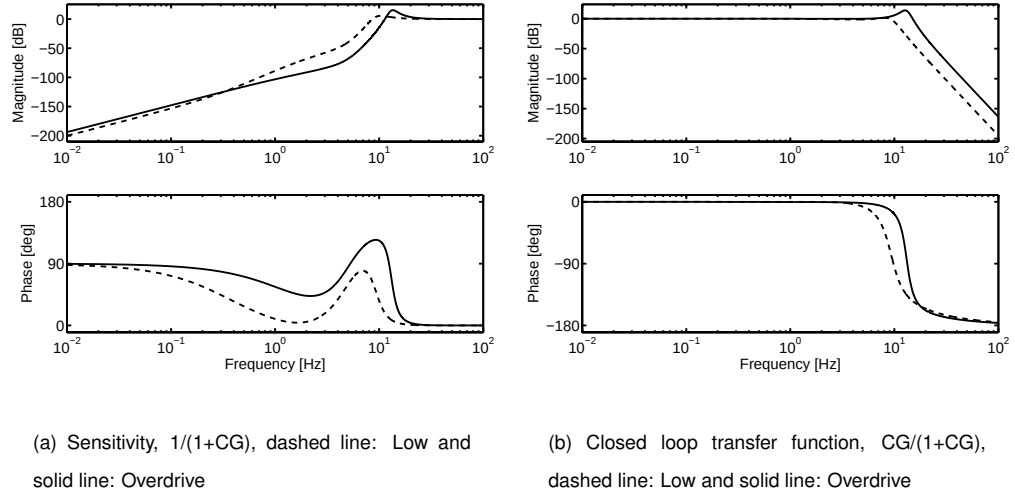


Figure 7.5: Bode plots of the sensitivity and the closed loop response

depend on many variables in this region. Therefore a maximum sensitivity of 4 dB is chosen in the controller synthesis method, which is relatively low. In the macro-slip region a maximum sensitivity of 12 dB is chosen, this is much higher since there are less model uncertainties in this region and this allows a lower sensitivity at low frequencies and hence a good load disturbance response. Additionally the worst-case values of the controller parameters were taken to ensure stability for every operating point.

For macro-slip the bode plots of the closed loop sensitivity $1/(1 + CG)$ (with $G = G_1 G_2$) is shown in Figure 7.5(a). It can be seen that for Low slightly less bandwidth is achieved than for Overdrive. The closed loop transfer function $CG/(1 + CG)$ is shown in Figure 7.5(b).

7.3. Experimental results

The results from the beltbox testrig are satisfactory, but they are not conclusive, because the hydraulic actuation system of the beltbox is not representative for a real automotive CVT. The actuation system in a production CVT will have a lower bandwidth than the actuators used on the beltbox testrig. Therefore the control method is tested in a production CVT.

The CVT used is the CK2 transmission from Jatco. The performance of the testrig however is limited in comparison to the beltbox testrig. The main limitation is the maximum bandwidth of the load disturbance. As can be seen from Figure 7.6, the ramp of the torque peaks is very limited. Although the results are promising, they are not conclusive with respect to the robustness of the CK2 variator.

Experiments were performed at fixed ratios and with a fixed engine speed of 200 rad/s . The eddy-current brake provided a constant torque high enough to reach a slip value at the transition between the micro and macro-slip region. Torque peaks were then introduced by suddenly engaging the disc

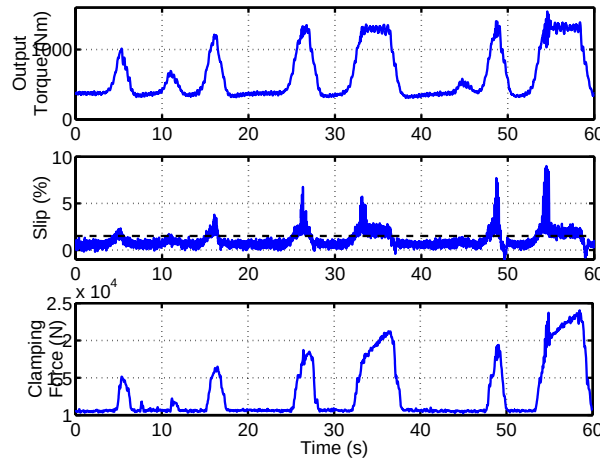


Figure 7.6: Slip controller performance with torque peaks acting on the driveline, measured for ratio 0.43 (low), at an engine speed of 200 rad/s .

brake. Limitations in the disk brake actuation system restricted the rise time of these torque peaks, however.

Figure 7.6 shows the result of one of these measurements. The figure shows that the torque peaks cause belt slip, which was expected at the transition of the micro- and macro region. The slip controller is able to deal with torque peaks of up to 1000 Nm in the drive shaft, although this causes the slip level to peak above 5% for short periods of time. Visual inspection however, showed that the belt was not damaged after such tests. This means that short peaks in slip do not cause belt damage. This was also shown by Van Drogen [89]. Additional tests should be performed to investigate if this is the case for all operating points of the CVT. These tests should also include faster disturbances in torque, as they may occur from road irregularities.

Another important aspect that should be considered is the long-term effect of slip control with respect to belt damage. If necessary, the bandwidth of the controller could be increased to get better load disturbance response, resulting in lower peak values of the belt slip. This can be achieved by improving the gain scheduling scheme with more operating points and using higher maximum sensitivities. If this is not sufficient, an alternative actuation system with a higher bandwidth should be used.

The efficiency when using the TCM is compared to the efficiency when using the slip controller. The efficiency comparison is carried out at fixed ratios and with a constant engine speed of 300 rad/s . The slip value is controlled between 1.5% for ratio 2.25 (overdrive) and 3% for ratio 0.43 (low). At these slip values the maximum efficiency of the CK2 is reached. The engine torque is gradually increased and plotted against the efficiency. Figure 7.7 shows that the efficiency improvement when using the slip controller is quite significant, especially for low engine torques. Since the average engine torque in normal drive cycles is usually relatively low, this is a very promising result. For ratios until $r_g \approx 1.4$, the efficiency improvement is a little lower than for low ratios, but still in the order of

10% to 5%. For approximately $r_g > 1.4$, the efficiency improvement will be lower. When driving in overdrive, there is hardly any improvement. This is caused by the minimum pressure level in the CK2 of 0.66 MPa , which results in a minimum clamping force of almost 10 kN . For normal engine torques hardly any slip will occur in overdrive with this clamping force level. Therefore the benefits of slip control cannot be fully exploited in the current CK2. Lower clamping forces are required for slip control in ratios near overdrive.

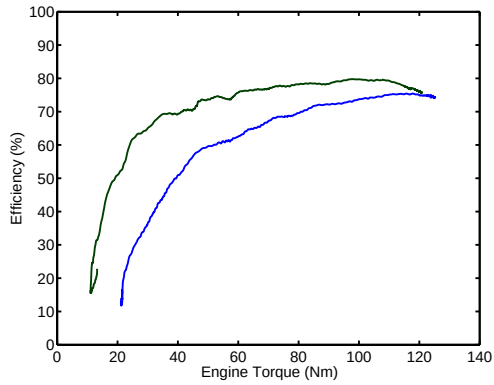
7.4. Conclusions and recommendations gain scheduled PI control

A stable control of slip was established using the gain scheduled PI control algorithm. The disturbance rejection of the closed loop system has been tested using the transmission testrig. The tests proved successful, but the abilities of the testrig were insufficient to simulate true life situations. Therefore it is necessary to do testing on a more sophisticated testrig or in a test vehicle (see Chapter 8).

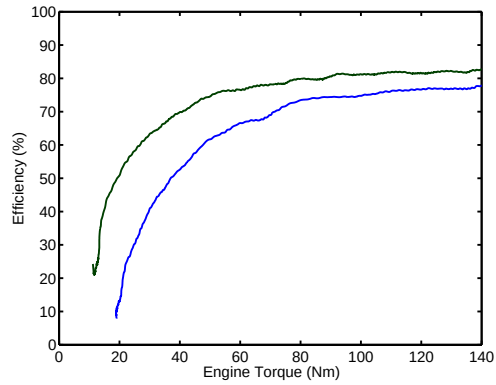
For the transmission testrig satisfactory results were found. It was not possible however to create disturbances with high bandwidth. This is a serious limitation to the testing results. The disturbances of up to 1000 Nm at the driveshaft (around 185 Nm at the secondary pulley) that were created were sufficiently resolved by the controller.

The developed slip controller shows efficiency improvements of the Jatco CK2 of up to 30% at low engine torques and in underdrive ratio. In the higher torque region, less improvement is possible, but still a significant improvement can be made of around 4%. Also for higher ratios the efficiency improvement is smaller in hydraulically actuated CVTs. The hydraulic system has a minimal pressure at which the system response is still satisfactory. Furthermore, the minimal pressure is also needed to drive the auxiliary components of the CVT. This minimal pressure is limiting the improvement for the higher ratios, because in these ratios very low clamping forces are needed. It should be considered to lower the minimal clamping force by reducing the minimal pressure, decreasing the pulley cylinder area or taking other measures to decrease this effect.

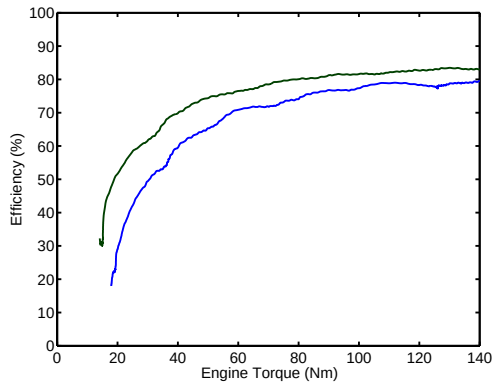
Unfortunately it was not possible to measure the fuel consumption at this testrig, so the total increase in fuel economy could not be evaluated. Even though this property could not be tested, it can be assumed that slip control will have a significant effect compared to the standard control scheme, because the efficiency is increased significantly.



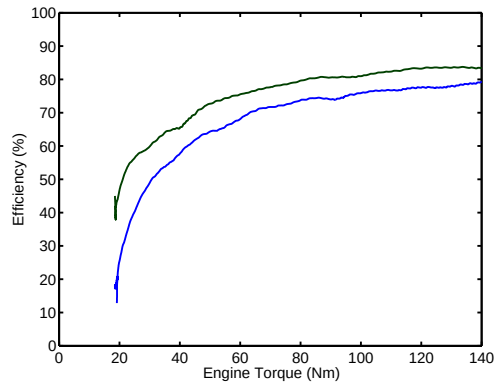
(a) Efficiency as a function of engine torque measured for ratio 0.43, at an engine speed of 300 rad/s



(b) Efficiency as a function of engine torque measured for ratio 0.64, at an engine speed of 300 rad/s



(c) Efficiency as a function of engine torque measured for ratio 0.86, at an engine speed of 300 rad/s



(d) Efficiency as a function of engine torque measured for ratio 1.11, at an engine speed of 300 rad/s

Figure 7.7: Comparison of efficiencies between TCM and slip control. (The lower curves represent the measured efficiency with TCM control and the upper curves represent the efficiency with slip control)

Implementation of slip control in a production vehicle

The slip control system has been evaluated on two test benches. These test benches are designed to simulate the driveline of a car, but have certain limitations. One of the limitations is the limited slope of the torque peaks that can be applied to the variator. In the beltbox testrig, the electric motors have a higher inertia than a similarly powerful internal combustion engine. This limits the realism of the tests performed on this testrig. The transmission testrig with the CK2 transmission uses an internal combustion engine, but has an eddy current brake that is able to simulate road-loads in normal driving situations, but not torque peaks generated by speed-bumps, slip events, ABS braking or other specific events requiring very fast transients of the brake torque.

To evaluate a slip controlled CVT under these circumstances a Nissan Primera is used as a test vehicle [12]. The Nissan Primera driveline is similar to the driveline in the transmission testrig, but with slight differences which will be explained in Section 8.1. Thereafter the results of the implementation in the vehicle will be discussed.

8.1. Nissan Primera

The Nissan Primera (shown in Figure 8.1) differs slightly from the testrig setup. The Nissan is equipped with a 2.3l gasoline engine instead of the 2.0litre engine. Furthermore, the transmission is a Jatco CK-Kai transmission, which is similar to the CK2 in most aspects, but can handle higher torques. This is necessary to handle the power from the larger engine.

8.1.1. Engine

The 4 cilinder 2.3litre IC engine has a maximum torque of 234 Nm at 4000 rpm and can deliver up to 116 kW of power at 5600 rpm . The maximum torque curve is shown in Figure 8.2(b). The torque output of the engine related to the throttle position has to be known. It is important to know



Figure 8.1: Picture of the Nissan Primera test vehicle at the Michelin Challenge Bibendum 2006.

the output torque of the engine with a given throttle position and engine speed, since the torque is not measured between the engine and the transmission.

8.1.2. Transmission control

The transmission control unit of the CK2 transmission uses a Motorola HC12 processor. This unit controls the ratio and line pressure of the variator and also controls the closing of the torque converter lock-up clutch. Input signals include the primary and secondary speed, engine speed, throttle position, ATF temperature and shifter position.

The ratio control strategy is shown in Figure 8.2(a). The aim of the ratio control unit is to minimize fuel economy while maintaining a good drivability. The ideal line for the 2.0litre engine is shown in Figure 7.2. The actual strategy for the CK-Kai transmission with the 2.3litre engine is shown in Figure 8.2(b). If these figures are compared it can be seen that the ratio strategy of the CK-Kai transmission tries to follow the optimal operation line of the IC engine.

8.1.3. Clamping force control

The clamping force control system based on slip discussed in Chapter 7 is used. The transmission used on the test bench setup is very similar to the transmission in the test vehicle. Therefore, no major changes were made to the control system.

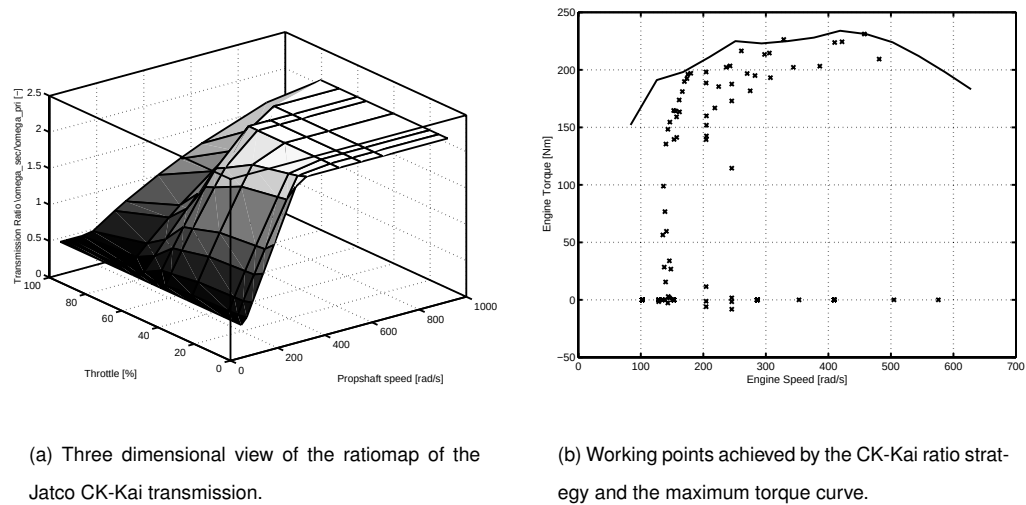


Figure 8.2: CK-Kai characteristics

8.1.4. Variator Properties

The main part of every CVT transmission is the variator. The layout of the transmission is shown in Figure 8.3. The CK-Kai transmission uses a pushbelt type variator from Van Doorne's Transmissie. For slip control it is important to know the relationship between slip and traction and for the optimization of the efficiency also the relationship between slip and efficiency is important.

Traction

The traction curve is measured by increasing the torque to the point that a maximum amount of slip is reached. In this case a 10% limit is used, because with this limit no damage will occur and higher values are not acceptable for a realistic CVT. A measured traction curve is shown in Figure 8.4(a). These results are comparable to the results given in Chapter 3. The maximum traction is in the same range as is the slip in the variator at the maximum traction.

8.2. Variator Robustness

The first responsibility of the clamping force actuation and control system is to allow the required torque transmission through the variator. For a pushbelt type variator this means that the clamping force must always be high enough for the torque load from both the road and the engine. High enough means in this context that the variator transmits the required torque without a large amount of slip, which would not only change the desired speed ratio, but could also potentially damage the variator.

If sudden changes occur in the driveline torque, the variator should be able to deal with these changes by transmitting these torques without causing slip beyond a certain threshold. These

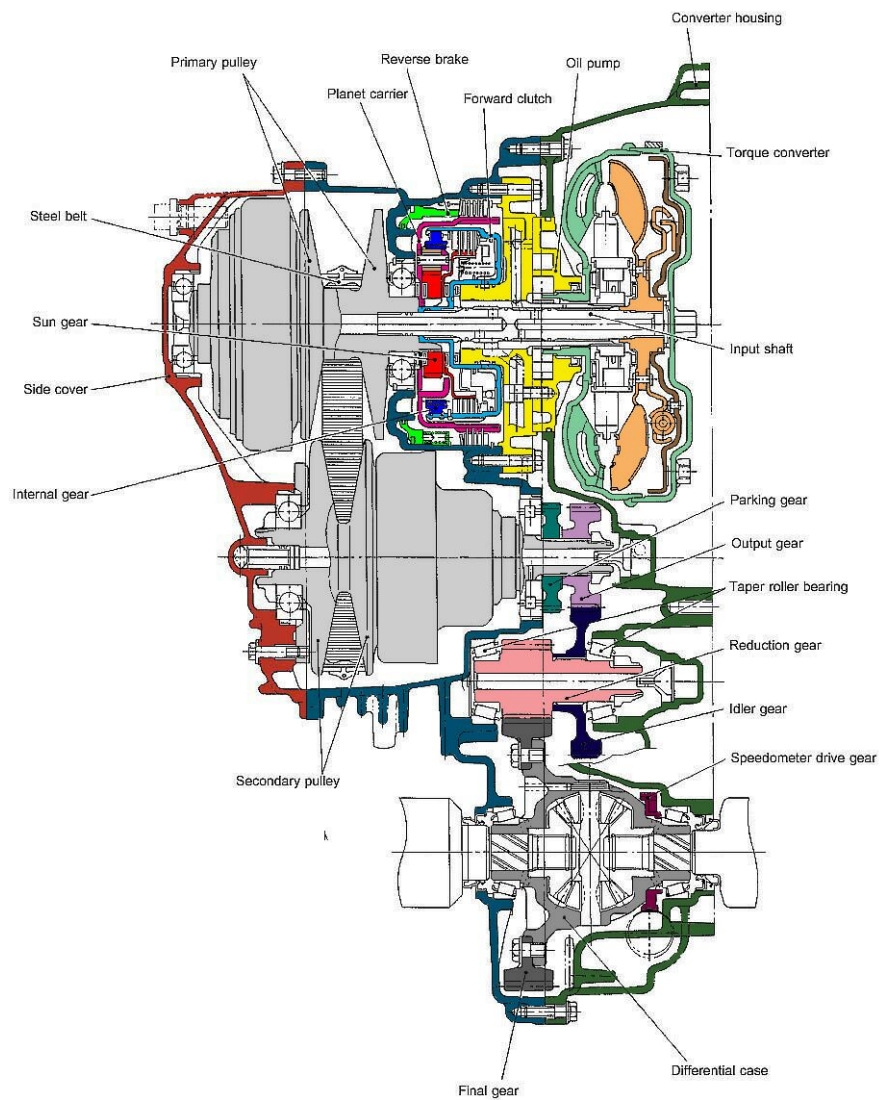
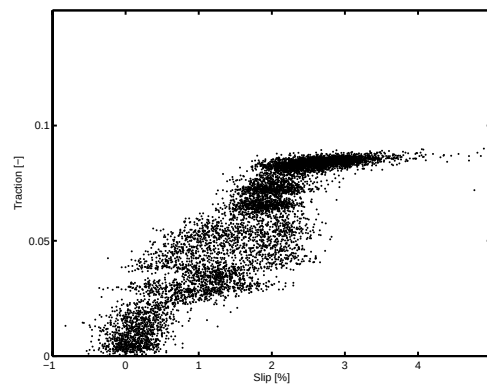
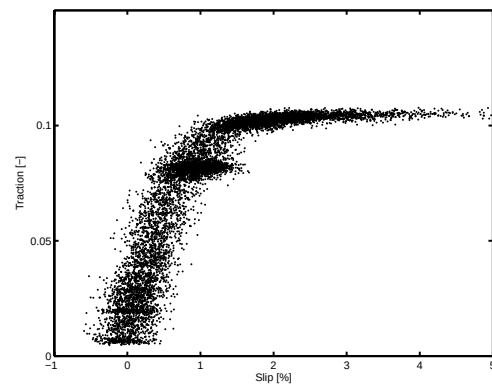


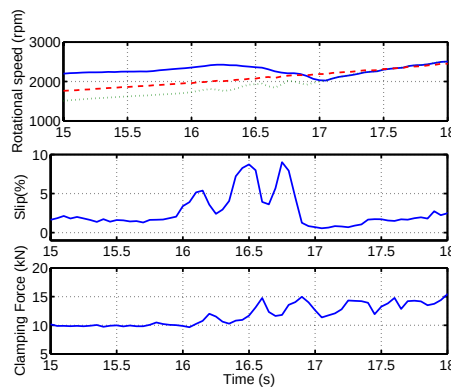
Figure 8.3: Layout of the CK2 transmission



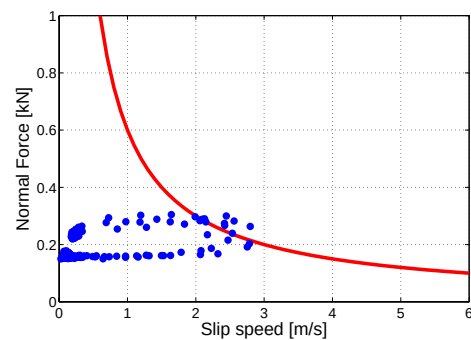
(a) Traction curve in LOW for the CK2 transmission



(b) Traction curve in MED for the CK2 transmission



(c) System response, slip, clamping force and speeds



(d) F-v diagram

Figure 8.4: Response to torque converter lockup

changes can be caused by the engine or by the road conditions. Focus will therefore be on peaks in the engine or driveline torque.

Test track results

The vehicle was tested on the Bosch proving grounds in Boxberg, Germany. Over 50 persons drove the car over a handling course to test the slip controller. These tests showed that under normal driving conditions the slip controller performed satisfactory. When driving in a more aggressive way, the slip controller was not always able to keep the slip level within acceptable levels. This resulted eventually in wear of the belt and pulleys. Figure 8.4 and 8.5 show two major problems that occurred during the test drive. In these figures both the response of the variator slip to the event, the clamping force and the shaft speeds are shown as is the F-v diagram for both incidents.

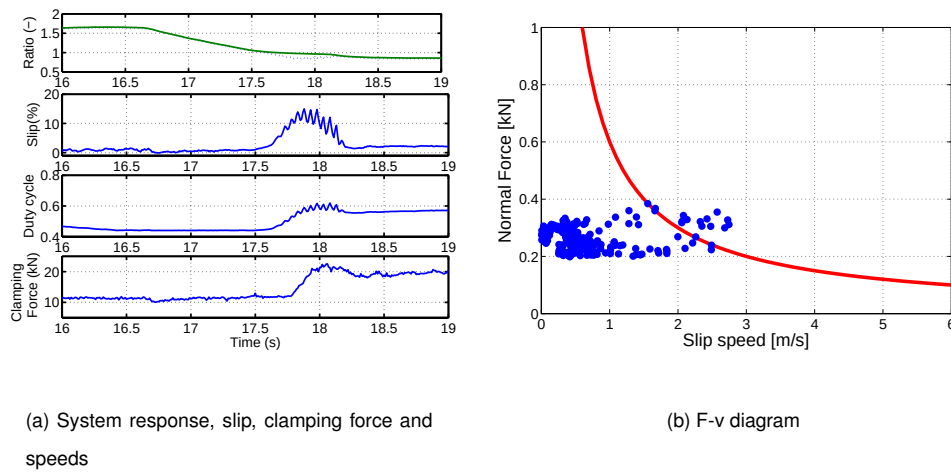


Figure 8.5: Response to a fast downshift

- Closing of the torque converter caused relatively large slip peaks and vibrations in the drive line. Because the slip controller increases the clamping force with increasing slip, these slip peaks did not cause damage to the belt and pulleys. The vibrations in the drive line however cause a very uncomfortable driving experience. Figure 8.4 shows an example of such a measurement. Opening and closing of the torque converter cause an important change in the dynamics of the drive train. This effect should be taken into account for further development, because it can damage the variator as can be seen from the F-v diagram.
- Large slip peaks occurred often with fast variator shifts. This is expected to result from the assumption that the ratio shifting behavior can be seen as quasi-static. With aggressive driving this assumption becomes invalid since large and fast downshifts occur with fast variations of the throttle pedal position. Shifting of the variator can trigger belt slip, as can be seen in Figure 8.5. Shifting dynamics should therefore be taken into account for future slip control development. This is especially interesting due to the observation reported by Ide [37], who found that the shifting dynamics is influenced by variator slip.

8.3. Comfort

Under normal circumstances it should not be possible for the driver to detect whether a CVT is using slip control or not. The comfort of driving should therefore be the same. Comfort is evaluated by the driveline shocks introduced or affected by the slip control system.

If the transmission is slipping there is a dampening effect on driveline vibrations due to the decoupling of the driveline speed and the engine speed. However, an abrupt slip to stick transition in the variator creates high torques to synchronize the primary and secondary shaft speeds. This type of behavior can be felt in the vehicle. These vibrations can be seen in Figure 8.5.

For comfort, slip peaks caused by torque peaks from the driveline are less important than slip caused by engine torque. This is due to the fact that torque peaks from the driveline always cause shocks in the vehicle. However, torque from the engine should be smoothly transferred to the drive-shaft to avoid drive discomfort.

During the tests it would occasionally occur that the drive torque from the engine caused slip in the variator. This could be felt by the driver. This can be caused by a poor engine torque estimation and by a slow response of the clamping force actuation system.

8.4. Conclusions and recommendations Nissan Primera tests

Unfortunately it was not possible to measure the torques and the fuel consumption in this test vehicle. However, the tests performed with this vehicle were very valuable to evaluate the real life performance of a CVT with slip control. The tests on the proving ground showed some limitations to the current implementation of slip control. One of these limitations was the torque induced by the torque converter lock-up clutch when closing. Also shifting proved a more difficult issue. The regulation of slip caused also some driveline vibrations that could be felt by the occupants of the car.

Conclusions and Recommendations

In this chapter conclusions will be drawn, based on this thesis and on the project as a whole. First, the general conclusions will be given, then some recommendations for future research are discussed.

9.1. Conclusions

Slip in a variator has long been looked upon as being very destructive to this variator and was to be avoided at all cost. New findings suggest that a pushbelt type variator can sustain small amounts of slip for prolonged periods of time without failing or being damaged [89]. With this knowledge in mind a method is proposed in this thesis to control slip in a variator.

In Chapter 2 several models have been described for the forces acting in the variator. With these models a fairly accurate prediction can be made for the stationary clamping force ratio in the variator. The added complexity of the pushbelt model did not give better results compared to the continuous belt model. For controller design and simulation this model will suffice.

For transient situations the CMM model gave the most accurate result compared to experimental results. Especially for low transmission ratios the results for the CMM model agreed more with the experimental results than the other models. In other operating points the Ide model will give similar results. The Shafai model does not take into account the dependency of the ratio changing behavior with the input shaft speed as can be seen from the experiments.

In Chapter 3 the traction of the variator was studied. A model of the traction in the variator was given and the result from this model was compared to measurements. Good agreement was found for a belt with increased play.

Measurements of the traction coefficient showed that the traction coefficient varies with the relative slip, ratio, speed and clamping force.

Chapter 9 Conclusions and Recommendations

A dynamic model of input speed, slip and transmission ratio in the variator was described in Section 3.2. From this model can be derived that the stability of slip in a variator mainly depends on the slope of the traction curve. If this slope is positive, then slip will be stable. If the slope of the traction curve is negative, then slip will be unstable.

In Section 3.3 several estimation methods of the geometric ratio and slip were compared. The position measurement of the pulley or the belt are accurate enough for estimation of slip and ratio. The other methods do not work in all required operating points, are not conclusive, or require additional research.

In Chapter 4 a model was described for the efficiency of the variator. Measurements were used to identify the torque loss in the variator. Using this torque loss an accurate prediction can be made for the efficiency of the variator.

From the measurements of the efficiency for different geometric ratios can be seen that the efficiency of the variator near $r_g = 1$ is higher than the efficiency in other ratios. This is caused by the absence of slip between bands and blocks in the belt.

The measurements of the efficiency of the variator showed that the optimal operating point of the variator is at a slip level higher than zero. If the efficiency curve is compared to the traction curve, than it can be seen that the optimal efficiency point is reached before the maximum of the traction coefficient is reached if slip is increased. However, if the actuation losses of a hydraulic system is taken into account the slip level of the optimal efficiency is very near the slip level of the maximum traction coefficient. This means that if the variator is operated at its maximum efficiency, the safety margin is almost zero.

In Chapter 5 the clamping force control problem is discussed. In the case of classic clamping force control methods only open-loop control is used. If slip control is used, the feedback loop can be used to stabilize the variator in operating points that would be unstable with the open-loop control. The simulation results showed that open-loop clamping force control is vulnerable for torque peaks caused by extreme events except for very high safety-factors. High safety-factors will reduce the efficiency of the variator as described in Chapter 4.

In Chapter 6 the results were shown of the first implementation of slip control in a pushbelt type variator. The test results showed promising results. The setup used however a far more sophisticated actuation system than the actuation systems used in production CVTs. Therefore an implementation is done on the CK2 transmission described in Chapter 7. In this chapter a slip control algorithm is tested in a production CVT. Slip control was implemented and tested in a test facility with an internal combustion engine simulating a complete driveline of a Nissan Primera. Although the tests were successful, they proved not completely conclusive, because of the limitations of the testrig.

Therefore slip control was implemented in a test vehicle, a Nissan Primera. The results from these tests were shown in Chapter 8. Several complications were found with the implementation in the Nissan Primera.

Important factors for successful implementation of slip control are:

- the bandwidth of the actuation system,
- quality of the measured slip signal,
- quality of the estimation of the road-load and the engine torque.

The improvements that can be reached using slip control in a CVT:

- 50% less clamping force compared to the CK2 transmission,
- Over 30% improvement in efficiency at low ratios and low torques,
- Around 5% improvement in efficiency at higher ratios and higher torques.

9.2. Recommendations

For further research a few suggestions are given to extend the knowledge gained in this project.

- The ratio changing behavior of the variator, especially under macro-slip conditions is not well known. This should be investigated further, because a slip controlled variator can have very different ratio changing dynamics when the reference value of slip is in the macro-slip area. The closed loop system might become unstable, because the ratio changing dynamics will change.
- To be able to use beltspeed for the estimation of slip or ratio, it is necessary to know on which pulley slip will occur. Measurements of the beltspeed combined with the pulley position can give this insight. This knowledge can also be helpful for improving the model of beltsip.
- To be able to estimate the geometric ratio or slip without adding extra sensors to the variator would make slip control more interesting for commercialization. Some methods are proposed in this thesis, but more research on this topic is necessary for useful results.
- To improve the efficiency of the pushbelt type variator a pushbelt should be developed without a difference in running radius between the bands and blocks, eliminating the friction in

the belt.

- The durability of the variator under slip control was not investigated. It would be of interest to know the consequences of slip control with respect to durability.
- The performance of the system could unfortunately not be tested for several drive cycles like the NEDC. This was due to the limitations of the test setups. It will be valuable to perform this test in future research.
- In Appendix A a decoupling and linearizing feedback is proposed to include the ratio changing behavior of the variator in the controller. With this method the influences of the nonlinearities and the coupling between ratio and clamping force control might be reduced significantly. This could potentially make the design of a controller for either the ratio or clamping force easier.

Slip control using Linearizing and decoupling feedback

For future implementations of slip control in CVTs it is suggested that attention is paid to linearizing and decoupling feedback. The phenomena encountered in the previous chapter can be avoided by using the information from the driver and the engine to predict the output torque and power of the engine. Furthermore, by choosing an appropriate feedback, the influences of the nonlinearities of the CVT can be minimized. This can be done for both the transmission ratio control loop as the slip control loop. The mutual influences of the ratio and slip control systems are reduced by the decoupling feedback.

First, the linearizing and decoupling feedback will be explained and then the implementation of this method in the controller design is described. Since this method is not implemented yet, no experimental results can be shown. However, simulations are shown to give an indication of its performance.

A.1. Linearizing and decoupling feedback

Linearizing feedback is the idea of using feedback to linearize a nonlinear system, i.e. the closed loop system behaves in a similar way to a linear system. Decoupling feedback is a method to use feedback control to minimize or remove the influences of one input signal on all but one output signal. Effectively a single multiple input, multiple output system is transformed into multiple single input, single output systems.

A.1.1. Linearizing feedback

Lets consider a nonlinear system in state space description of the form: $\dot{x} = f(x) + g(x)u$. If it is assumed that the system is fully controllable, then applying an appropriate control law of the form:

$$u(x, w) = \alpha_c(x) + \beta_c(x)w \quad (\text{A.1})$$

makes the system linear with respect to the new control input w , so that linear control methods can be used to regulate this system [39].

A.1.2. Decoupling feedback

Decoupling state feedback was introduced by Gilbert [32]. A system is considered with two inputs and two states. If this system is linear, then the following state space description can be obtained:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad (\text{A.2})$$

If the diagonal terms of a_{12} and a_{21} and/or b_{12} and b_{21} are not zero then the input u_1 has influence on both x_1 and x_2 . If an appropriate control law of the form:

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = - \begin{bmatrix} \frac{a_{12}}{b_{11}} x_2 \\ \frac{a_{21}}{b_{22}} x_1 \end{bmatrix} - \begin{bmatrix} \frac{b_{12}}{b_{11}} u_2 \\ \frac{b_{21}}{b_{22}} u_1 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad (\text{A.3})$$

is applied, so that the influence of w_1 on x_2 is zero, then effectively a decoupled system is obtained of the form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & 0 \\ 0 & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_{11} & 0 \\ 0 & b_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad (\text{A.4})$$

This method can in certain situations be extended to nonlinear applications. Modeling uncertainties and measurement errors reduce the decoupling effect.

A.2. Controller design

In the previous chapters the control design was based on the assumption that the ratio changing is quasi-static. If the ratio changing is not quasi-static, the control problem becomes more complicated, because of the interactions between the ratio dynamics and the slip dynamics. If a decoupling and linearizing feedback can be designed, the control design can be simplified. The ratio control has no longer an influence on the slip dynamics and vice versa. Linear control design techniques can be used.

If we assume that F_s is smaller than F_p , ω_p and ω_s are measured, an estimate exists for the geometric ratio, $\hat{r}_g$, and that estimates $\hat{T}_e$, $\hat{T}_d$ are available for the disturbances, control inputs F_p and F_s can be chosen such that for inputs u_1 and u_2 an almost linear system with respect to r_g and ν , the controlled variables, is found.

Therefore a control law should be designed such that:

1. the slip ν , with unknown initial error, converges as fast as possible to a prescribed (constant) value ν_d and
2. the ratio r_g converges as fast as possible to the prescribed value r_d .

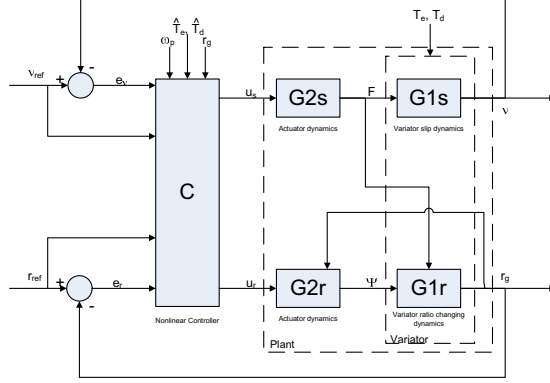


Figure A.1: Block diagram of the variator control scheme

To meet the second demand, we choose:

$$\ln\left(\frac{F_p}{\Psi F_s}\right) = \frac{1}{\omega_p \kappa(r_g)} (\dot{r}_d + u_1) \quad (\text{A.5})$$

with a new input u_1 . With this choice follows:

$$\dot{r}_g - \dot{r}_d = u_1 \quad (\text{A.6})$$

To meet the first demand we choose:

$$\frac{2\mu(\nu, r_g)R_s(r_g)}{\cos\beta} \frac{J_e + (1-\nu)r_g^2 J_s}{J_e J_s} F_s = (1-\nu)\omega_p(\dot{r}_d + u_1) + (1-\nu)\frac{r_g}{J_e}\hat{T}_e + \frac{1}{J_s}\hat{T}_d - u_2 r_g \omega_p \quad (\text{A.7})$$

With this new input u_2 follows:

$$\omega_p r_g \dot{\nu} = \omega_p r_g u_2 + (1-\nu)\frac{r_g}{J_e}(T_e - \hat{T}_e) + \frac{1}{J_s}(T_d - \hat{T}_d) \quad (\text{A.8})$$

Reconstructing the variator inputs F_p and F_s from these control inputs gives:

$$F_s = \frac{\cos\beta}{2\mu R_s} \frac{J_e J_s}{J_e + (1-\nu)r_g^2 J_s} \left[(1-\nu)(\omega_p \dot{r}_d + r_g \frac{\hat{T}_e}{J_e}) + \frac{\hat{T}_d}{J_s} + ((1-\nu)u_1 - r_g u_2)\omega_p \right]$$

$$F_p = \Psi F_s e^{\frac{\dot{r}_d + u_1}{\omega_p \kappa(r_g)}}$$

For the situation that $F_p < F_s$ a solution can be found in a similar way.

The first order differential equation with $x = [\omega_p, \nu, r_g]^T$ is now given by:

$$\dot{x} = \begin{bmatrix} \frac{T_e}{J_e} - \frac{J_s}{J_e + (1-\nu)r_g^2 J_s} \left[(1-\nu)(\omega_p \dot{r}_d + r_g \frac{\hat{T}_e}{J_e}) + \frac{\hat{T}_d}{J_s} + ((1-\nu)u_1 - r_g u_2)\omega_p \right] \\ u_2 + (1-\nu)\frac{1}{J_e \omega_p}(T_e - \hat{T}_e) + \frac{1}{\omega_p r_g J_s}(T_d - \hat{T}_d) \\ u_1 + \dot{r}_d \end{bmatrix}$$

As output is chosen:

$$y = \begin{bmatrix} \omega_p \\ \nu \\ r_g \end{bmatrix} \quad (\text{A.9})$$

Both inputs are independent and have a linear relation to the controlled variable. Using linear control techniques a suitable controller can be designed for u_1 and u_2 .

A.3. Conclusions and recommendations linearizing and decoupling feedback

In this chapter a linearizing and decoupling feedback has been designed. This feedback method can be used to minimize the influences of the ratio control on the slip control system and vice versa. It should improve the effectiveness of the slip control system. However, further investigation is necessary into the feasibility and possibilities of the method. So far, no tests have been done with this control strategy. For the successful implementation of this method the modeling uncertainties should not be too large and the estimations of the disturbances, T_d and T_e , must be reasonably accurate.

Equations

B.1. System matrices for the linearized model

The derivatives D_p of the primary radius R_p and D_s of the secondary radius R_s with respect to the ratio r_g can be determined from the geometric relations in Chapter 2. If the elongation of the belt is neglected it follows that:

$$D_p(r_g) = \frac{d}{dr_g}(R_p(r_g)) = \frac{\pi - 2\phi(r_g)}{(\pi - 2\phi(r_g)) + (\pi + 2\phi(r_g))r_g} \cdot \frac{R_p(r_g)}{r_g} \quad (\text{B.1})$$

$$D_s(r_g) = \frac{d}{dr_g}(R_s(r_g)) = \frac{\pi + 2\phi(r_g)}{(\pi - 2\phi(r_g)) + (\pi + 2\phi(r_g))r_g} \cdot R_s(r_g) \quad (\text{B.2})$$

With respect to the experimentally determined functions $\mu = \mu(\nu, r_g)$, $\Psi = \Psi(\nu, r_g)$ and $\kappa = \kappa(\nu, r_g)$ it is assumed that they are continuous and at least once differentiable with continuous partial derivatives, *i.e.*

$$M_\nu(\nu, r_g) = \frac{\partial}{\partial \nu}(\mu(\nu, r_g)); \quad M_r(\nu, r_g) = \frac{\partial}{\partial r_g}(\mu(\nu, r_g)) \quad (\text{B.3})$$

$$P_\nu(\nu, r_g) = \frac{\partial}{\partial \nu}(\Psi(\nu, r_g)); \quad P_r(\nu, r_g) = \frac{\partial}{\partial r_g}(\Psi(\nu, r_g)) \quad (\text{B.4})$$

$$K_\nu(\nu, r_g) = \frac{\partial}{\partial \nu}(\kappa(\nu, r_g)); \quad K_r(\nu, r_g) = \frac{\partial}{\partial r_g}(\kappa(\nu, r_g)) \quad (\text{B.5})$$

With these derivatives the system equations, given by (see Section 3.2:

$$\dot{\underline{x}} = \underline{f}(\underline{x}, \underline{u}) + \underline{L}(\underline{x})\underline{w}, \quad (\text{B.6})$$

can be linearized around a stationary working point, characterized by the state $\underline{x}_0$, the input $\underline{u}_0$ and the disturbance $\underline{w}_0$. In state space the result can be written as:

$$\delta \dot{\underline{x}} = \underline{A}_0 \delta \underline{x} + \underline{B}_0 \delta \underline{u} + \underline{L}_0 \delta \underline{w} \quad (\text{B.7})$$

where the state matrix $\underline{A}_0$ and the input matrix $\underline{B}_0$ follow from:

$$\underline{A}_0 = \underline{A}(\underline{x}_0, \underline{u}_0); \quad \underline{A}(\underline{x}, \underline{u}) = \frac{\partial}{\partial \underline{x}^T}(\underline{f}(\underline{x}, \underline{u})) \quad (\text{B.8})$$

$$\mathbf{B}_0 = \mathbf{B}(\underline{x}_0, \underline{u}_0); \quad \mathbf{B}(\underline{x}, \underline{u}) = \frac{\partial}{\partial \underline{u}^T} (f(\underline{x}, \underline{u})) \quad (\text{B.9})$$

whereas the disturbance matrix $\mathbf{L}_0$ is given by:

$$\mathbf{L}_0 = \mathbf{L}(\underline{x}_0) \quad (\text{B.10})$$

With $x = [\omega_p, \omega_s, r_g]^T$, $u = [F_p, F_s]^T$ and $w = [T_e, T_d]^T$ the matrices are found to be:

$$\mathbf{A}_0 = \frac{2F_0}{\cos \beta} \begin{bmatrix} -\frac{R_{s0}k_{1i}r_{s0}}{J_e \omega_{p0}} & \frac{R_{s0}k_{1i}}{J_e \omega_{p0}} & -\frac{R_{s0}(k_{1i}+k_{2i})}{J_e} \\ \frac{R_{s0}k_{1i}r_{s0}}{J_s \omega_{p0} r_{g0}} & -\frac{R_{s0}k_{1i}}{J_s \omega_{p0} r_{g0}} & -\frac{R_{s0}k_{1i}\omega_{s0}}{J_s \omega_{p0} r_{g0}^2} \\ \frac{\cos \beta \kappa_0 \log(\frac{F_p}{\Psi F_s})}{2F_0} & 0 & 0 \end{bmatrix} \quad (\text{B.11})$$

$$\mathbf{B}_0 = \begin{bmatrix} -\frac{2R_{p0}\mu}{J_e \cos \beta} & 0 \\ 0 & \frac{2R_{s0}\mu}{J_s \cos \beta} \\ \frac{\kappa_0 \omega_{p0}}{F_s} & \frac{\kappa_0 \omega_{p0}}{F_p} \end{bmatrix} \quad (\text{B.12})$$

$$\mathbf{L}_0 = \begin{bmatrix} \frac{1}{J_e} & 0 \\ 0 & -\frac{1}{J_s} \\ 0 & 0 \end{bmatrix} \quad (\text{B.13})$$

B.2. Iterative calculation of the wrapped angle and running radii

For calculating the wrapped angle of the belt on the pulley ϕ must be known. The relation between the running radii and the wrapped angle are implicit. The following algorithm can be used to calculate the value of ϕ using an iterative procedure for a given transmission ratio r_g .

$$L = R_p(\pi + 2\phi) + R_s(\pi - 2\phi) + 2a \cos \phi \quad (\text{B.14})$$

$$R_p - R_s = a \sin \phi \quad (\text{B.15})$$

$$r_g = \frac{R_p}{R_s} \quad (\text{B.16})$$

$$\rho_g = \frac{1 - r_g}{1 + r_g} \quad (\text{B.17})$$

Using Equations B.16 and B.17 we can rewrite Equation (B.14) to:

$$f(\phi) = \left(\frac{\pi}{2} + \rho_g \phi\right) \sin \phi + \rho_g \cos \phi - \rho_g \frac{L}{2a} \quad (\text{B.18})$$

$$f(\phi) = 0 \quad (\text{B.19})$$

Differentiating $f(\phi)$ with respect to ϕ gives:

$$f'(\phi) = \left(\frac{\pi}{2} + \rho_g \phi\right) \cos \phi \quad (\text{B.20})$$

If ϕ_i is a reasonable estimation of the zero of $f(\phi)$, then a new approximation ϕ_{i+1} can be determined by:

$$\phi_{i+1} = \phi_i - \frac{f(\phi_i)}{f'(\phi_i)} \quad (\text{B.21})$$

B.3 Quadratic approximation of the wrapped angle and running radii

If this iteration is stopped when $\left| \frac{f(\phi_i)}{f'(\phi_i)} \right| < \epsilon$, then a solution has been found that has a relative error smaller than ϵ . Using Equations B.14 and B.16 also the values for R_p and R_s can be found.

B.3. Quadratic approximation of the wrapped angle and running radii

Equations (B.14) and (B.16) can be approximated with a quadratic function using the second order Taylor expansion of the cosine and sine functions:

$$\cos \phi \approx 1 - \frac{1}{2}\phi^2 \quad (\text{B.22})$$

$$\sin \phi \approx \phi \quad (\text{B.23})$$

Rewriting Equation (B.14) gives:

$$\begin{aligned} L &= R_p \left(\pi + 2 \frac{R_p - R_s}{a} \right) + R_s \left(\pi - 2 \frac{R_p - R_s}{a} \right) + 2a \left(1 - \frac{1}{2} \left(\frac{R_p - R_s}{a} \right)^2 \right) \\ \phi &= \frac{R_p - R_s}{a} \end{aligned} \quad (\text{B.24})$$

This equation can be solved if the ratio r_g is known or one of the running radii is known. Using Equation (B.24) the wrapped angles can be calculated.

Robust PI Control

Panagopoulos [61], [60] and Solyom [76] proposed a method for PI controller synthesis with certain robustness properties which minimizes the effect of a load disturbance by maximizing the integral gain. The closed-loop stability is guaranteed by two constraints posed on the system. They showed this method for a class of systems with a cone bounded nonlinearity in the control loop.

For optimal load disturbance attenuation, the maximum value of the integral gain is determined from the figure. The proportional gain is then determined graphically.

The PI control synthesis problem can be seen as the following optimization problem:

$$\begin{aligned} & \max && k_i \\ & \text{subject to} && f(k_p, k_i, \omega_1) \geq R_s^2 \quad \forall \omega_1 > 0 \end{aligned} \quad (\text{C.1})$$

$$g(k_p, k_i, \omega_2) \geq 0 \quad \forall \omega_2 > 0 \quad (\text{C.2})$$

$$k_p > 0, k_i > 0 \quad (\text{C.3})$$

Constraints (C.1) and (C.2) can be plotted as ellipses in the k_p - k_i plane. In this way the optimization problem can be visualized using this graphical representation. The graphical representation is shown in Figure C.1.

Constraint (C.1) is the function:

$$f(k_p, k_i, \omega_1) = |C_s + C(i\omega)G(i\omega)|^2 \quad (\text{C.4})$$

By choosing R_s and $C_s = 1$ the controller will have a maximum closed loop sensitivity of $1/R_s$. This is called the sensitivity constraint.

Constraint (C.2) is called the robustness constraint. This constraint guarantees the stability of the closed loop system for all systems within the uncertainty bounds. With the closed loop plant $P = G_1(s)/(1 + C(s)G_1(s)G_2(s))$, where $G_1(s)$ is the linear plant and $G_2(s)$ is the nonlinear feedback, the robustness constraint can be written as:

$$\text{Re} \left\{ \frac{1 + \beta P(i\omega)}{1 + \alpha P(i\omega)} \right\} > 0 \quad (\text{C.5})$$

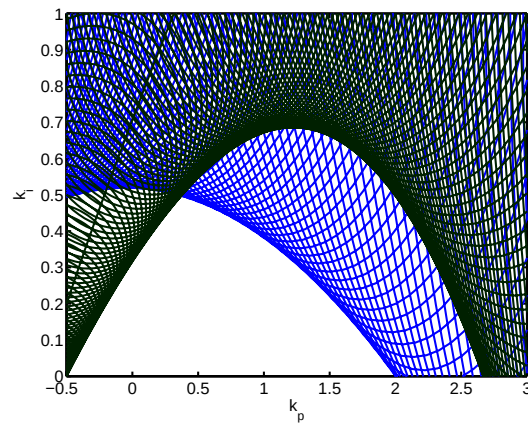


Figure C.1: Combined criteria

In this equation α and β are the uncertainties parameters of the (unknown) nonlinearity.

Constraint (C.3) further guarantees that the controller does not have unstable zeros. For proofs is referred to Solyom [75].

Drive cycles

It is very difficult to compare vehicles in terms of fuel economy and emissions on the basis of real world driving. Field tests are not reproducible and also measured real life driving situations are mostly too dynamic to reproduce in a laboratory. Therefore standardized drive cycles are used. For measurement purposes easy to follow drive cycles are used like the NEDC cycle [19] and for simulation purposes also more complex drive cycles can be used. For comparison reasons it is useful if results can be compared on the basis of standardized cycles. The type of drive cycle has a large influence on the performance of a drivetrain. Therefore several drive cycles are compared and their advantages and disadvantages are discussed.

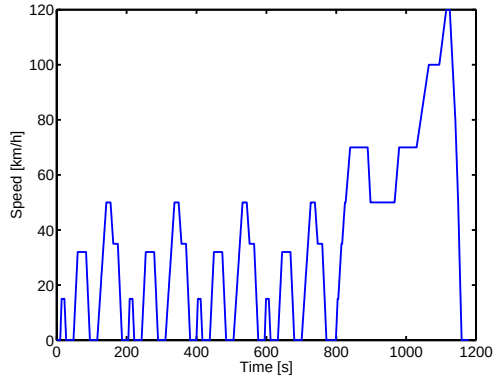
D.1. NEDC

The New European Drive Cycle combines three city cycles with one highway cycle. This cycle is not a very realistic drive cycle, because it contains several constant velocity parts and constant acceleration parts, but it can be reproduced in experiments. Other cycles, for example cycles measured in real traffic, are very difficult or almost impossible to reproduce accurately. Therefore the NEDC drive cycle is ideal for laboratory testing of a car.

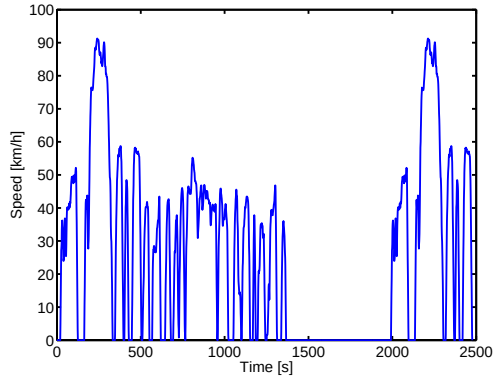
The NEDC cycle is plotted in Figure D.1(a). The required accelerations are plotted in Figure D.1(c). The accelerations are more or less constant, but are realistic for real driving. For a midsize car a histogram of the power needed to run the NEDC cycle is plotted in Figure D.1(e). From these figures can be seen that the power needed for this cycle is relatively low.

Drivetrain concepts that are optimized for partload conditions will perform much better than traditional drivetrain concepts. Furthermore, from Figure D.1(a) can be seen that a lot of the cycle the vehicle is standing still. This is an advantage for concepts that use a start-stop strategy.

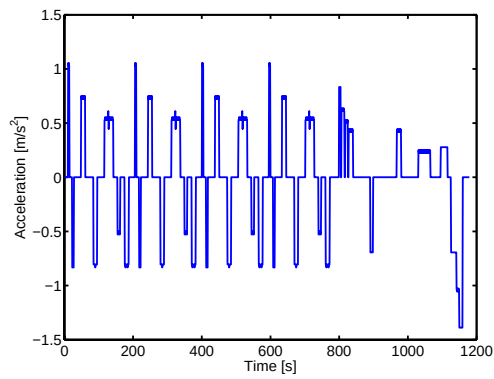
Appendix D Drive cycles



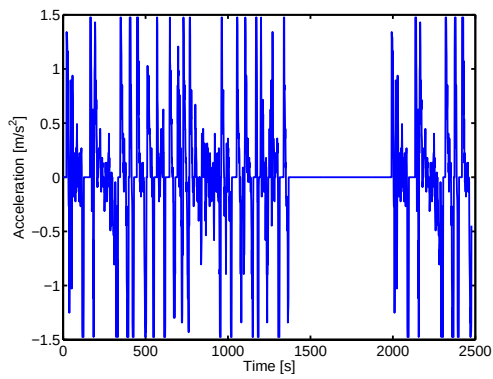
(a) New European Driving Cycle



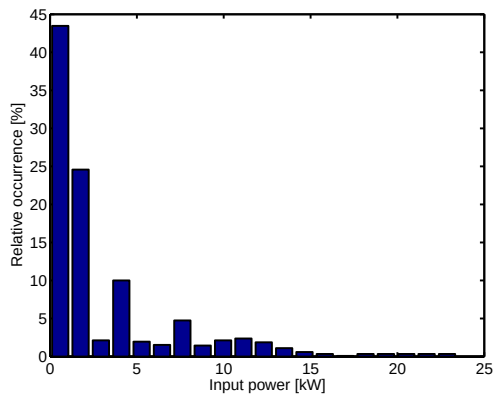
(b) FTP75 cycle



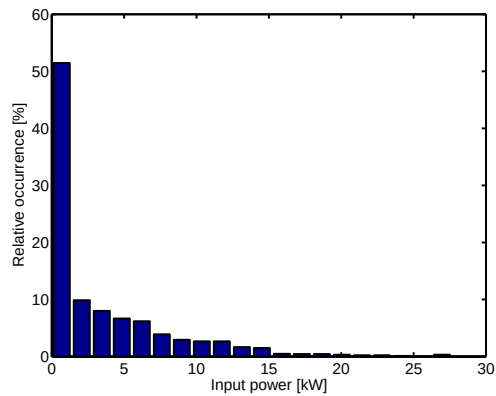
(c) Accelerations in the New European Driving Cycle



(d) Accelerations in the FTP75 driving cycle



(e) Histogram of the power used in the New European Driving mid-size vehicle



(f) histogram of the required power during the FTP75 cycle

D.2. FTP75

The FTP75 cycle [28] is the official driving cycle of the US federal government. FTP stands for Federal Test Procedure and is used for fuel economy and emission testing. The FTP75 cycle is more dynamic than the NEDC cycle. This cycle is used for simulation of drivetrain performance in city and rural road traffic and is mostly used in the USA. High speeds are not reached. The FTP75 cycle is plotted in Figure D.1(b).

In Figure D.1(d) the accelerations are shown for the FTP75 drivecycle. From this graph can be seen that the accelerations are about 1.5 times higher than the accelerations in the NEDC cycle. This results in more required power needed to drive the cycle with the same automobile. However, the FTP75 cycle has a relatively long standing still period. This means that the overall dependence of the total fuel economy on the higher loads is small. In Figure D.1(f) gives the relative occurrences of different loads during the cycle.

Ratio Setpoint Strategy

The ratio setpoint determines for a large part the engine efficiency and the 'sportiness' and comfort feeling of the car. Careful consideration therefore has to be given to the choice of setpoints. This is meant by ratio setpoint strategy.

One way of operating the CVT is consistently choosing the ratio setpoint that minimizes the fuel consumption [50]. Another is to maximize the torque reserve to have optimal performance. The first approach, Optimal Operation Line tracking, or OOL tracking, gives rather poor performance because most engines have their maximum efficiency at or near the maximum torque and if the torque reserve is maximized, fuel economy is compromised [13].

In the next section the OOL strategy is discussed, thereafter driveability issues are discussed and an compromise between the driveability and fuel economy is sought.

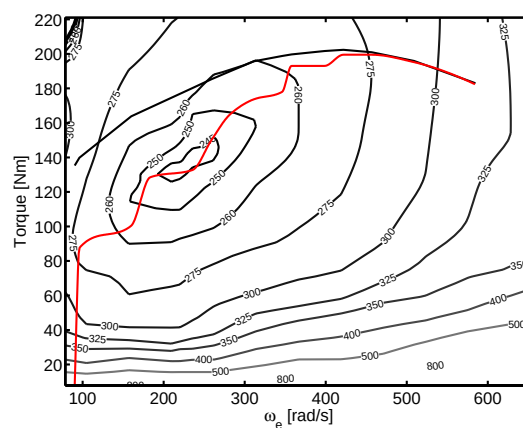


Figure E.1: Optimal Operation Line

E.1. Optimal Operation Line

OOL tracking is the most fuel economical way to operate the driveline. The OOL as shown in Figure E.1 can be calculated from the engine map by minimizing the fuel consumption in g/kWh for a set of output power values. For a given output power setting the relation between the engine torque and the engine speed is given by:

$$T_{engine} = \frac{P_{required}}{\omega_{engine}} \quad (E.1)$$

On this iso-power curve the point with the minimum fuel consumption per unit of energy is taken. If this process is repeated for a set of possible power settings, each giving one point on the OOL, an estimate of the OOL is found.

The torque output of the engine is not linear with the throttle position. Because the torque of the engine is not known, but the throttle position is, the throttlemap is used to estimate the torque output of the engine. This throttlemap is shown in Figure E.2.

Using this throttlemap and the OOL, a ratio control strategy can be derived by:

$$r(x_{throttle}, \omega_{driveshaft}) \quad (E.2)$$

The OOL can be approximated using a third order polynomial. The result is shown in Figure E.3. The smoothing of the OOL improves the smoothness of the shifting and decreases the number of datapoints in the embedded controller. On the other hand it will increase the fuel consumption of the vehicle.

E.2. Drivability

From the OOL the optimal engine speed for a give throttle position can be obtained. If this strategy is used, the drivability of the vehicle is lowered, because the response to sudden throttle movements is slow. Since drivability is also an important issue for ratio control design, a new line has to be designed that finds a good combination of fuel economy and drivability. In this paper drivability will be evaluated using simulation [34]. The ratio control strategy is evaluated for fuel economy, acceleration times of 0-100, 50-80 and 80-120.

A cost function is designed that is minimized:

$$J = l_{fuel}\gamma_1 + t_{0-100}\gamma_2 + t_{50-80}\gamma_3 + t_{80-120}\gamma_4 + a_{min}\gamma_5 + a_{max}\gamma_6 \quad (E.3)$$

In this equation γ_{1-6} are the weighting factors for the different measurement values.

The NEDC cycle is used to evaluate the fuel economy [97], acceleration simulations are used to evaluate the drivability.

The fuel consumption map of the IC engine used is defined for a certain torque/engine speed combination. To obtain the relation of the engine torque with the throttle position the map in Figure E.2

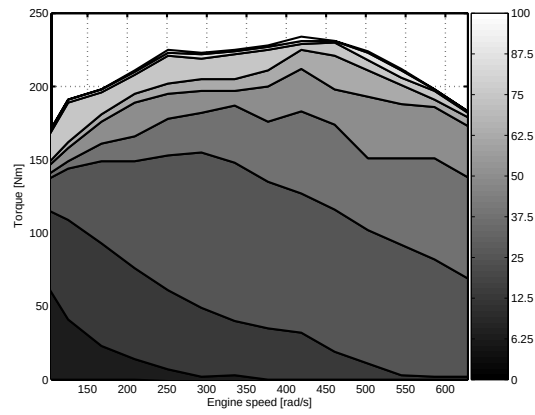


Figure E.2: Throttle map

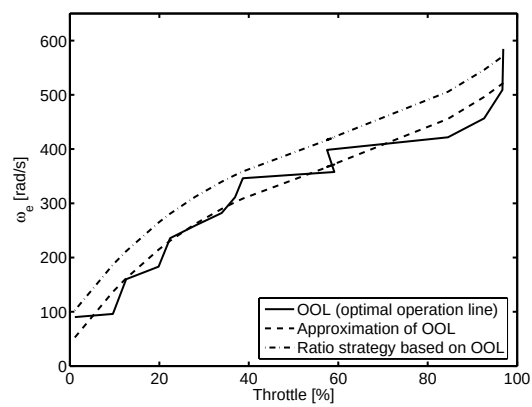


Figure E.3: Optimal Operation Line strategy compared to approximated OOL lines

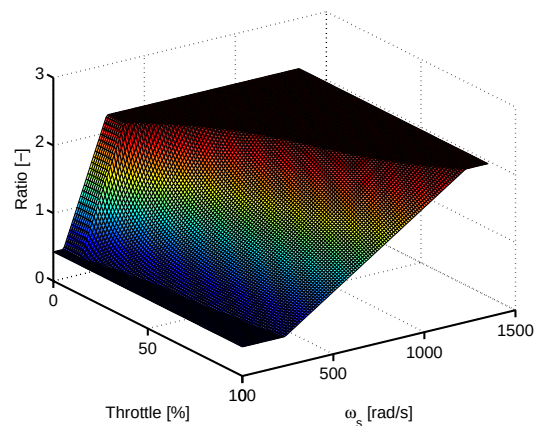


Figure E.4: Optimal Operation Line

Appendix E Ratio Setpoint Strategy

is used.

The ratio control strategy can be calculated from:

$$\omega_{engine} = f(x_{throttle}, \omega_{driveshaft}) \quad (E.4)$$

The engine speedmap gives a function of the throttle position and the driveshaft speed. For the OOL tracking this would look like the solid line in Figure E.1.

An alternative line is designed using the following method:

$$\omega_{engine} = \min(\omega_{max}, \max(\omega_{min}, f(x_{throttle}, \omega_{driveshaft}))) \quad (E.5)$$

Where $f(x_{throttle}, \omega_{driveshaft})$ is a third order approximation of the Optimal Operation Line (OOL).

A factor is added to increase the torque reserve of the engine, thereby increasing the responsiveness of the vehicle. This value is of course a compromise between agility and fuel consumption. Furthermore, the engine speed is increased with increasing driveshaft speed to enhance the feeling of the driver. The amount with which this is done is given by: ω_{d2} .

Three variables are subject to tuning: $[\omega_{d1}, \omega_{d2}, \omega_{d3}]$.

Dynamic programming is used to determine the best compromise. For all possible values of the design variables a simulation is done and the results are used to calculate the cost of each combination of the design variables [7].

Implementation is done by calculating a 2D lookup table of desired ratios for each vehicle speed and throttle position combination. This is stored in a matrix in the TCM.

Because of several reasons mentioned earlier a stepped ratio approach is investigated. For each of the possible combinations of the design variables also a stepped ratio control strategy is derived. For this purpose the continuous ratio strategy as stored in the 2D lookup table is quantized in a fixed number of evenly distributed steps.

System constraints are:

RATIO		
minimum ratio:	(LOW)	0.45
maximum ratio:	(OD)	2.25

ENGINE SPEED	
minimum engine speed:	80 rad/s
maximum engine speed:	550 rad/s

E.2.1. Shiftspeed

The reference of the transmission ratio from the ratio strategy can change as fast as the movement of the gaspedal. If the ratio of the CVT would change this fast very nervous drivetrain behavior

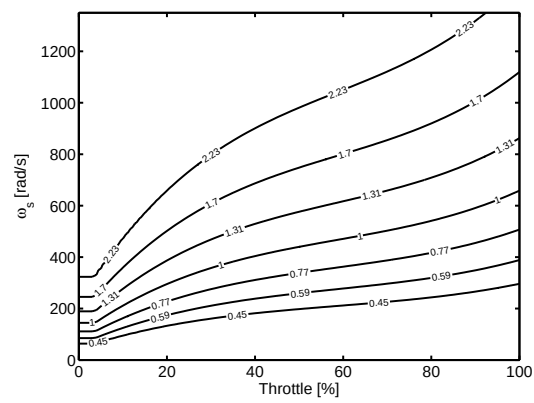


Figure E.5: CVT Ratio control strategy

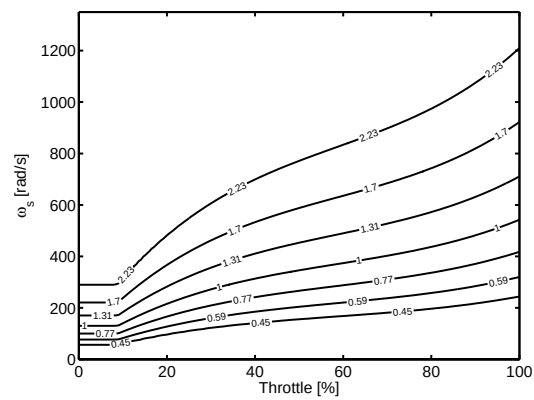


Figure E.6: CVT Ratio control strategy based on the Optimal Operation Line

would result. Moreover, due to inertial effects unwanted accelerations would occur. These effects are detrimental to the driving experience.

To avoid the nervous behavior of the variator, the reference value can be filtered. Another approach is to apply a rate-limiter. Because the variator behaves like a first order system (see Chapter 3) the latter can be used without problems.

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Nomenclature

Definitions

SYMBOL	DESCRIPTION
Beltbox	Test transmission containing only the variator
EMPAct	Electro-Mechanical Pulley Actuation
Macro-slip mode	Unstable slipping occurs, the friction coefficient decreases with increasing slip
Micro-slip mode	Stable, creep like, slipping occurs, the friction coefficient increases with increasing slip
Overdrive	Transmission ratios greater than one
Pushbelt	Metal belt that uses compression forces to transmit the driving forces instead of tension forces
Variator	The ratio changing device of a Continuously Variable Transmission
Underdrive	Transmission ratios lower than one
V-belt	Metal or Rubber belt that has slanted edges to fit in a V-shaped groove.

Acronyms

SYMBOL	DESCRIPTION
AE	Belt surface times Young's modulus
ATF	Automatic Transmission Fluid
BTS	Bedrijfs Technologische Samenwerking
CK2	Jatco CVT Transmission type
CK-Kai	Jatco CVT Transmission type
CMM	Carbone Mangialardi Mantriota
CVT	Continuously Variable Transmission
DNR	Drive-Neutral-Reverse gears
ECM	Engine Control Module
FDR	Final Drive Reduction
FTP	Federal Testing Procedure
LOW	Lowest transmission ratio
LVDT	Linear Variable Differential Transformer
MED	Transmission ratio one
MIMO	Multi Input Multi Output
NEDC	New European Drive Cycle
OD	Highest transmission ratio (overdrive)
OOL	Optimal Operation Line
PID	Proportional-Integral-Differential controller
RPM	Rotations Per Minute
SISO	Single Input Single Output
TC	Torque Converter
TCM	Transmission Control Module
TU/e	Technische Universiteit Eindhoven
UT	Universiteit Twente
VDT	Van Doorne's Transmissie

Symbols

SYMBOL	UNIT	DESCRIPTION
A	$[m^2]$	Area
E	$[N/m^2]$	Young's modulus
F_D	$[N]$	Normal force in band-block contact point
F_N	$[N]$	Normal force in pulley belt contact point
F_c	$[N]$	Clamping force
F_{clamp}	$[N]$	Clamping force
F_p	$[N]$	primary clamping force
F_r	$[N]$	radial force
F_s	$[N]$	secondary clamping force
F_t	$[N]$	tangential force
F_w	$[N]$	friction force
J_e	$[kgm^2]$	Inertia of the engine
J_p	$[kgm^2]$	Inertia primary shaft
J_s	$[kgm^2]$	Inertia secondary shaft
K_ν	$[-]$	partial derivative with respect to ν of κ
K_r	$[-]$	partial derivative with respect to r_g of κ
L	$[m]$	Length of belt
M_v	$[kg]$	Mass of the vehicle
M_ν	$[-]$	partial derivative with respect to ν of μ
M_r	$[-]$	partial derivative with respect to r_g of μ
P_{in}	$[W]$	Power input into the system
P_{out}	$[W]$	Output power of the system
P_h	$[W]$	Mechanical power consumed by the hydraulic pump
P_r	$[-]$	Partial derivative with respect to r_g of Ψ
P_ν	$[-]$	Partial derivative with respect to ν of Ψ
Q	$[N]$	Compression force
Q_0	$[N]$	Compression force at the pulley entry
Q_1	$[N]$	Compression force at the entry of the primary pulley
Q_2	$[N]$	Compression force at the exit of the primary pulley
R	$[m]$	Running radius
R_{mp}	$[m]$	Minimum running radius of belt on pulley on the primary side
R_{ms}	$[m]$	Minimum running radius of belt on pulley on the secondary side
R_p	$[m]$	Running radius of belt on pulley on the primary side
R_s	$[m]$	Running radius of belt on pulley on the secondary side
S	$[N]$	Tension force

Nomenclature

SYMBOL	UNIT	DESCRIPTION
S_0	[N]	Tension force at the pulley entry
S_1	[N]	Tension force at the entry of the primary pulley
S_2	[N]	Tension force at the exit of the primary pulley
S_f	[-]	Safety factor
T_d	[Nm]	Torque on the secondary side caused by the road-load
T_e	[Nm]	Engine torque
T_h	[Nm]	Input torque of the hydraulic pump
T_{fr}	[Nm]	Friction torque of the hydraulic pump
T_{input}	[Nm]	Input torque of the variator
T_{loss}	[Nm]	Torque loss in the variator
T_{max}	[Nm]	Maximum input torque that can be transmitted by the variator
T_p	[Nm]	Torque transmitted by the variator on the primary side
T_s	[Nm]	Torque transmitted by the variator on the secondary side
T_{sf}	[Nm]	Safety margin
a	[m]	Axial distance between pulleysets
a_0	[-]	Coefficient of the Stribeck curve approximation
a_1	[-]	Coefficient of the Stribeck curve approximation
b	[m]	Perpendicular distance between fixed pulley sheaves.
c_{fl}	[cc]	Flow per rotation constant of the pump
c_0	[-]	viscous damping coefficient
c_r	[-]	viscous damping coefficient of Shafai's model
d	[m]	Width of belt
d_e	[m]	Width of belt element
m_r	[kg]	moving pulley mass
p_l	[bar]	Line pressure
p_0	[bar]	Minimal line pressure
p_{cf}	[bar]	Line pressure needed for clamping force
q_N	[N/m]	distributed normal force
q_r	[N/m]	distributed radial force
q_t	[N/m]	distributed tangential force
q_w	[N/m]	distributed friction force
r	[-]	transmission ratio
r_g	[-]	geometric transmission ratio R_p/R_s
r_t	[-]	torque transfer transmission ratio T_p/T_s
r_s	[-]	transmission speed ratio ω_s/ω_p
r_{s0}	[-]	transmission speed ratio at zero load

SYMBOL	UNIT	DESCRIPTION
v	[m/s]	Velocity
v_{belt}	[m/s]	Longitudinal velocity of the belt
v_e	[m/s]	Belt element velocity
v_p	[m/s]	Velocity of the pulley at the running radius
v_s	[m/s]	Slip velocity
v_1	[m/s]	Static to sliding friction transfer velocity
v_2	[m/s]	Belt velocity
x	[m]	Moveable pulley position.
x_p	[m]	Moveable primary pulley position.
x_s	[m]	Moveable secondary pulley position.
Ψ	[-]	clamping force ratio in stationary conditions
α	[rad]	active part of the wrapped angle
α_p	[rad]	active part of the wrapped angle on the primary pulley
α_s	[rad]	active part of the wrapped angle on the secondary pulley
β	[rad]	Pulley wedge angle
γ	[rad]	Slip angle of belt element on pulley
δm	[m]	Average gap between the belt elements
δo	[m]	Initial gap in the belt
δt	[m]	Sum of the gaps between the belt elements
ϵ	[-]	Strain in the belt
ϵ_w	[-]	Parameter determining the slope of the static part of the continuous friction model
ϵ_0	[-]	Strain in the belt on the pulley entry side
η	[-]	transmission efficiency (P_{out}/P_{in})
κ_{cmme}	[-]	Experimentally determined factor for the CMM transient variator model
κ_{ide}	[-]	Experimentally determined factor for Ide's transient variator model
μ	[-]	friction coefficient
μ_c	[-]	Coulomb friction coefficient
μ_{eff}	[-]	traction coefficient
μ_{max}	[-]	global maximum of the traction coefficient
μ_s	[-]	static friction coefficient
ν	[-]	relative slip
ϕ	[rad]	Angle of contact point of belt and pulley
ϕ_p	[rad]	Angle of contact point of belt and pulley on the primary side

Nomenclature

SYMBOL	UNIT	DESCRIPTION
ϕ_s	[rad]	Angle of contact point of belt and pulley on the secondary side
τ	[-]	Torque ratio
θ	[rad]	Position on wrapping arc
ω_e	[rad/s]	Engine speed
ω_p	[rad/s]	Speed primary axis
ω_s	[rad/s]	Speed secondary axis
ω_d	[rad/s]	Driveshaft speed

Summary

Continuously Variable Transmissions (CVT) are becoming increasingly popular in automotive applications. What makes them attractive is the ability to vary the transmission ratio in a stepless manner without interrupting the torque transfer. This increases comfort by eliminating the discrete shifting events and increases performance by choosing the most suitable transmission ratio for every driving situation. Using a CVT could potentially save more than 15% of fuel consumption compared to manually shifted vehicles. This figure however is never met, because of the internal losses in the CVTs in production today.

If the losses in a CVT can be lowered, then the overall fuel economy of a CVT equipped vehicle will be improved with the same amount. With current CVTs ranging around 80% efficiency, an improvement of around 10% is possible compared to currently available CVTs if an optimal actuation and control system is used. This thesis is about the optimization of the control system of the CVT by using slip as the control variable. This is part of a larger project focussing on the entire actuation and control system. Also a CVT with Electro-Mechanically Pulley Actuation (EMPA_{ct}) is developed aiming to reduce the power consumption of the CVT actuation system. Combined, these two projects aim to improve the fuel economy of the CK2 transmission from Jatco with 10%.

Models for the clamping forces and traction in the variator are compared. The continuous belt model is compared with a pushbelt model. A parameter study shows the influence of the model parameters on the outcome of the models. The output of the models are also compared to measured values.

A nonlinear dynamic model for slip in the variator is derived. This model can be linearized in certain operating points. This model can be used for the design of a control system, simulation of slip in the variator or for analysis.

Measurement of slip directly is not possible, therefore a good estimation method is needed. Several estimations of slip in the variator are compared. The position measurement of the pulley is used in the measurements shown in this thesis.

Measurements on a beltbox testrig are given that clearly show a relation between slip and efficiency and slip and traction. This relation changes as a function of other parameters like speed, ratio, clamping force etc. Estimation of the efficiency potential of the pushbelt variator shows that a

Summary

potential of between 5% for high torques and 20% for low torques exists.

A slip control system is developed to show the possible efficiency improvement. First, a beltbox setup is used to test a simplified slip controlled variator. Ratio changing is not taken into account in this setup. After successful tests with this setup another setup is used that incorporates a Jatco CK2 transmission and an internal combustion engine. This test setup is more realistic, but therefore also more complicated to control. A gain scheduled approach is used to compensate for the slower actuation system. This system is then also applied to a testing vehicle.

Samenvatting

Continu Variabele Transmissies (CVT) worden steeds populairder in automotive toepassingen. Wat ze aantrekkelijk maakt is mogelijkheid om de overbrengverhouding traploos te veranderen, zonder dat er een onderbreking van de koppeloverdracht is. Dit verhoogt het comfort door het ontbreken van discrete schakelingen en verhoogt de prestaties doordat de beste overbrengverhouding voor elke omstandigheid kan worden gekozen. Het gebruik van een CVT kan potentieel meer dan 15% besparing opleveren in het brandstofverbruik in vergelijking met handgeschakelde autos. Dit getal wordt echter niet gehaald, omdat de interne verliezen in hedendaagse CVTs te hoog is.

Als de verliezen in een CVT verlaagd kunnen worden, dan zal het brandstofverbruik van een auto met een CVT evenredig dalen. CVTs die momenteel op de markt zijn halen een efficiëntie van ongeveer 80%. Als een optimaal actuatie en regelsysteem wordt gebruikt kan deze efficiëntie met 10% verbeterd worden. Dit proefschrift gaat over de optimalisatie van het regelsysteem van de CVT door het gebruik van slip als de regelgrootte. Dit is een onderdeel van een groter project dat als doel heeft het gehele actuatie en regelsysteem van de CVT te verbeteren. In dit kader is ook de Electro-Mechanical Pulley Actuation CVT (EMPACT CVT) ontwikkeld, welke het rendement van het actuatie systeem van de CVT verbeterd. De combinatie van slip regeling en EMPACT CVT levert een besparing van 10% op.

Modellen voor de knijpkracht en tractie in de variator worden vergeleken. Het continue band model wordt vergeleken met het duwband model. De invloed van de parameters van de beide modellen worden bekeken. Beide modellen worden gefit op gemeten waarden.

Een niet-lineair dynamisch model voor slip in de variator wordt afgeleid. Dit model kan gelin-eariseerd worden in bepaalde werkpunten. Dit model kan gebruikt worden voor het ontwerp van een regelsysteem, simulatie van slip in de variator of voor analyse van het systeem.

Het meten van slip kan niet direct gebeuren. Daarom is er een schattingsmethode nodig. Meerdere methoden voor het schatten van slip in de variator worden vergeleken. Het meten van de positie van de poelie is gebruikt bij de metingen die in dit proefschrift worden getoond.

Metingen op de L-bak proefstand worden getoond. Deze metingen laten zien dat er een duidelijke relatie bestaat tussen slip en efficiëntie en tussen slip en tractie. Deze relatie verandert als functie van parameters als snelheid, overbrengverhouding, knijpkracht enz. Schattingen van het rende-

Samenvatting

mentspotentieel van de duwband variator laten zien dat er een potentile verbetering mogelijk is van tussen de 5% voor hoge koppels tot meer dan 20% voor lage koppels.

Een regelsysteem is ontwikkeld om aan te tonen dat het mogelijk is het rendement te verbeteren door slip in de variator te regelen. Eerst wordt er een regeling gecomplementeerd op de L-bak. Hierbij wordt een versimpelde slip regelaar gecomplementeerd. Het schakelen wordt hier buiten beschouwing gelaten. Na succesvolle testen wordt de regeling geïmplementeerd op de CK2 transmissie van Jatco. Deze transmissie is gekoppeld aan een verbrandingsmotor. Deze proefstand is realistischer, maar daarmee ook moeilijker te regelen. Een gain-scheduling aanpak wordt gebruikt om de tragere respons van het actuatiesysteem te compenseren. Dit systeem is daarna ook toegepast op een testvoertuig.

Curriculum Vitae

Bram Bensen was born on June 3rd 1974 in Amersfoort, the Netherlands. After graduating high school he studied a year abroad in the USA. In this period he participated in the Experiential Learning Program and in regular classes at Ramapo College of New Jersey. After that year he started studying Mechanical Engineering at the University of Twente in the Netherlands. In 2000 he obtained his Masters degree from prof. dr.ir. J.B. Jonker. His specialization was Mechanical Automation and he graduated on a modal reduction method for the dynamic analysis and simulation of flexible mechanisms and manipulators.

In this period he also worked for Limis B.V. in Enschede. At this company he developed a job-shop scheduling module for the Shopfloor Planning and Registration system under development by Limis B.V.

After he graduated from the University of Twente he started working as a consultant for Cap Gemini Ernst & Young. Here he worked on implementing SAP software at several companies.

In 2002 he started his Ph.D. studies at the Technische Universiteit Eindhoven under the supervision of prof. dr. ir. Maarten Steinbuch, dr. Bram Veenhuizen and prof. ir. Nort Liebrand. The project was conducted in cooperation with ir. T.W.G.L. Klaassen and ir. K.G.O. van de Meerakker. The project details and results are discussed in this thesis.